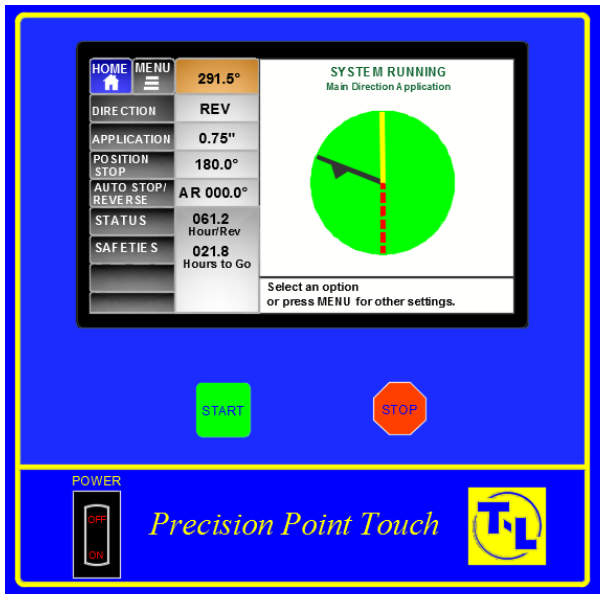
T-L IRRIGATION CO.

Hastings, Nebraska



<u>P</u>recision <u>P</u>oint <u>T</u>ouch

PPT – TOUCH SCREEN BASIC

OPERATORS MANUAL

CD90442

July 17, 2025
Red 1-Chip = v276
Green 2-Chip = v256 Main,v256 Display
v256 Release Notes, revised GPS Comm Lost Testing

CONTENTS

IMPORTANT: USE THIS MANUAL IN CONJUNCTION WITH: CD90440 T-L PIVOT OPERATORS MANUAL.

CONTENTS

T-L Irrigation Company
151 E HWY 6 & AB Rd
Hastings, NE 68901
(402)-462-4128
www.tlirr.com

FIRMWARE RELEASE NOTES

See Sections 6.7, 6.8 & 6.9 for information on how to update the firmware.

RED 1-CHIP BOARD VERSION:

PPT – v276 Basic Release Notes (April 11, 2025) (Major changes in red font)
- Fixed issue with Com 2 and RS485 Full Duplex driver enable line not going high. (v272)
- Telemetry $STA string changed so Run State = 1 during Soft Start and Start Movement Delay. (v273)
- Eliminated the ability to bypass Soft Start by pressing Start. Now you must wait 20 seconds. (v273, 274)
- Fixed an issue with the Red 1-Chip Board not recognizing some FAT32 UBS's. (v274)
- Fixed an issue with bypassing the Start Delay Timer and the Time to Start stays on screen. (v275)

PPT – v271 Basic Release Notes (Dec. 2, 2024, received at T-L 09/26/24)
- Fixed issue with Watchdog not enabled.
- Note, v270 released Nov. 6, 2024 on Dealer Point (received at T-L 9/26/24 for testing) was actually v271 as shown on the Title and Status Screen. Changed filename.

PPT – v270 Basic Release Notes (received at T-L 09/24/24, not released)
- Fixed issue with Log File being scrambled with Telemetry set to Off, removed $STA string from Log.

PPT – v269 Basic Release Notes (June 24, 2024)
- Fixed issue with not being able to adjust speed when running with ET Sensor Safety Disabled.

PPT – v268 Basic Release Notes (June 5, 2024)
- Fixed issue with GPS Not Set Error sometimes at startup. (v268)
- Fixed issue with Date/Time access being really slow. (v267)
- Fixed issue with using GPS Antenna to set Pivot Point. (v266)
- Changed ET Sensor Safety Times, Max = 240 secs, Default = 120 secs. (v265)

GREEN 2-CHIP BOARD VERSION:

PPT – v256 Main & v256 Display Release Notes (July 14, 2025)
- Telemetry $STA string changed so Run State = 1 during Soft Start and Start Movement Delay. Fixes issue where FieldWise Computer Controller will disengage relays 10 secs after start.
- Eliminated the ability to bypass Soft Start by pressing Start. Now you must wait 20 seconds.

PPT – v253 Main & v253 Display Release Notes (June 5, 2024)
- Changed ET Sensor Safety Times, Max = 240 secs, Default = 120 secs.

PPT – v251 Main & v251 Display Release Notes (June 13, 2023)
- Fixed issue if using Telemetry and last End Gun Area on the PPT Panel is set to 359.5 degrees.

PPT – v250 Main & v250 Display Release Notes (Jan. 16, 2023)
- Fixed occasional issue with safeties not getting logged in datalog file or on the telemetry portal.
- Fixed occasional issue with angle not reading correctly when placed in Run.
- Added entire telemetry status string to the log file.
- NOTE: MUST update FieldWise firmware at the same time updating PPT from v248 or prior.

PPT – v249 Main & v249 Display Release Notes (July 22, 2022)
- Work with FieldWise on inadvertent telemetry application change to PPT where PPT goes to min. spd.
- Now the PPT Panel must see the Application change command from FieldWise for 3 consecutive times.
- NOTE: Update suggested if using FieldWise Computer Controller and MUST update FieldWise firmware at the same time updating PPT to v249.

1 INTRODUCTION

1.1 *General Information*

- The Precision Point Touch Manual is to be used in conjunction with the standard T-L PIVOT OPERATORS MANUAL. Please read these manuals completely before operation of the system. If any questions arise, please contact your T-L Dealer.

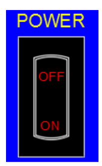
1.2 *Operation Guidelines*

- If the Precision Point Touch panel is to be powered from a “spark ignition engine”, the following must be used to reduce electronic noise:
  - Resistor Plugs
  - Suppresser Plug Wires

1.3 *Control Panel Power – PPT Basic*

NOTE: PPT Panel will only be powered when the Engine or Electric Motor is running.
- Power to the PPT Panel can be controlled by the “Power Switch” on the PPT Panel.

Power Switch on Front of Panel



Control Notes:
- The PPT Basic does NOT control the start of the Hydraulic Pump (engine or electric) or the Water Pump.
- The engine or electric motor driving the Hydraulic Pump must be started first, then the PPT Basic can be powered and used.
- The PPT Basic will then provide full control of the pivot direction, application rate (speed), end guns, etc.

1.4 *Power Check – Initial Start*

Verify 28VDC power to the PPT Panel when switched ON.

Gnd	= Term 4 & 5
+28VDC	= Term 6 & 7

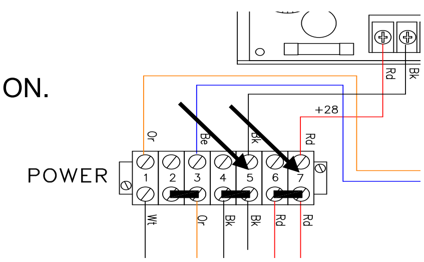


Fig. 1 Power Check.

1.5 *What to Do First - MENU Items to Set*

Enter settings for each of the following items as they pertain to your particular system.
- System
- End Gun 1
- End Gun 2
- Auxiliary Application
- Time & Date
- Position Sensor

The following 3 MENU items are for troubleshooting or service help.
- Diagnostics
- USB
- Display

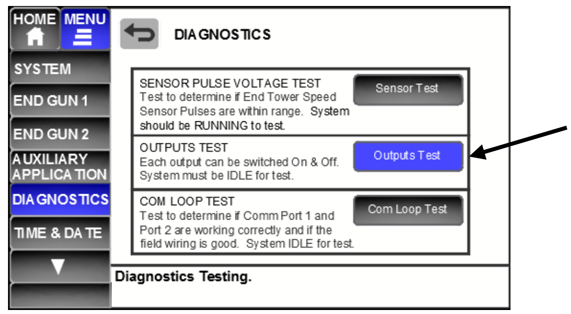
1.6 What to Do First - HOME Items to Set

Enter settings for each item for specific operation.
- Direction
- Application
- Position Stop
- Auto Stop/Reverse

The following 2 Menu items are for information and troubleshooting or service help.
- Status
- Safeties

1.7 What to Do First – Outputs Test

Using "Outputs Test" under the MENU – DIAGNOSTICS option, each output can be cycled On and Off to test operation.



1.8 What to Do First - Operational Check of Safeties

Safeties should be checked at initial startup to ensure they are operational.

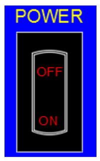
Safety	Description	How to check:
ET Sensor Safety	No pulses from ET Sensor Motor for XX sec. (Adjustable 30 to 240 sec. in HOME-SAFETIES.) (v268, v253 changed to 240s max.)	Remove Blue Span Cable wire (#5) from END TOWER Term Block in PPT Panel.
Overwater Safety	System has not moved 1.0 degree In amount of time set.	Change Overwater Time to 10 minutes in HOME - Safeties, then run system as slow as possible. Watch the Overwater Timer count down on the STATUS screen.
Auto Stop Shutdown	Auto Stop angle set on HOME Screen has been met if set "By Angle". Or End Tower Switch has been activated.	Activate End Tower Switch (if used). Set Auto Stop by Angle, run system to that angle, make sure AS/AR Angle Ignore Delay Timer has counted down.
Position Stop Shutdown	Position Stop angle set on HOME Screen has been met.	Set Postion Stop Angle, run system to that angle, make sure AS/AR Angle Ignore Delay Timer has counted down.
GPS Comm Lost (if GPS)	GPS Antenna cannot be detected on Com Port 1 for 40 secs (v239) at start (after Soft Start or Delay Timer has counted down.)	Unplug Com Port 1 and start system. Bypass "Soft Start" and "Delay Timer" by pressing Start.
Encoder Comm Lost (if Encoder)	Encoder cannot be detected on Com Port 1 for 6 seconds at start (after Soft Start or Delay Timer has counted down.)	Unplug Com Port 1 and start system. Bypass "Soft Start" and "Delay Timer" by pressing Start.

2 CONTROL PANEL OPERATION & FEATURES

2.1 *Control Panel Power – PPT Basic*

NOTE: PPT Panel will only be powered when the Engine or Electric Motor is running.
- Power to the PPT Panel can be controlled by the “Power Switch” on the PPT Panel.

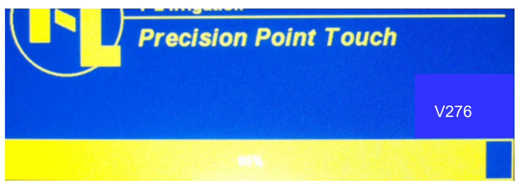
Power Switch on Front of Panel

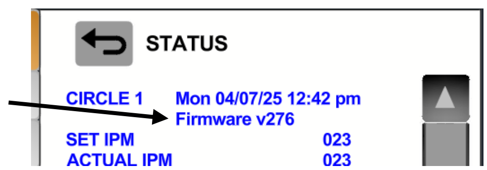


Control Notes:
- The PPT Basic does NOT control the start of the Hydraulic Pump (engine or electric) or the Water Pump.
- The engine or electric motor driving the Hydraulic Pump must be started first, then the PPT Basic can be powered and used.
- The PPT Basic will then provide full control of the pivot direction, application rate (speed), end guns, etc.

2.2 *Control Panel Power Up – Title Screen – Firmware Version*

The **Title Screen** shown at power up will display the Firmware Version(s) in the lower right.
NOTE: Firmware Version can also be viewed on the Status Screen - added in v248, 5/25/22.
- Green Board 2-Chip version will have 2 firmwares displayed.
- Red Board 1-Chip will only have 1, example v276.





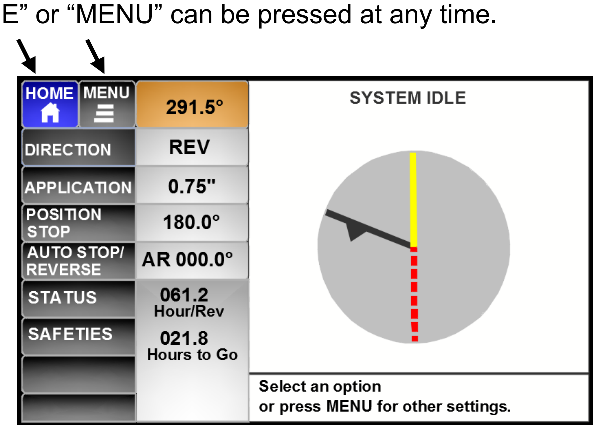
2.3 *Control Panel Power Up – Safety Screen*

If a Safety has occurred causing the system to lose power, the Safety Message will be displayed at power up after the Title Screen.
- Press the “STOP” key to acknowledge the safety and continue to the HOME screen.

2.4 *Control Panel Power Up – HOME Screen*

See **Section 3** for more information

“HOME” or “MENU” can be pressed at any time.



2.5 HOME Screen – Pivot Circle & Line Colors

<u>Pivot Circle Colors</u>

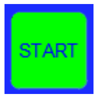
Pivot is Stopped - Idle	Gray	
Idle with a Safety Disabled	Gray Red Center	
Start Delay	Orange	
Running	Green	
Running with a Safety Disabled	Green Red Center	
Pivot Stopped - Safety Active	All Red	
Auxiliary Application Area	Magenta	
End Gun 1 Area On	Blue	
End Gun 2 Area On	Cyan	

<u>Pivot Line Colors</u>

Pivot Angle Location	Black	
Direction of Travel	Black Arrow	
Position Stop Location (In Field Stop)	Red Dashed	
Auto Stop Angle Location (at Field Boundary)	Red	
Auto Reverse Angle Location (at Field Boundary)	Yellow	

2.6 *START Key*

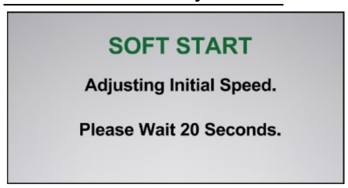
The START key can be used while in any HOME screen, will <u>not</u> work if "GPS Not Set" is shown.



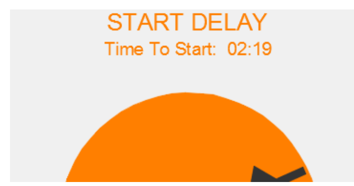
When the START key is pressed, one of 2 screens will appear.

- Which appears depends if a "Start Delay" time has been set in MENU - System setup.
- Note the Soft Start Screen will appear on the Start Delay Screen for 20 seconds.

"**Soft Start**"
- if no Start Delay Time



"**Start Delay**"
- if Start Delay Time set

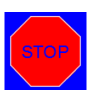


2.7 *STOP Key*

The STOP key can be pressed at any time to stop the system.

When pressed once, a pop up confirmation screen will appear.

- either press the STOP key again, or press the "Yes" button on the screen to stop.

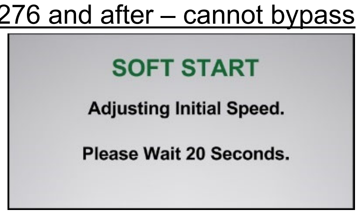


2.8 *Soft Start*

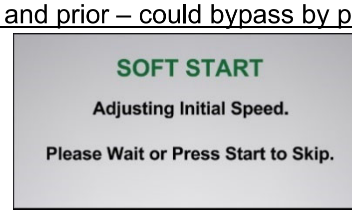
"Soft Start" is used to make sure the system always starts at a slow speed.

- 20 seconds of operation:
  Close MFC for 15 secs, open for 5 seconds.
  Then energize Direction Control, etc.

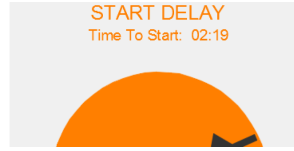
<u>v276 and after – cannot bypass</u>



<u>v272 and prior – could bypass by pressing Start</u>.



2.9 Start Delay



"Start Delay" is used to delay all start functions for a set time.
- This delay is useful with a Corner system using GPS Navigation.
- This allows time for an "RTK Fix" to be established so the Safety Solenoid at the LRDU will open and allow the end tower to move.

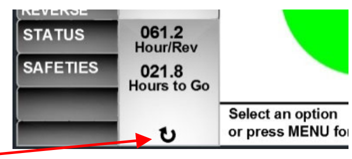
If Start Delay Timer at Start:
Performs "Soft Start" during Start Delay Time.
At the end of Delay Timer, continue with Start sequence:
Energize Direction Control, etc. as described in the "Soft Start" section above.
WARNING: The START key can be pressed to bypass the Start Delay, but the End Tower MFC may be in a fully closed position per the "Soft Start", depending at which point during the Start Delay the START key was pressed. If a very slow speed is set, an ET Sensor Safety may occur.

If Auto Restart is set to Yes, then the Start Delay will be active when the system restarts.

2.10 Auto Restart

"Auto Restart" can be set to Yes or No in MENU – Setup. This option is typically used in electric drive systems equipped with Auto Restart/Load Control.

Set to YES:
- If a loss of power occurs while the system is running, the system will resume running when power is restored.
- Upon power-up, the PPT panel will start operation with either the "Soft Start" or "Start Delay".
- The Auto Restart icon will appear on the Home Screen.



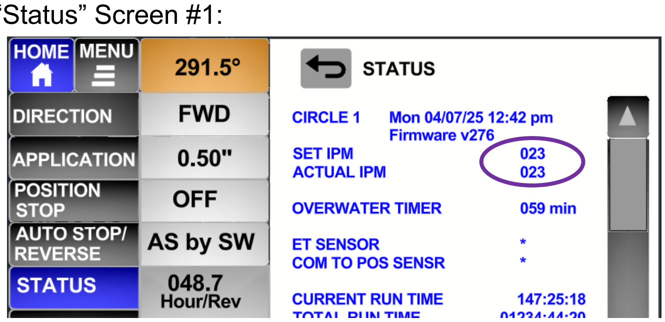
Set to NO:
- If a loss of power occurs while the system is running, the system will <u>not</u> resume running when power is restored and will return to System Idle.

2.11 Automatic Speed Control (v214 and after)

The following occurs in the Speed Control Loop:
- Checks "Actual Speed" for 20 seconds. (speed pulses)
- Then compares "Actual Speed" to "Set Speed". If ACTUAL is different than SET by more than 1.0 IPM, then make an MFC adjustment. Conversely, if ACT is within 1.0 IPM of SET, do not adjust.
- Speed adjustment if needed (increase or decrease speed) determined by a proportional formula.
  - amount of adjustment time is determined by "Spd Adj Rate MFC" in System Setup.
- Repeat, check "Actual Speed" for 20 seconds, compare and make adjustments if necessary, etc. every 20 secs.

Anytime the Application (speed) or Direction is changed while Running, it will reset the Automatic Speed Control and start again.
- Check "Actual Speed" for 20 seconds, compare and make adjustments, etc.

"Status" Screen #1:



SET IPM = calculated end tower speed for the Application set.
- Will show "DISABLED" in place of number if Spd Sensor Disabled.

ACTUAL IPM = end tower speed from speed pulses.

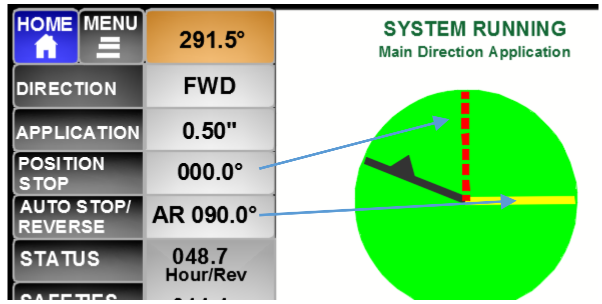
"+" or "-" will show after 20 second loop if adjusting speed.

2.12 Position Stop vs. Auto Stop/Reverse by Angle

See HOME SCREEN – OPTIONS Section for further information.

POSITION STOP is used to quickly set an in-field stop location by Angle on the screen.
- system will shut down immediately when the set angle is reached. The "Auto Stop/Reverse Delay" does NOT apply to the Position Stop.

AUTO STOP/REVERSE is typically used for "wiper" part circle end of field stops or reverse locations and can be set By Angle or By Switch.
- The "Auto Stop/Reverse Delay" timer will be applied when the set angle is reached.

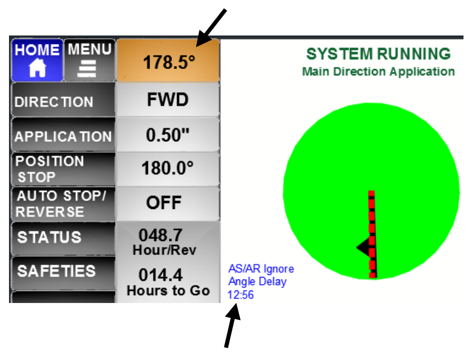


Both can be set, but whichever is reached first is the action that will be taken.

2.13 Angle Delay / Ignore Timer

Angle Delay Timer Set in MENU – SYSTEM to "Auto Calculate" or "Fixed 5 minutes".

<u>Important Note if Stop or Reverse set By Angle:</u>
- System WILL NOT STOP OR REVERSE AT AN ANGLE if the "AS/AR Ignore Angle Delay Timer" is active.
- Do **NOT** program the Position Stop or Auto Stop or Auto Reverse "By Angle: **within 3 degrees** of the current position **when starting**. The Angle Ignore Timer may be active and the action will be ignored.
- This timer is used to ignore the stop/reverse set angle so the system does not stop immediately if for example it is at 180.0 and wanting to go full circle back to 180.0.
- If either Position Stop or AS/AR by Angle is changed while running, the Angle Delay timer will be reset.



Example: System started FWD at 178.5º and Position Stop set at 180^0. If 180^0 is reached before the Delay timer is finished, system will NOT stop.

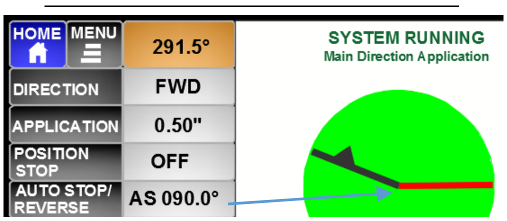
2.14 Auto Stop/Reverse Operation – By Angle vs By Switch

With Auto Stop/Reverse, the Pivot can be automatically stopped or reversed in one of 2 ways:

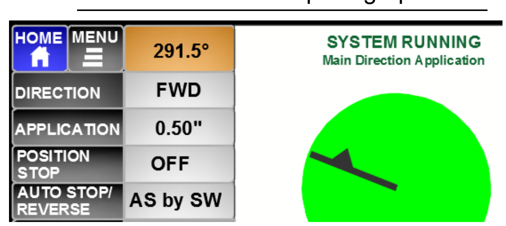
A. Set by Angle.

B. Using a Reversing Switch at the End Tower or at the Pivot Point wired into Terminal 7 of the 12 Terminal Block for the End Tower .

AS/AR **by Angle**
Angle Line shown on pivot graphic.



AS/AR **by Switch**
No line shown on pivot graphic.



<u>Auto Stop/Reverse Operation:</u>
- Reversing Switch activated or set Angle reached, end tower direction changes and moves the opposite direction for 25 seconds, and no changes on the panel can be made during those 25 seconds.

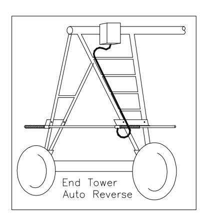


Fig. 2 End Tower Auto Reverse

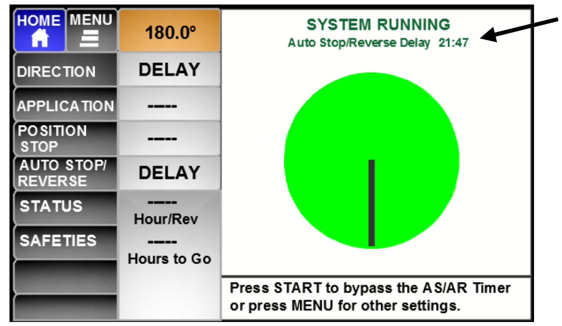
2.15 Auto Stop/Reverse Operation – ET Switch Activated but Set By Angle

If Auto Stop/Reverse is set “By Angle”, but the End Tower Switch is activated before the set angle is reached, the system will shut down with an “Auto Stop Safety” since this is an unexpected action.

2.16 Auto Reverse Delay Timer

- If an Auto Stop/Reverse Delay Time is set then the Delay Timer will show when the Auto Stop/Reverse is activated either “By Angle” or “By Switch”.
  - Press Start key to bypass the timer.

- If Auto Stop is set the system will shutdown when the Delay timer counts down to 0.
- If set to Auto Reverse, system will resume operation.

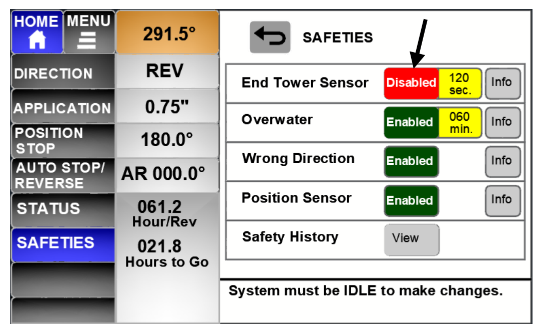


2.17 Disable “ET Sensor Safety”

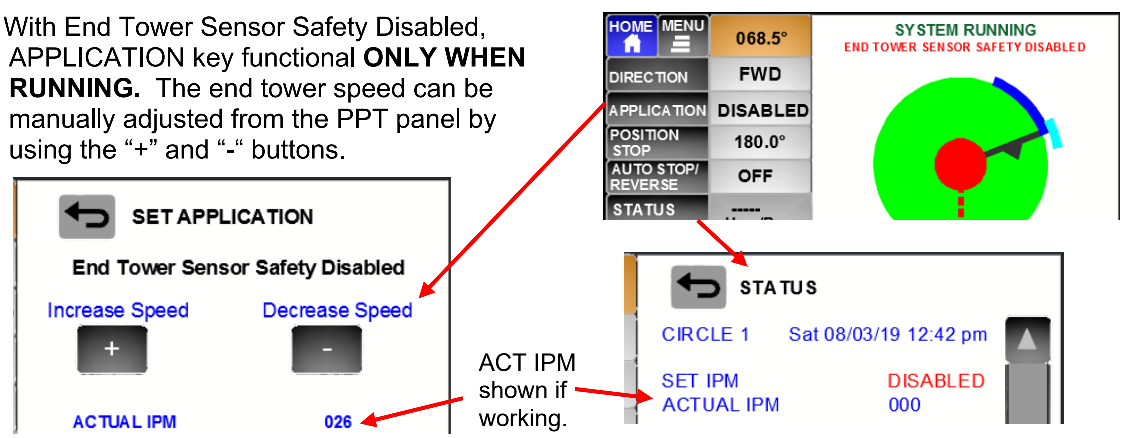
An “ET Sensor Safety” occurs when the PPT Board has not received pulses from the End Tower Sensor Motor in the time set in SAFETIES.

This safety can be disabled in HOME-SAFETIES.

Operation is then allowed as follows:
- ET Sensor Safety will not occur.
- Automatic Speed Control is not active. System will only run at one speed.
- Auxiliary Application or Web VRI not functional.



Note: With End Tower Sensor Safety Disabled, APPLICATION key functional **ONLY WHEN RUNNING.** The end tower speed can be manually adjusted from the PPT panel by using the “+” and “-“ buttons.



2.18 “Back” Button and Keypad “Esc”

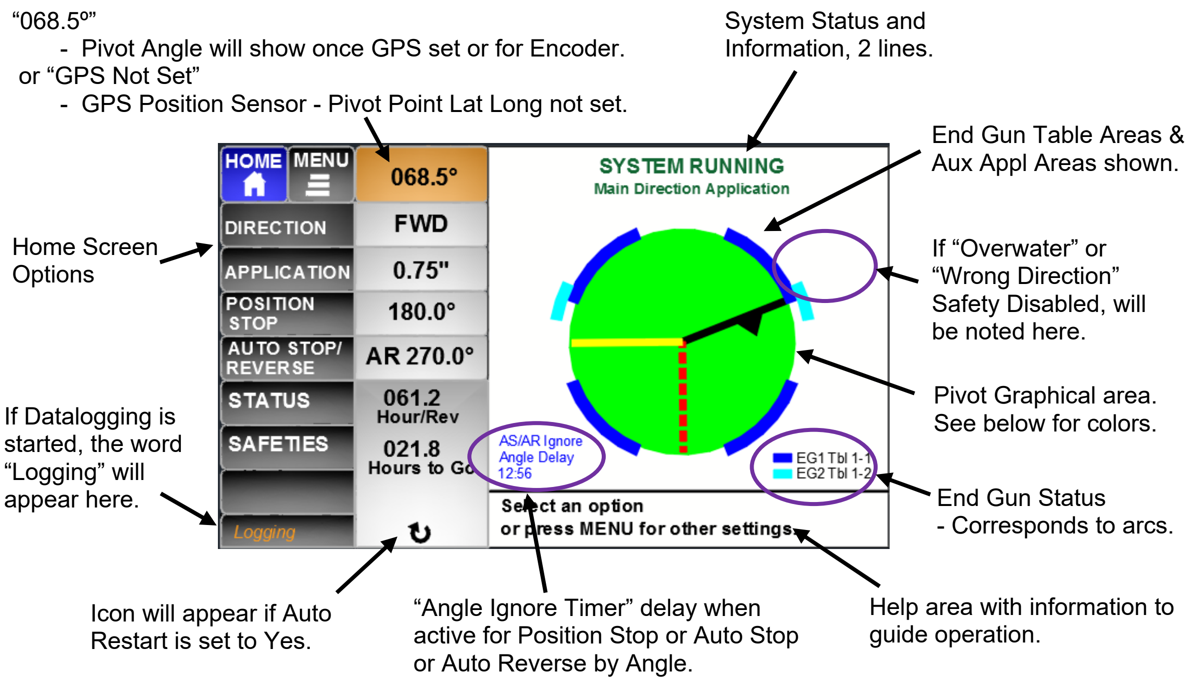
- The “Back” button will take you back to the previous “function” you were in.
- Keypad “Esc” button returns to what was being set without changing the value.

2.19 Web Based Telemetry

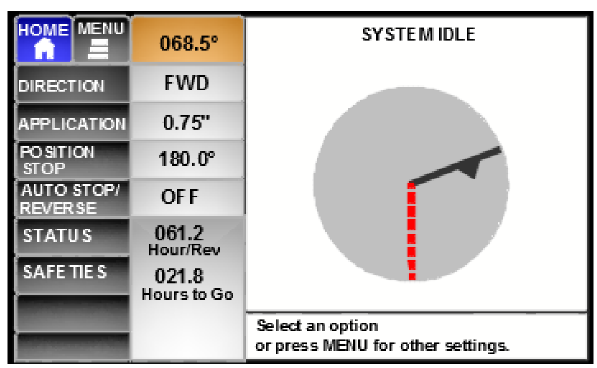
The PPT can be connected to the following devices for web/tablet/phone access.
FieldWise Computer Controller or AgSense Precision Link

3 HOME SCREENS – IDLE & RUNNING

3.1 HOME Screen Information:

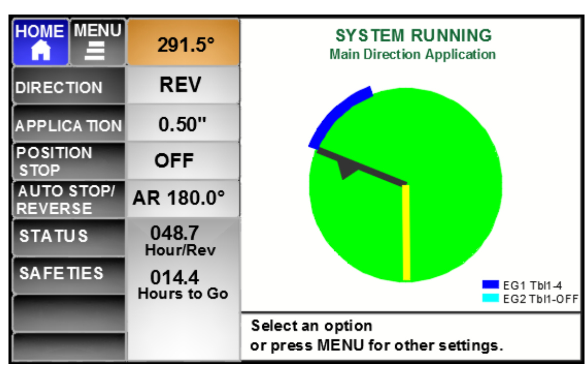


3.2 HOME Screens - Examples



<u>System Idle:</u>
At 68.5º
Set to move FWD at 0.75" application
Position Stop set at 180.0º

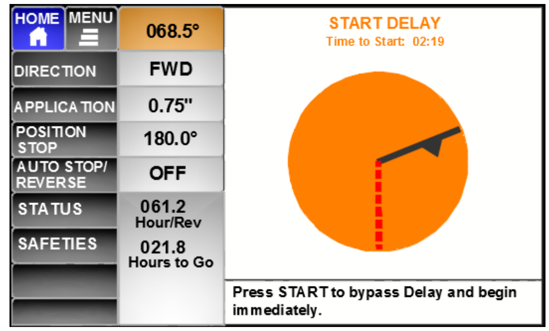
Position Stop is meant for an in-field stop.



<u>System Running:</u>
At 291.5º
Set to move REV at 0.50" application
Set to Auto Reverse at 180.0º
End Gun 1 is On, using Table 1 Area 4
End Gun 2 is Off, using Table 1

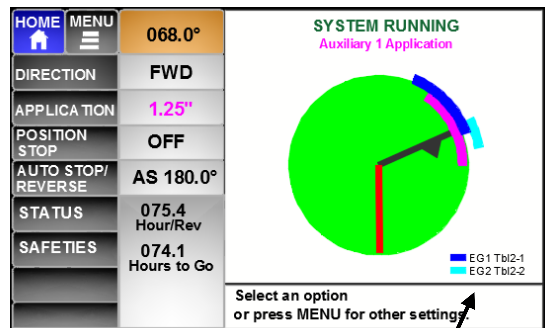
Start Delay Timer

Start Delay has been set
Start Button pressed.
System movment is delayed.
2 mins 19 secs left in Delay Time
Once Delay Time has elapsed:
System will move Fwd at 0.75” appl.



System Running – End Guns On, Auxiliary Application On:

At 68.0°
Set to move FWD at 1.25” application:
Aux Application set and running
in the area shown in the purple arc.
Set to Auto Stop at 180.0°
System is on Circle 2, denoted by EG Tables.



End Gun Status:

2 End Guns can be used.
3 Tables possible per End Gun.
Table used corresponds to Circle Number set in Menu - System.
Ex. Circle 2 selected, Table 2 for EG 1 & EG 2 will be used.

End Guns can be set to:

Use Tables:
Table number used shown and:
- if in Area to be on, Area number shown and arc for that area is shown on the pivot circle.
- if in Area to be off, then “OFF” is shown

EG1 – using Table 2 (Circle 2), Area 1
- set to **Use Tables** - Table number is shown.

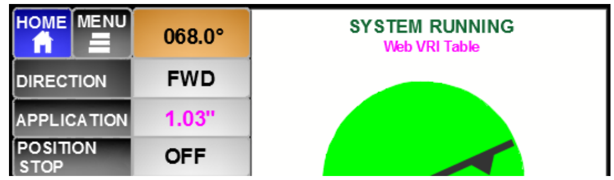
EG2 – using Table 2 (Circle 2), Area 2
- set to **Use Tables** - Table number is shown.

Always Off:
“OFF shown (ex. EG1 OFF)
No Table number shown, no arc shown.

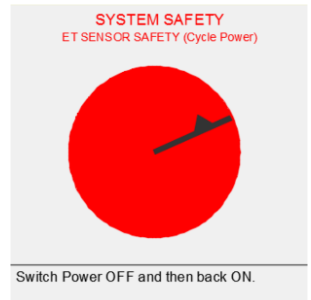
Always On:
“ON” shown (ex. EG2 ON)
No Table number shown.
Full arc around the pivot circle shown.

EG1 – set to Always Off
- Table number not shown.
EG2 – set to Always On
- full arc shown.

<u>System Running – Web VRI Table active:</u>
At 68.0º
Web VRI Table is Enabled.
VRI angle area set to 1.03” application.
VRI angles are Not shown graphically.

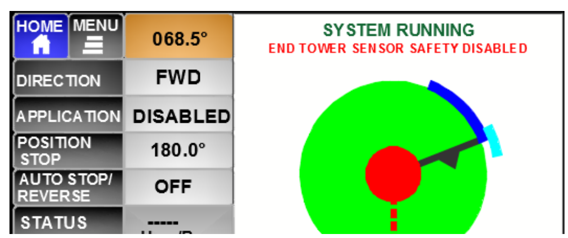


<u>System Safety – before Power loss:</u>
At 068.5º
ET Sensor Safety occurred.
- If the system does not automatically shut down and lose power, this screen will show.
Power <u>must</u> be cycled to the PPT.

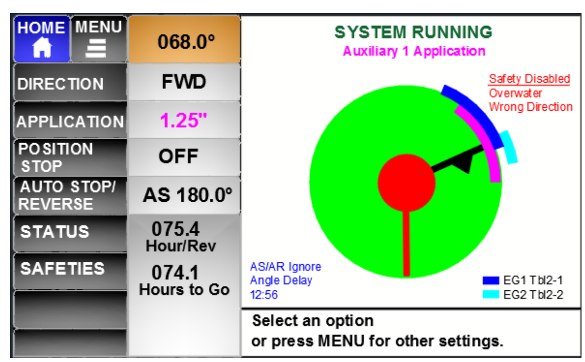


<u>System Safety – after Power cycled:</u>
At 068.5º
ET Sensor Safety screen at power up.
Press STOP key to clear Safety.

<u>System Running with a Safety Disabled:</u>
Red center circle.



<u>Combination of Events:</u>
Auxliary Application running
Angle Delay Timer active
End Gun 1 & End Gun 1 On
Overwater Safety Disabled
Wrong Direction Safety Disabled



4 HOME SCREEN OPTIONS:

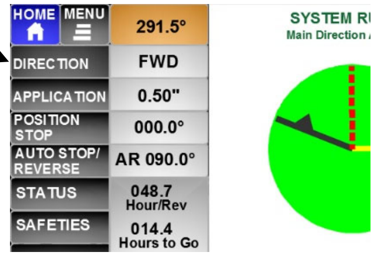
4.1 HOME – Set DIRECTION:

Press DIRECTION to change direction.

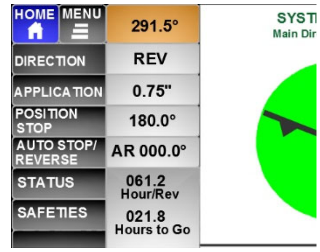
Can be changed when IDLE or RUNNING.
Arrow on Pivot Line of graphical display on HOME screen will show set direction.

Direction Button pressed while running:
Confirmation pop-up screen.

Forward



Reverse

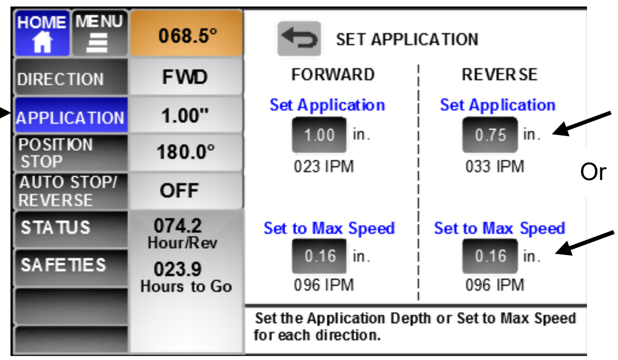


= If Yes, 5 second delay on Relay change = 5 seconds before moving other direction.

4.2 HOME – Set APPLICATION:

Press APPLICATION.
The "Set Application" Screen will appear.

NOTE: Application set in 0.01" increments from 0.01" to 2.00". Above 2.00" set in 0.05" increments to a maximum of 4.75".
Maximum Application is at 6 IPM minimum speed or at higher speed if 4.75" is the limit.
Minimum Application is at Max Speed set in MENU – SYSTEM Set Up.

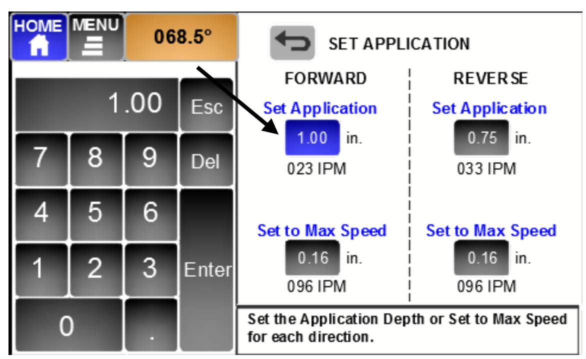


Or

Two ways to set Application (Speed):

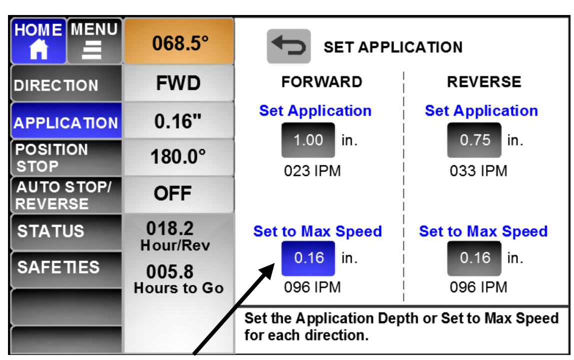
<u>Option 1: Set Application to a specific value:</u>

- Press the numerical Application button for the direction to be set.
- Use the keypad to set the Application and press Enter.
- Set in 0.01" increments up to 2.00".
- Set in 0.05" increments above 2.00".
- Depending on water flow, the Maximum Application will be at 6 IPM speed or 4.75".

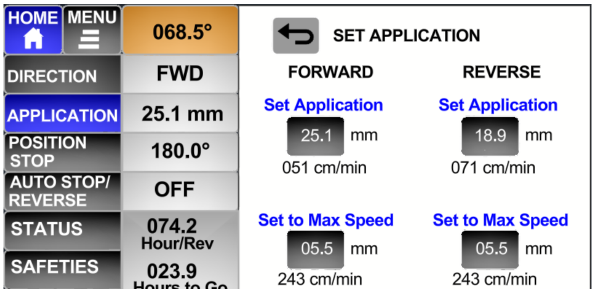


<u>Option 2: Set Application to Max Speed:</u>

- Press the "Set to Max Speed" numerical button for the direction to be set.
- This can be used to quickly set the speed to move the system out of the way for tillage or harvest work.



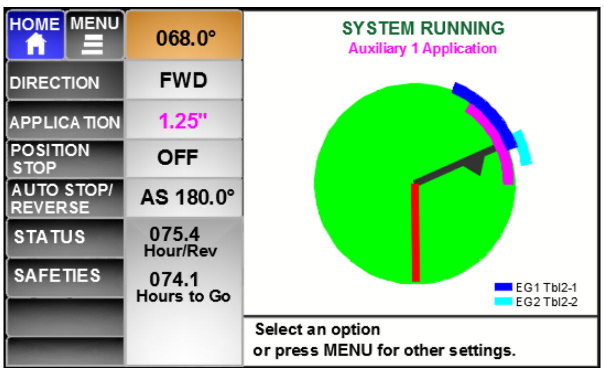
Application Screen if Metric Units:



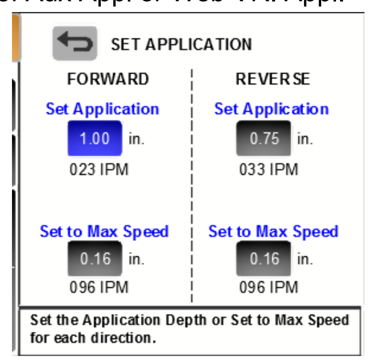
4.3 *HOME – APPLICATION if Auxiliary or Web VRI Table active:*

If the pivot is in an area where the Auxiliary Application is active, or if the Web VRI Table is active, then the "Main Direction APPLICATION" can still be changed. The change will take effect when the pivot moves out of the Auxiliary Application or Web VRI Table area.

Auxiliary Application
or Web VRI Table active



Main Direction Application can be changed but not in effect until out of Aux Appl or Web VRI Appl.

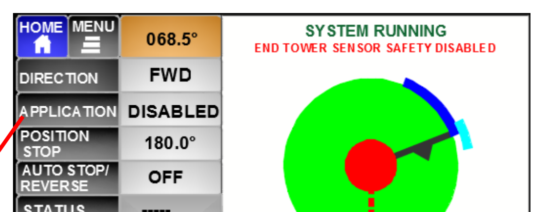


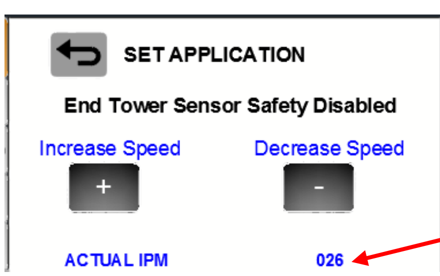
4.4 *HOME – APPLICATION, SET SPEED if END TOWER SENSOR SAFETY is DISABLED:*

System is <u>RUNNING</u>, with "End Tower Sensor Safety" Disabled.
The end tower speed can be manually adjusted using the PPT Control Panel.

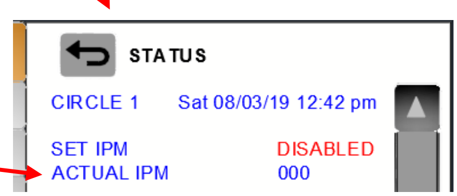
Note: With End Tower Sensor Safety Disabled, the end tower speed can be manually adjusted from the PPT panel **ONLY WHEN RUNNING** by pressing the "Application" button and using the "+" and "-".

If APPLICATION selected while "SYSTEM IDLE, nothing will happen.





ACT IPM shown if working.



4.5 HOME – POSITION STOP:

POSITION STOP is used to quickly set an in-field stop location by Angle.

- system will shut down immediately when the set angle is reached. The "Auto Stop/Reverse Delay" does NOT apply to the Position Stop.
- Position Stop will unlatch the Murphy 117 and display "Position Stop Shutdown".

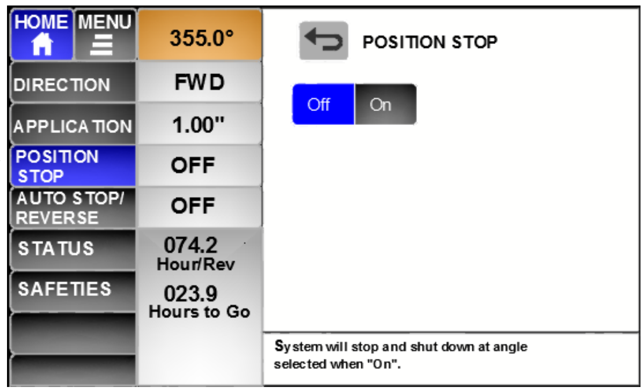
AUTO STOP/REVERSE explained in the next section is typically used for "wiper" part circle end of field stops or reverse locations and can be set By Angle or By Switch.

- The "Auto Stop/Reverse Delay" will be applied when the set angle or switch closure is reached.

Both can be set, but whichever is reached first is the action that will be taken.

Select POSITION STOP
The "Position Stop" Screen will appear.
The Position Stop can be set to Off or On.

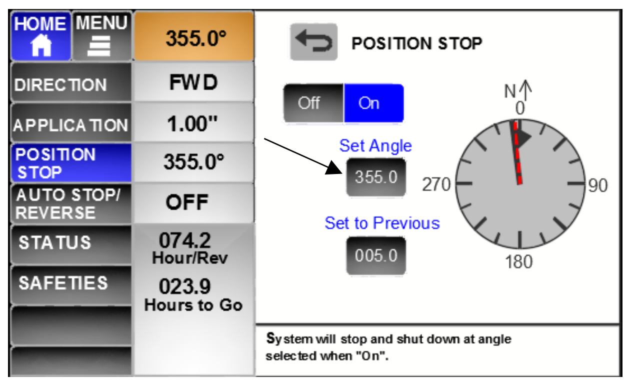
POSITION STOP can be changed when Idle or when Running.



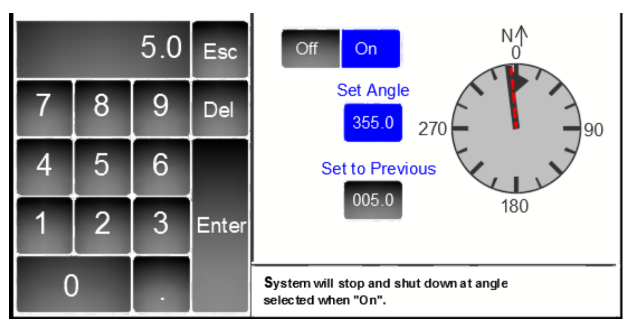
Set POSITION STOP to ON.
Two ways to set the Angle for Position Stop

<u>Option 1:</u> Set Stop Angle to a specific value:

- Press the numerical Set Angle button.

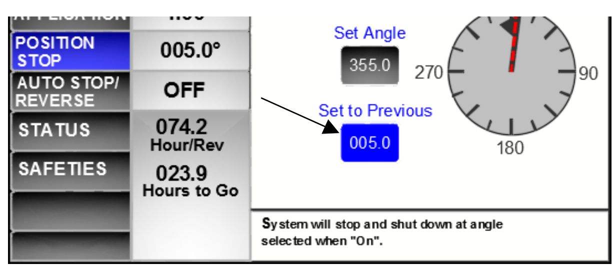


- The Keypad will appear. Use the keypad to set the angle.
- Once "Enter" is pressed, the Position Stop line will reflect the number in the entry box.



<u>Option 2:</u> "Set to Previous" Angle used:

- Press Set to Previous button.
- The "Set Angle" becomes "Previous" and the "Previous" becomes "Set".
- This can be used to quickly set the Angle to one previously used.



4.6 *HOME – AUTO STOP/REVERSE:*

AUTO STOP/REVERSE is typically used for “wiper” part circle end of field stops or reverse locations and can be set By Angle or By Switch.

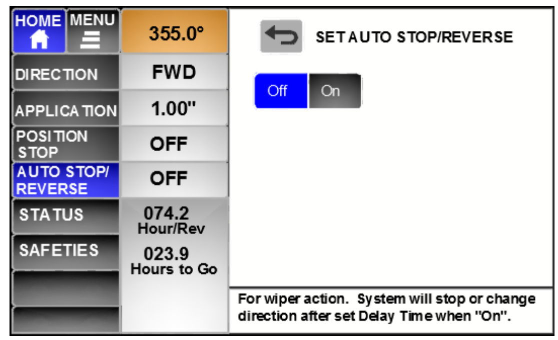
- The “Auto Stop/Reverse Delay” will be applied when the set angle or switch closure is reached.
- If “Auto Stop” is selected, when activated By Switch or By Angle, it will unlatch the Murphy 117 and display “Auto Stop Shutdown”.

POSITION STOP explained in the previous section is used to quickly set an in-field stop by Angle.

Both can be set, but which ever is reached first is the action that will be taken.
If Auto Stop/Reverse is set “By Switch”, an angle line will **not** appear on the pivot graphic Home screen.

Select AUTO STOP/REVERSE
The “Set Auto Stop/Reverse” Screen will appear.
Auto Stop/Reverse can be set to Off or On.

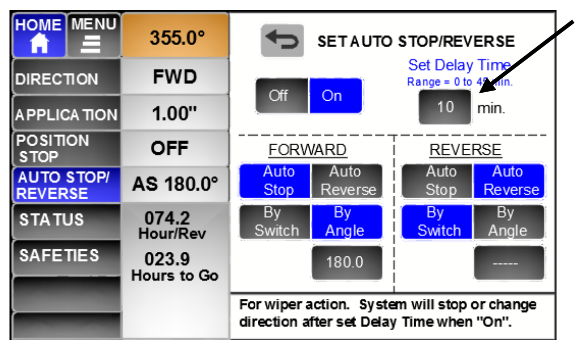
AUTO STOP/REVERSE can be changed when Idle or when Running.



Set AUTO STOP/REVERSE to ON.
Change settings as needed.

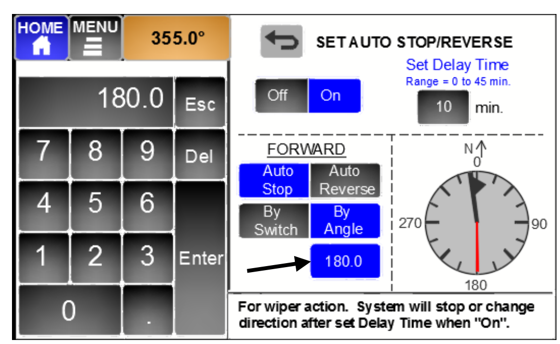
A. Set the Delay Time.
Will be applied to both Auto Stop or Auto Reverse when the set angle or switch is reached.

Range is 0 to 45 minutes.

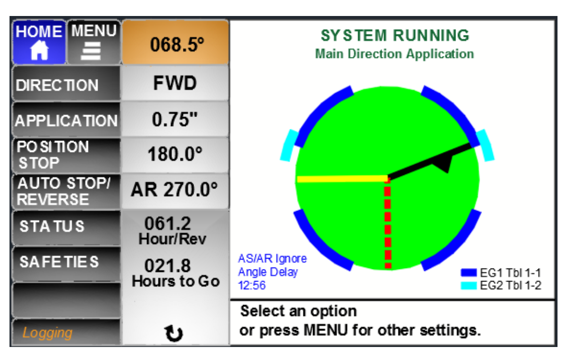


B. For each direction, select the desired action.
Auto Stop or Auto Reverse
By Switch or By Angle

If By Angle, set the Angle location.
- use the Keypad to enter the angle.
- the Auto Stop or Auto Reverse line will reflect the number in the entry box.
- Yellow line for Auto Reverse.
- Red solid line for Auto Stop.
- Press Enter to accept:

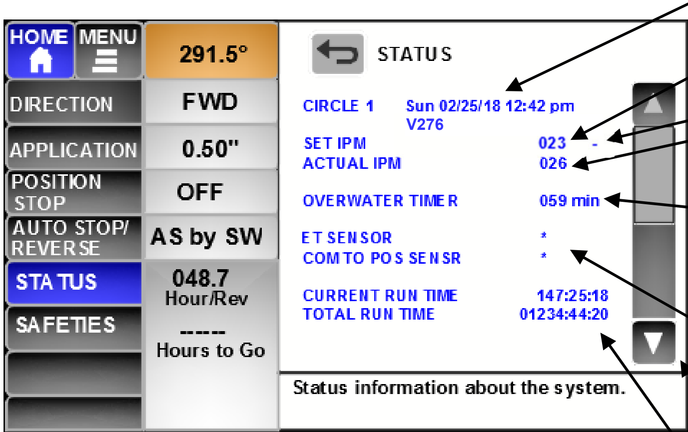


Example with both POSITION STOP and AUTO REVERSE by Angle set.
- when started in Forward, system will stop at 180 degrees due to Position Stop.



4.7 *HOME – STATUS*

STATUS SCREEN 1:
If Encoder Position Sensor, only Status Screen 1 is available.



Circle 1 or Circle 2 or Circle 3. Current Date & Time. Firmware version. (added in v248 2-chip 5/25/22)

SET IPM, for set Application.
Disabled if ET Spd Sensor Disabled in Safeties.
+ or - should show when adjusting speed.
ACT IPM = end tower speed from speed pulses.

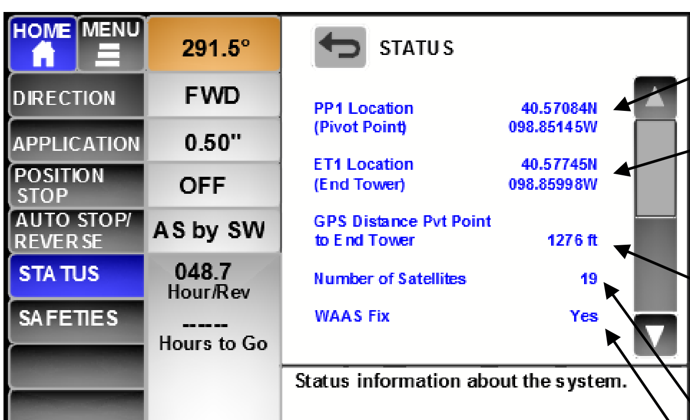
Overwater Timer countdown when Enabled and System Running.
DISABLE if Overwater Timer Disabled in Safeties.

End Tower Sensor Pulses.
Comm to Pos Sensor flashing when good comm.

Arrow Down will show if GPS Position Sensor so can arrow down to Status Screen 2.

Current Run Time = Time since system Started.
(does not include Start Delay in time.)
Total accumulated Run Time of system.

STATUS SCREEN 2 if GPS Position Sensor:



PP1 or PP2 or PP3 Lat-Long per Circle selected in Setup.
If respective PP not set, will show all 000's.

ET1 or ET2 or ET3 Lat-Long from GPS Antenna
If at startup, may show "**AQUIRING**".
If GPS Pos Sensor Disabled will show "**GPS Disabled**".

Calculated Distance of PP1 to ET1 (or PP2 to ET2, or PP3 to ET3)
"ERR" if greater than 9999 ft.
If GPS Pos Sensor Disabled show "**GPS Disabled**".

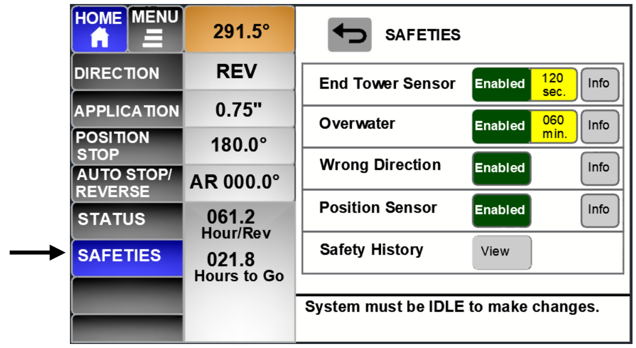
Number of Satellites from GPS Antenna.

WAAS Fix – Yes or No.

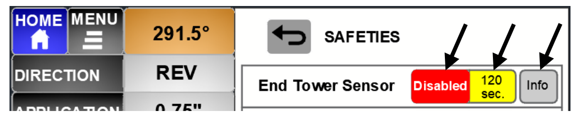
4.8 HOME – SAFETIES

Select SAFETIES
The "Safeties Screen will appear.

System MUST BE IDLE to make changes.



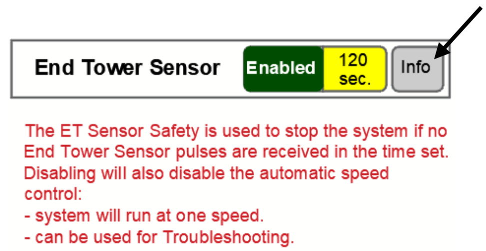
Press "Enabled" to change to "Disabled"
Press "Time" button to change time.
Press "Info" button for more information.



"End Tower Sensor Safety"

Time range can be set from 30 to 240 secs.
- select the time number button to change.
- Default is 120 sec. (changed in v268)

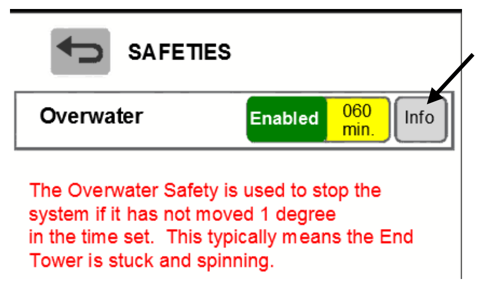
<u>If Disabled</u>:
- The system will still operate but at one speed.
- The speed can be manually adjusted.



"Overwater Safety"

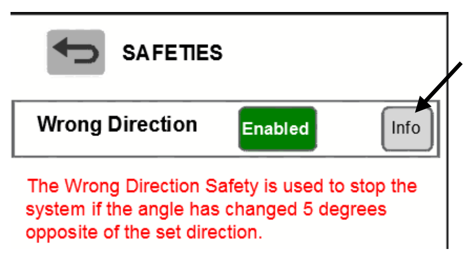
Time range can be set from 10 to 120 minutes.
- select the time number button to change.
- Default is 60 min.

<u>If Disabled</u>:
- The system will still operate but will not shut down if the End Tower gets stuck.



"Wrong Direction Safety"

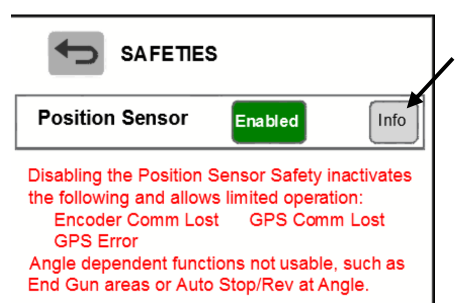
<u>If Disabled</u>:
- The system will still operate but will not shut down if moved 5 degrees the wrong way.



"Position Sensor Safety"

Deals with 3 different safeties related to the Position Sensor being used.
- See later section "SAFETY MESSAGES" for full explanation.

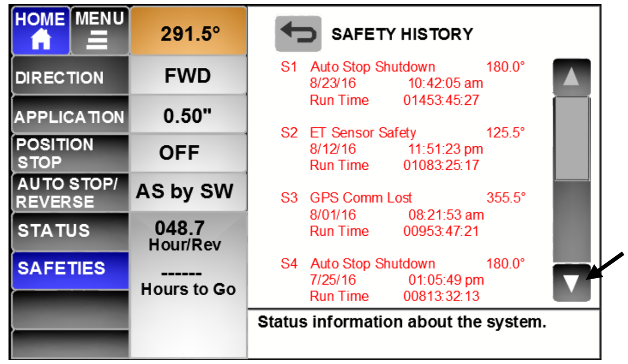
<u>If Disabled:</u>
- System can still be operated, but any function related to angles will not function. The End Tower Reverse Switch will still be operational to Auto Stop or Auto Reverse by Switch.



"Safety History"

Press "VIEW" to review the last 8 Safety Messages.

- Page 1 shows the 4 most recent safeties and Page 2 shows the 4 prior to that.
- The most recent safety is S1.
- Use "Down Arrow" to advance to Page 2.
- Use the "Up Arrow" to scroll back to Page 1.

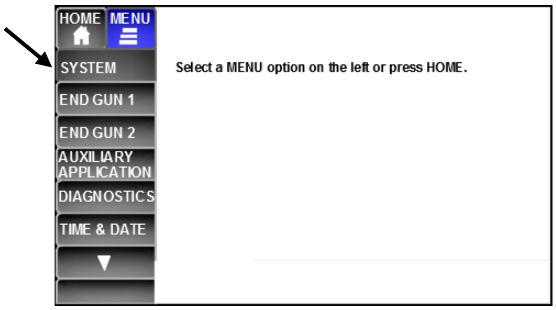


Information given:

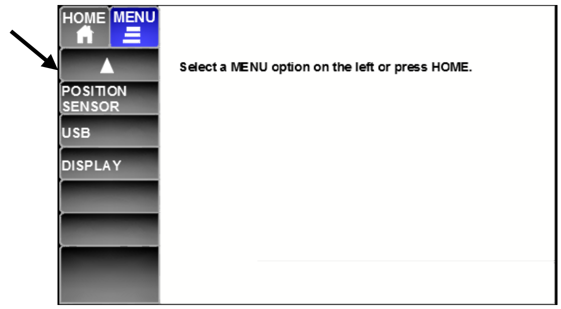
- Safety number
- Safety message
- Angle at which safety occurred.
- Date & Time of safety.
- Total Run Time when safety occurred.

5 MENU SCREENS

MENU SCREEN 1:



MENU SCREEN 2:

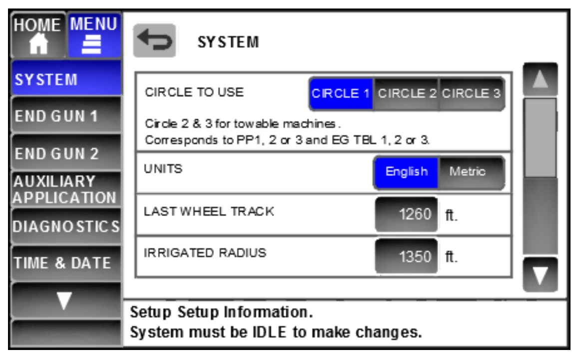


5.1 *MENU – SYSTEM Set Up*

NOTE 1: System must be IDLE to make changes. Settings can be reviewed if system is running.
NOTE 2: After changing SYSTEM Set Up values, cycle power to the PPT so the Application and Speed values are correct for the new settings. Or make sure to set the Application values for each direction.

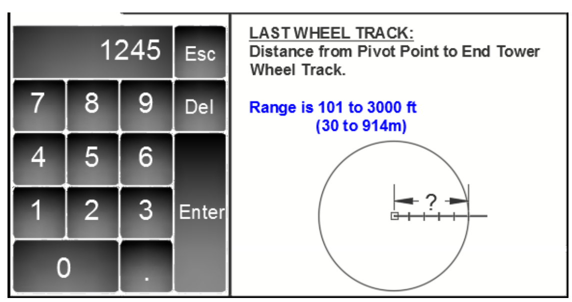
<u>SYSTEM Set Up Page 1</u>

CIRCLE TO USE
- Circle 1 is default.
- Circle 2 & 3 are for Towable machines and allow different GPS Pivot Point locations to be set. PP1 = EG Table1 will be used. PP2 = EG Table 2 will be used, etc. for PP3.

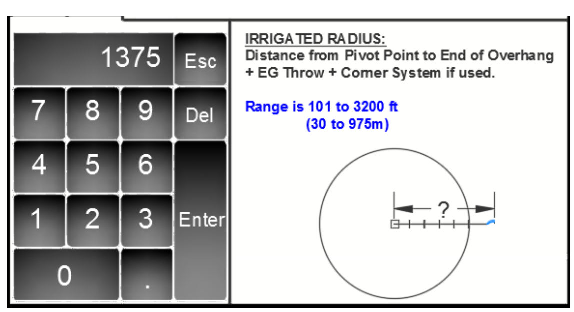


UNITS
- English (ft and inches) or Metric (meters)

LAST WHEEL TRACK
- Range is 101 to 3000 ft
  or 30 to 914 meters
- Use number pad to enter.



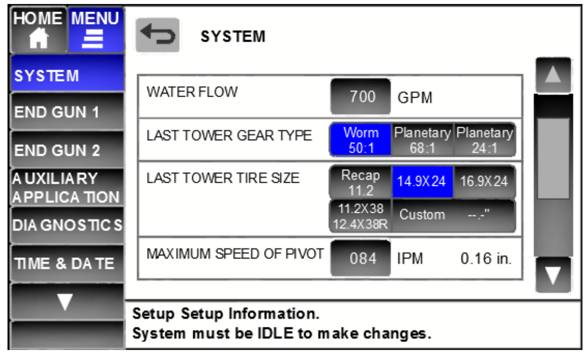
IRRIGATED RADIUS
- Range is 101 to 3200 ft
  or 30 to 975 meters
- Use number pad to enter.



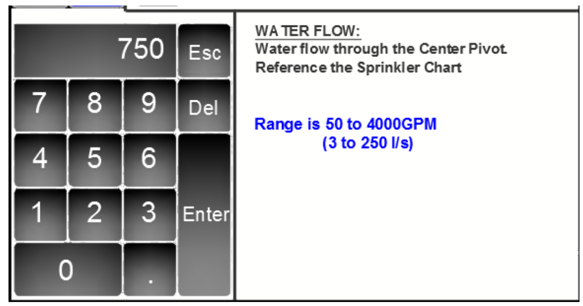
Irr. Radius - For Corner System w/Booster
- add the following to the Last Wheel Track

187' = +355 ft (175 + 80 + 100)
197' = +364 ft (184 + 80 + 100)
207' = +373 ft (193 + 80 + 100)

SYSTEM Set Up Page 2



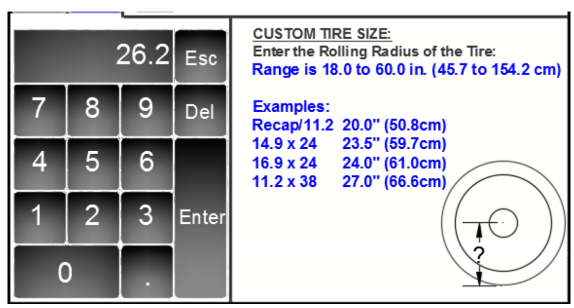
WATER FLOW
- Range is 50 to 4000 GPM
  or 3 to 250 l/s (litres per sec)
- Use number pad to enter.

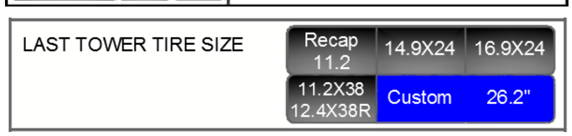


LAST TOWER GEAR TYPE
- Select one of those shown

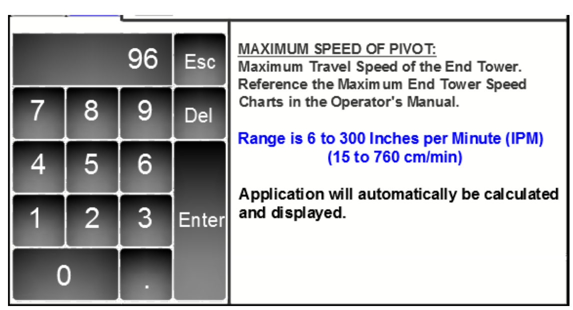
LAST TOWER TIRE SIZE
- Select one of those shown:
- NOTE: For 12.4 X 38R, use "Custom"

If "Custom" selected
- enter the Rolling Radius
- for 12.4 x 38R = 28.5" (72.4 cm)



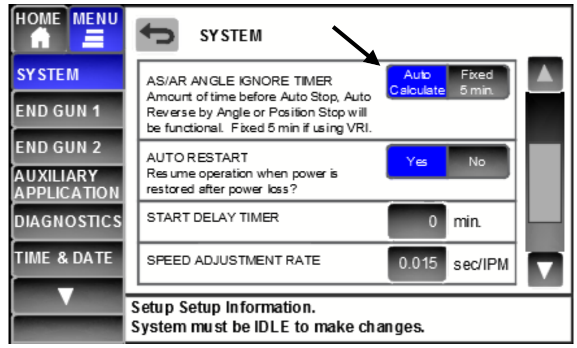


MAXIMUM SPEED OF PIVOT
- Range is 6 to 300 IPM
  or 15 to 760 cm/min
- Use number pad to enter.

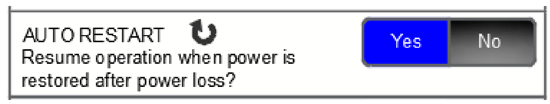


SYSTEM Set Up Page 3

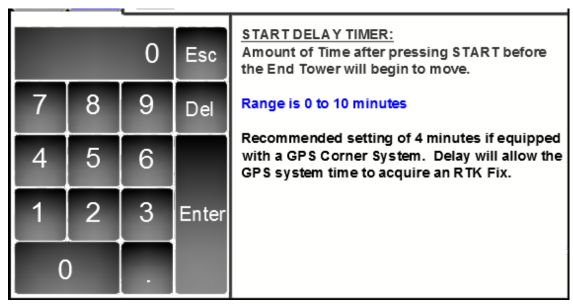
AS/AR ANGLE IGNORE TIMER
- Used for full circle systems to allow a full circle 360 degrees to be made where the Position Stop or Auto Stop/Reverse Angle is the same as the current angle of the pivot. Ex. at 000º and set to go to 000º.
- The set speed and the length of the system are both used in the “Auto Calculate” formula.
- If using VRI and changing speeds every 3 degrees, set to “Fixed 5 min.”



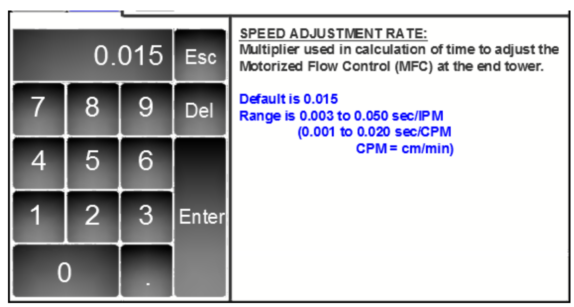
AUTO RESTART
- If set to “Yes”, the system will resume operation when power is restored after a power loss, providing it lost power under normal operation and not because of a PPT safety. (Set to “YES” for Electric Drive with Power Company Load Control.)



START DELAY TIMER
- Amount of time the system should wait to begin operation after the Start button is pressed, or after an Auto Restart.
- Set to at least 3 minutes if a GPS Navigation corner is attached to the pivot to allow time for an RTK Fix to be established.
- Range is 0 to 10 mins.
- Use number pad to enter.

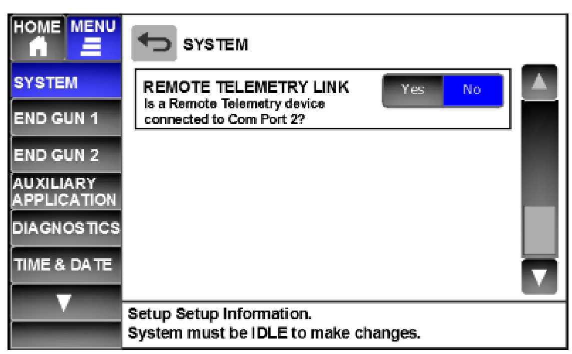


SPEED ADJUSTMENT RATE
- Multiplier used to calculate time for MFC to adjust.
- Range is 0.003 to 0.050 sec/IPM
- Use number pad to enter.



SYSTEM Set Up Page 4

REMOTE TELEMETRY LINK
- Set to “Yes” if a remote telemetry device such as AgSense Precision Link or FieldWise is connected to Com Port 2.
- Otherwise set to “No”.

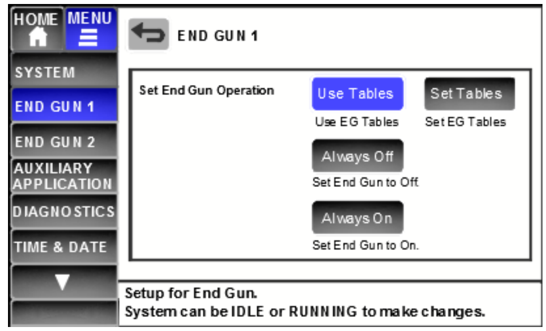


5.2 *MENU – END GUN 1 and END GUN 2*

Settings can be changed while the system is Idle or when Running.

Use Tables = use the End Gun Tables as set.
Set Tables = Set the End Gun Table areas.
Always Off = End Gun will always be off.
Always On = End Gun will always be on.

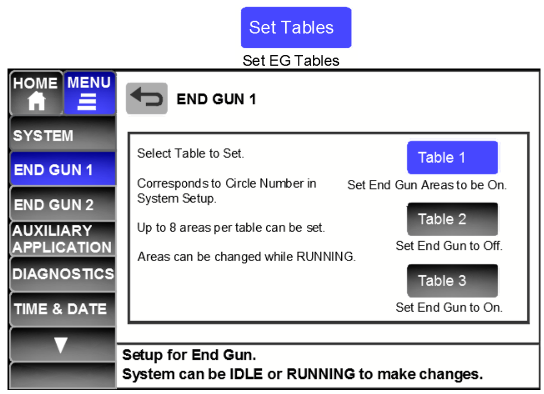
Default is "Always Off" at first power up after programming.



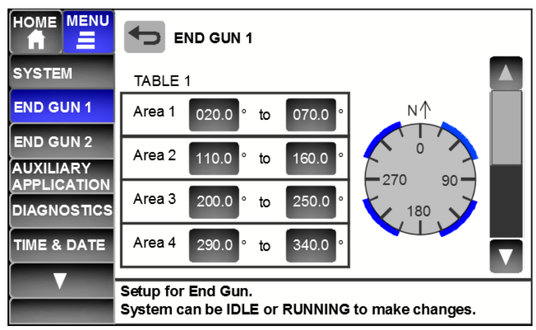
"Set Tables"

Select which Table to set for End Gun 1.

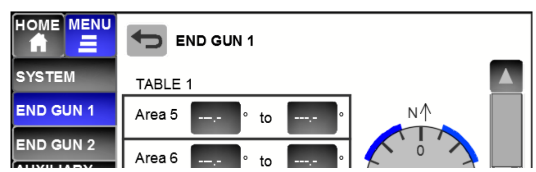
- Table 1, or Table 2, or Table 3
- Table Number corresponds to the Circle Number set in System Setup. For towable systems so different End Gun Areas can be set at different sites.
- Ex. If set for Circle 1, then EG Table 1 will be used for End Gun control.
- Up to 8 areas per Table can be set.
- Areas can be changed when Running.



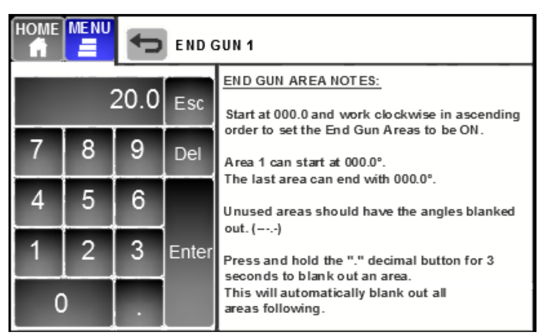
END GUN 1, TABLE 1, Areas 1 to 4
Set areas will be shown graphically.
Set the Areas to be ON.
Will be the same for TABLE 2 & TABLE 3.



END GUN 1, TABLE 1: Areas 5 to 8
Set areas will be shown graphically.
Will be the same for TABLE 2 & TABLE 3.



Keypad and End Gun Area Notes.
- Do NOT program an area across 000.0º.
  Example: 320.0º to 030.0º is not correct.
  Area 1 should be 000.0 to 030.0.
  Last Area should be 320.0 to 000.0
  Fill in the areas in between as needed.
  Press "." decimal button for 3 seconds:
  - will clear that area and all areas following.



Example End Gun Areas for Corner Systems

	Left Trail	Right Trail
Area 1	040º to 058º	033º to 050º
Area 2	130º to 148º	123º to 140º
Area 3	220º to 238º	213º to 230º
Area 4	310º to 328º	303º to 320º

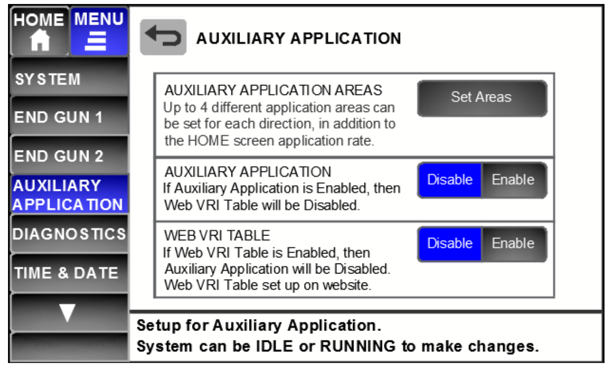
5.3 *MENU – AUXILIARY APPLICATION*

Settings can be changed while the system is Idle or when Running.

Set Areas = Set areas for different applications. (more detail below)

Auxiliary Application: Disable or Enable
Disable = Do not use the Set Areas.
Enable = use the Set Areas. If Enabled, the Web VRI Table will be Disabled.

Web VRI Table: Disable or Enable
Disable = Do not use the Web VRI Table..
Enable = use the VRI Table downloaded from a website, such as for the AgSense Precision Link. If Enabled, the Auxiliary Application Set Areas will be Disabled.



"Set Areas"

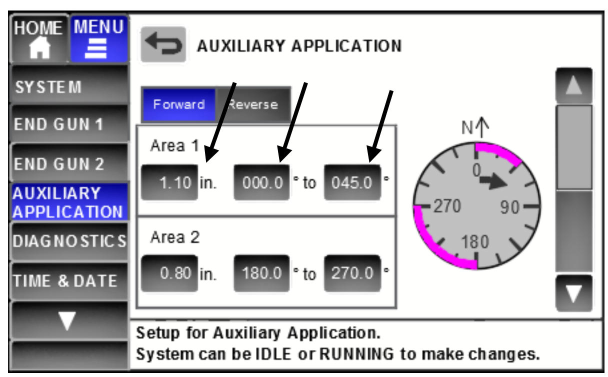
Up to 4 different application areas can be set for each direction. (4 in Forward, 4 in Reverse.)

- These areas should have a different application than the Home Screen for that direction.
- In total, there can be up to 5 different applications set in each direction.

Select the direction, either Forward or Reverse.

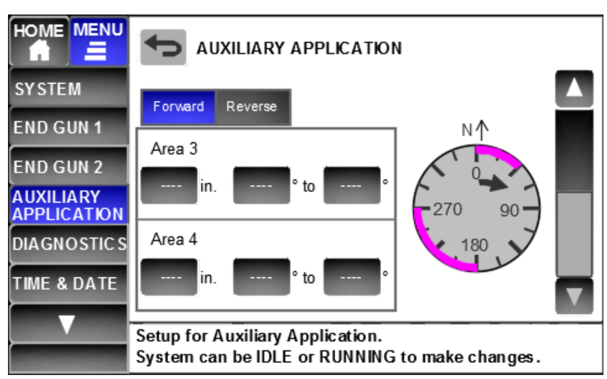
FORWARD:
Areas 1 and 2

- Set the Application
  Ex. 1.10 in, or 27.9 mm.
- Set the Area for that application.
  Forward: Program in ascending order (CW).
  Ex. 000.0 to 045.0, increasing.
  0.80 in from 100.0 to 175.0, etc.
- Set areas will be shown graphically.



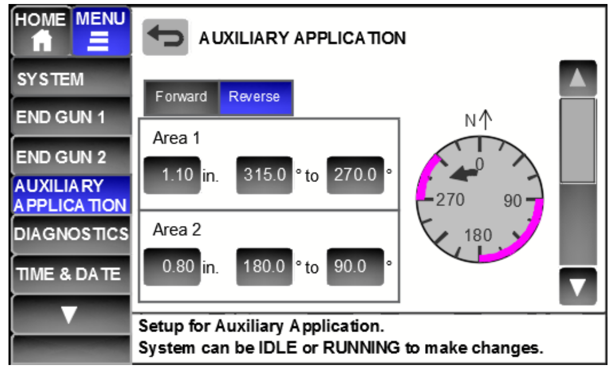
Areas 3 and 4 for Forward

- Set the Application & Areas.



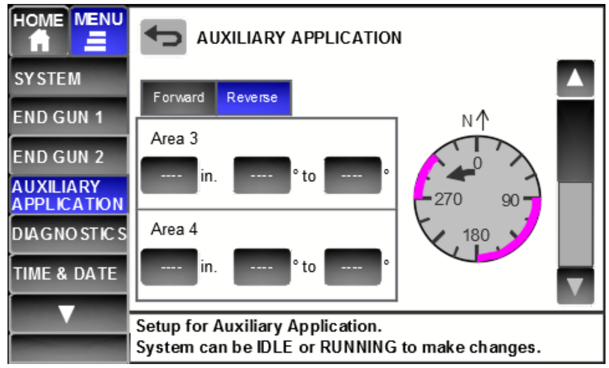
REVERSE:
Areas 1 and 2

- Set the Application
  Ex. 1.10 in, or 27.9 mm.
- Set the Area for that application.
  Reverse: Program **DESCENDING (CCW).**
  Ex. 315.0 to 270.0, decreasing.
  0.80 in. from 180.0 to 090.0, etc.
- Set areas will be shown graphically.

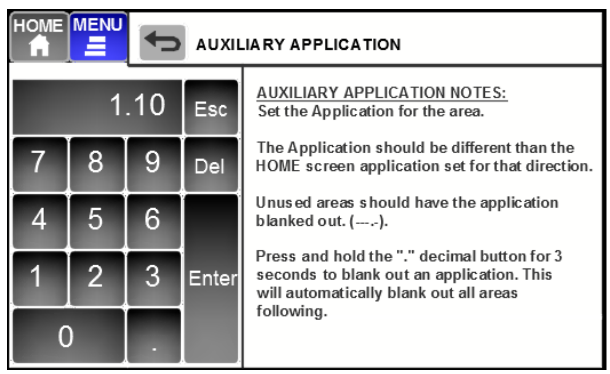


Areas 3 and 4 for Reverse

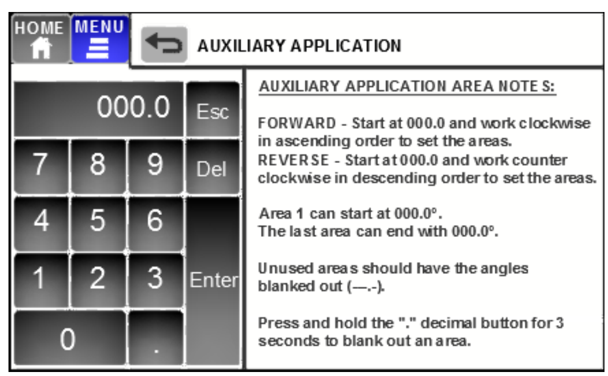
- Set the Application & Areas.



Set Application Keypad and Notes:

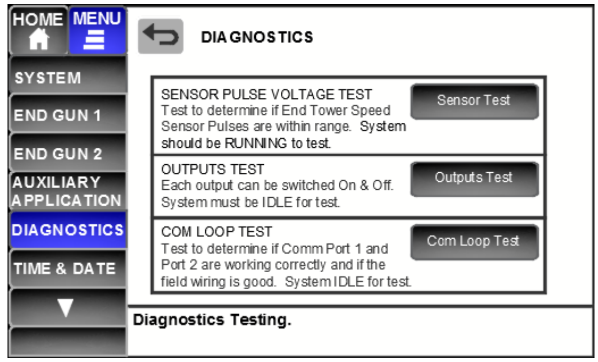


Set Areas Keypad and Notes:



5.4 *MENU – DIAGNOSTICS*

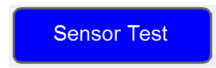
Diagnostics can be used to check or troubleshoot the PPT system.



"Sensor Pulse Voltage Test"

NOTE: This does not replace the test using an actual voltmeter to verify the voltages! It is for a quick check.

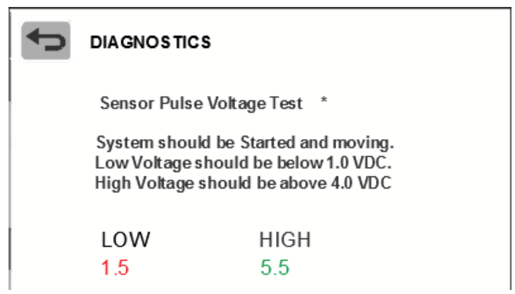
The system should be **<u>started and moving</u>** for this test.



Low = the lowest voltage detected on the pulse line.
If above 1.0, then additional grounding is needed.

High = the lowest voltage detected on the pulse line.
If equal to the low, verify the system is moving.
If moving and equal:
- could be a wiring issue.
- could be a cable or sensor issue

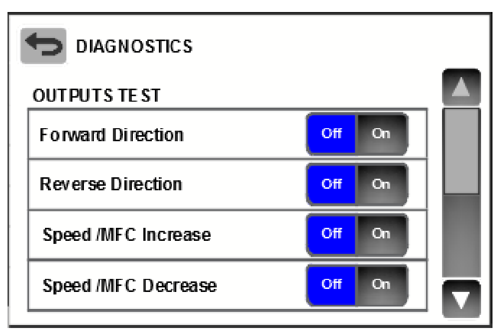
See Troubleshooting – End Tower Sensor Safety.



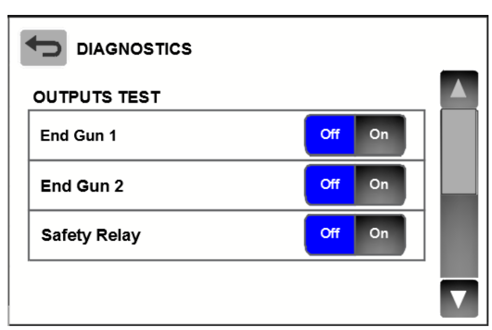
"Outputs Test"

The system must be <u>Idle</u> to test Outputs.
- Only one output at a time can be set to On.
- Setting another output to On will set any others to Off.
- When exiting Outputs Test, all outputs will be set to Off.

<u>Outputs Test Page 1</u>



<u>Outputs Test Page 2</u>



"Com Loop Test"

The system should be Idle for the Com Loop Test.

The Com Port Jumpers on the circuit board must be set to the RS232 pins.
- see diagram to the right.

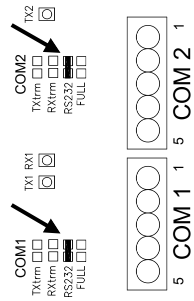


Fig. 3. Com Loop Test Jumpers.

A loop must be wired from Terminal 2 to Terminal 4 of each Com Port for the test to work..
Confirm this has been done, set to Yes and ENTER.

Loop Wiring:
- first start at the Com Port.
- insert a jumper wire from 2 to 4.

The Swivel Cable and / or Span wires can also be looped to check the integrity of those wires.
See below drawing for GPS or Encoder wire loops.

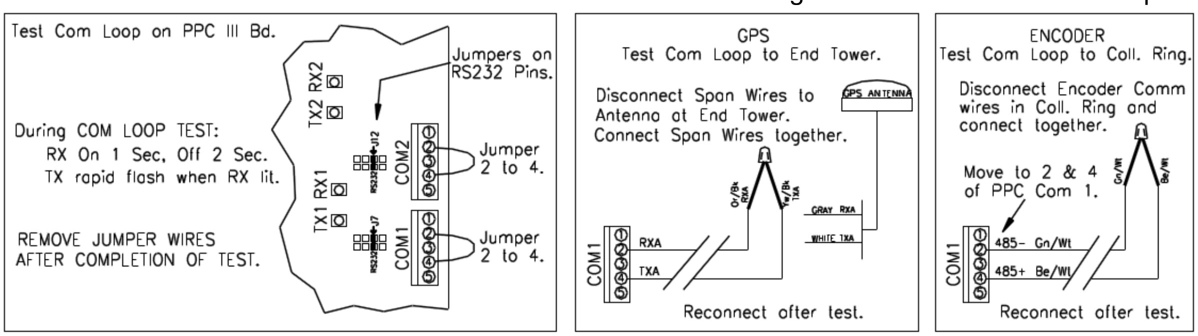
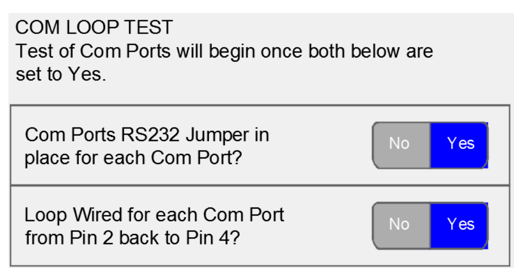
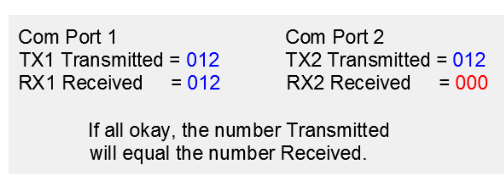


Fig. 4. Com Loop Test Wiring.

The Com Loop Test will begin when both are set to Yes:



- Packets of information will be Transmitted and then checked by the board to see if they have been Received.
- The number Received should equal the number Transmitted.
- If not equal, the Received number will be red.
- If Red, there is either a wiring or board problem.



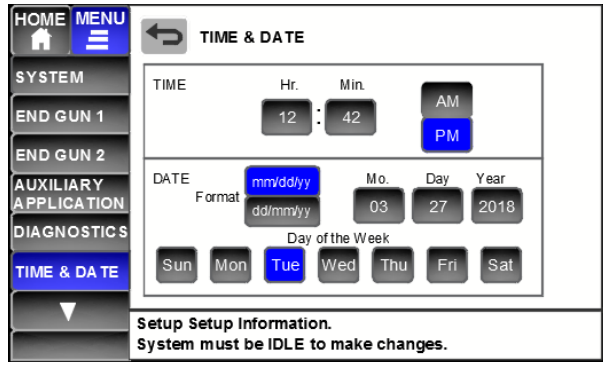
5.5 MENU – TIME & DATE

TIME
- Use Keypad to enter Hr. and Min.
- Set to AM or PM.

DATE
- Date Format Options:

  mm/dd/yy
  Example 03/27/18 = March 27, 2018

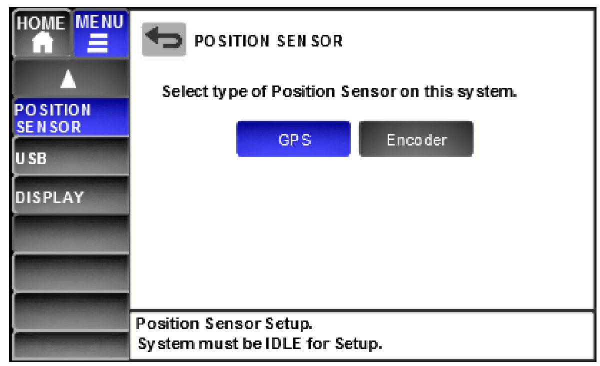
  dd/mm/yy
  Example 27/03/18 = 27 March, 2018



5.6 MENU – POSITION SENSOR

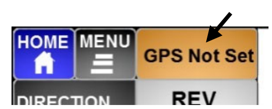
Set the type of Position Sensor used on this system.

The system should be Idle to set the Position Sensor.

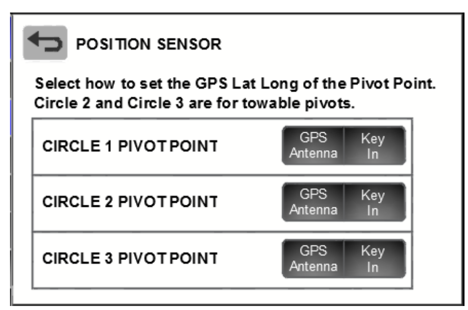


IF "GPS" selected:

NOTE: The HOME Screen will show "GPS Not Set" if the Pivot Point Lat Long is not entered for the Circle Number selected in MENU – System.



For the respective Circle Number Pivot Point, select how the Pivot Point Lat-Long is to be set.
- Up to 3 different Pivot Point GPS locations can be stored in the case of a towable system.
- This then corresponds to the Circle Number to use selected in MENU - Setup



<u>If set by GPS Antenna</u>:

- The GPS Antenna must be wired into the Control Panel and set at the center of the Pivot Point.
- See Figure below for wiring.
  Set to Yes when Wired & Ready

POSITION SENSOR

Circle 1 Set Pivot Point using GPS Antenna

GPS Antenna Wired & Ready?
Select Yes when ready. Yes

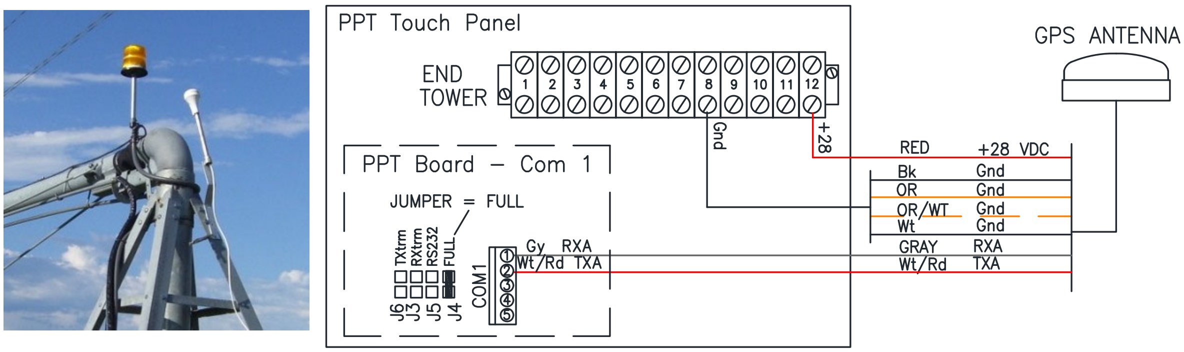


Figure 4.1. GPS Antenna Wiring at Pivot Point.

If the GPS Antenna cannot be detected, a warning message will appear.

Circle 1 Set Pivot Point using GPS Antenna

GPS Antenna Not Detected.
Return to MENU using button above.

If the GPS Antenna is detected, then information is shown about the GPS position as follows:
- The number of satellites being seen.
- WAAS Fix or No Fix
- The Lat Long location of the GPS Antenna.

Circle 1 Set Pivot Point using GPS Antenna

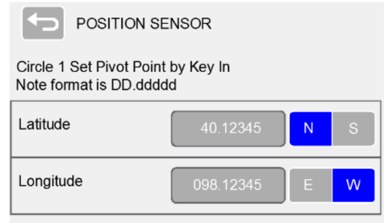
GPS Antenna Detected.
Waiting for WAAS Fix or for 10 Minutes
Number of Satellites 05
WAAS FIX No
Pivot Point Lat Long
40.12345N
098.12345W
Press START to accept Lat Long before WAAS Fix or 10 minute time period.

The Lat Long location will be set one of three ways:
- After 10 minutes.
- Before 10 minutes if it has a WAAS Fix.
- Before 10 minutes if Enter is pressed.

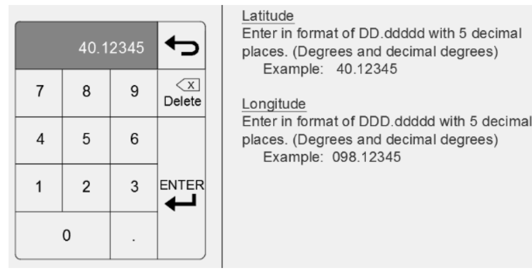
Once set, the Pivot Point location will be displayed.
Record this Lat Long position for future reference.

Remember to write down the Pivot Point Lat Long for reference in the future.

Pivot Point Lat Long
40.12345N
098.12345W



If set by Key In:
- Repeat the following as needed for Circle 2 and 3.
- Set each item of the Latitude and Longitude.



- Use the Keypad to enter the Latitude and Longitude of the Pivot Point.
- **NOTE THE PPT FORMAT: DD.dddddº**
  - Decimal degrees, to 5 decimal places.
- **NOTE THE PPC III FORMAT was:** DD°MM.MMMM'
  - Degrees & Decimal minutes

Conversion example from DD°MM.MMMM

DD°MM.MMMM' = 40°34.8488'

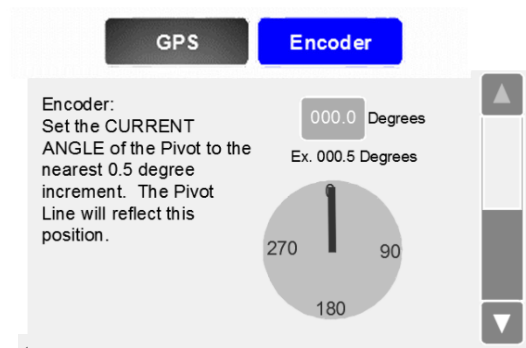
Divide the MM.MMMM by 60

34.8488/60 = 0.58081

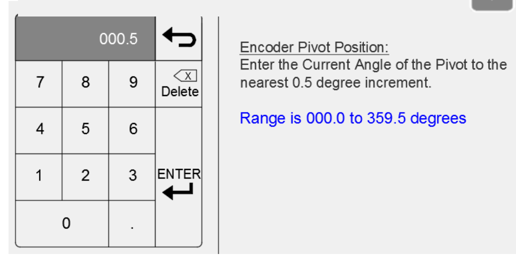
Example Latitude to enter into PPT Touch Panel:

DD.dddddº = 40.58081°

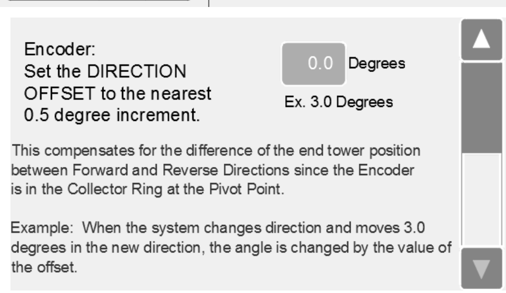
IF "Encoder" selected:



Set the CURRENT ANGLE of the Pivot Point.
- Set to the nearest 0.5 degree increment.
- The Pivot Line will reflect the number set.

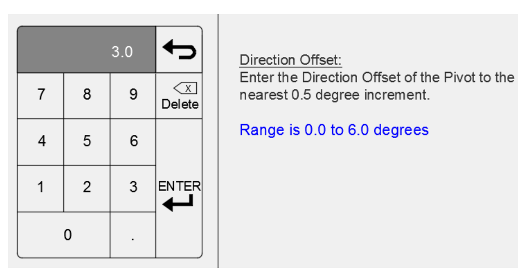


Use the Keypad to enter the Angle.
- Range is 000.0 to 359.5 degrees.



Set the DIRECTION OFFSET of the Pivot.
- Set to the nearest 0.5 degree increment.

See screen for explanation.

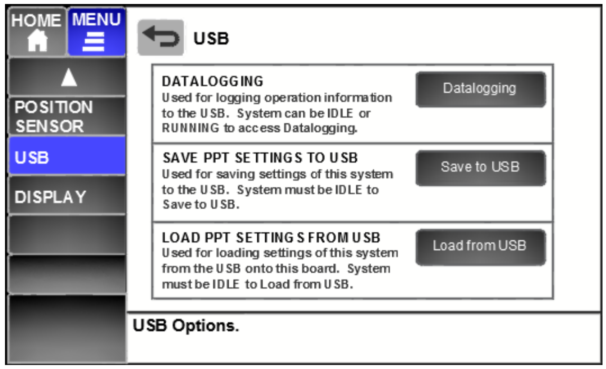


Use the Keypad to enter the Direction Offset.
- Range is 0.0 to 6.0 degrees.

5.7 *MENU – USB*

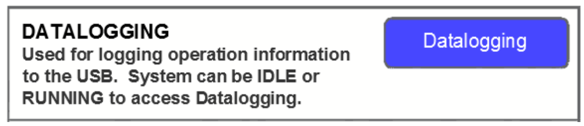
NOTE: The USB functions utilize the “Main” USB port.

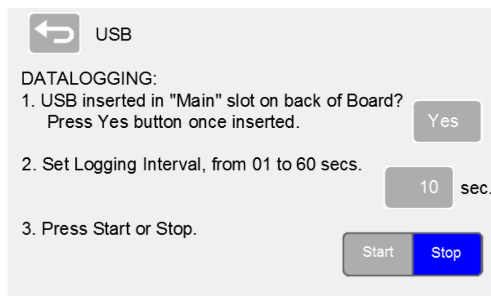
Check the on screen instructions for each option to see if the system should be Idle or can be Running.



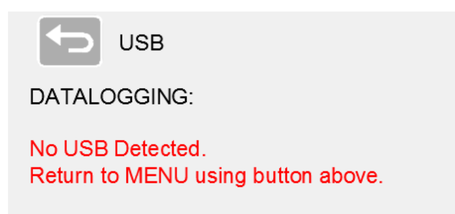
USB must be formatted “FAT32”.
Insert into “Main” USB Port on Basic Board.

“Datalogging”
Datalogging will create a file on a USB drive and log several system parameters at the interval specified.

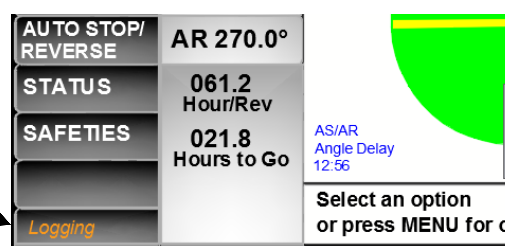




If USB not detected, an error message will appear.



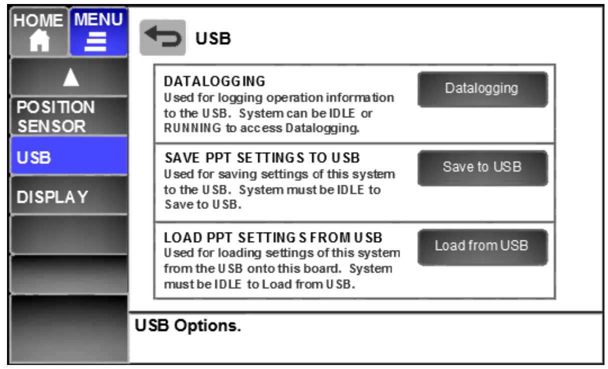
With Datalogging active, the word “Logging” will appear in the bottom left of the screen on both the MENU and HOME Screens.



The Log file will be saved on the USB with a number and date and a .LOG extension.
Ex. 01080319.LOG
1st file saved on Aug. 3, 2019.
Use Notepad or Excel to view as a Text File.

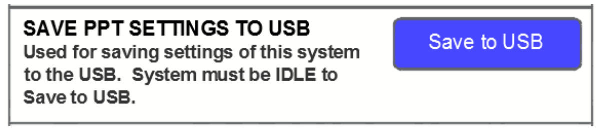
Parameters being Logged v251 (GPS Position Sensor):
Firmware version
Total Run Time (at time Logging started)
Date Time Dir Angle Appl Set IPM Act IPM EG1 EG2 AUX RUN Circle PP Lat PP Long ET Lat ET Long WAAS # Sats AR Delay Pos Stop AS/AR State App Code (v270 removed STA string)

MENU - USB

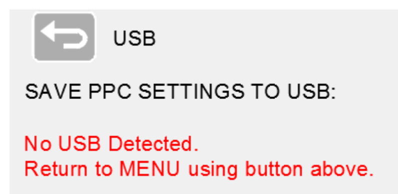


"Save PPT Settings to USB"

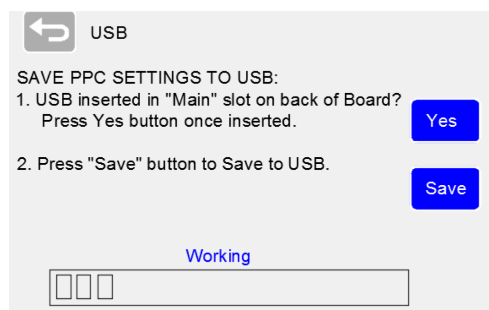
If a firmware update is required, you may be told to save the settings to the USB before performing the update, especially if "Restore Settings" will need to performed. This will save all setting specific to this system such as:
Setup, End Gun Areas, Auxiliary Application Areas, GPS Pivot Point locations, etc.



If USB not detected, an error message will appear.

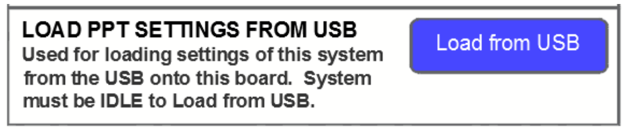


With USB detected and Save button pressed, the Progress Bar will show.

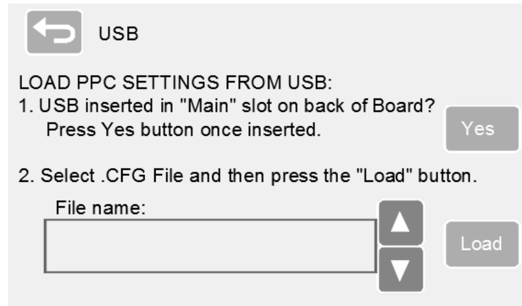


The Configuration file will be saved on the USB with a number and date and a .CFG extension.
Ex. 01110816.CFG
1st file saved on Nov. 8, 2016.
Use Notepad or Excel to view as a Text File.

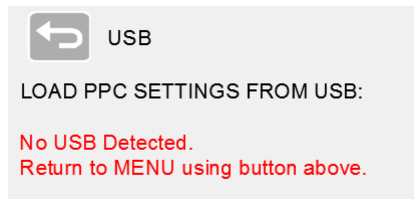
"Load PPT Settings from USB"



If a firmware update is required, you may be told to save the settings to the USB before performing the update. This file can then be Loaded back into the PPT to restore the settings such as:
Setup, End Gun Areas, Auxiliary Application Areas, GPS Pivot Point locations, etc.

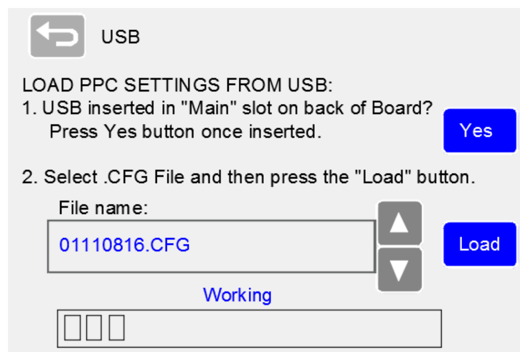


If USB not detected, an error message will appear.

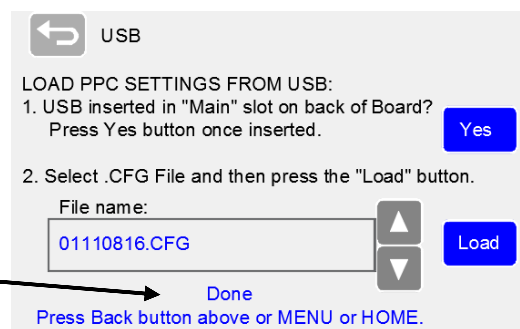


The Configuration file to be loaded from the USB must have a .CFG extension.
Ex. 01110816.CFG

Select the correct .CFG file and press "Load". Progress Bar will show.



When finished loading, the word "Done" will appear under the file selection box.

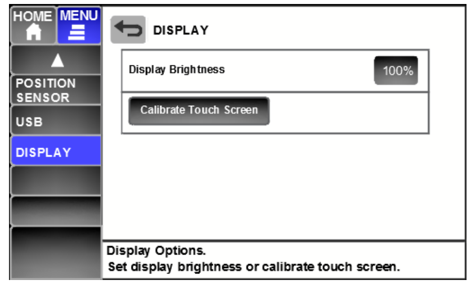


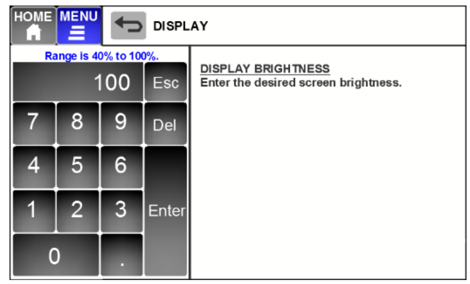
5.8 *MENU – DISPLAY*

“Display Brightness”

Set the Display Brightness, typically at 100% for daylight readability.

- can be set lower if using for an extended period at night.

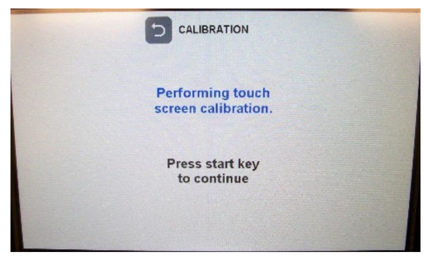




“Calibrate Touch Screen”

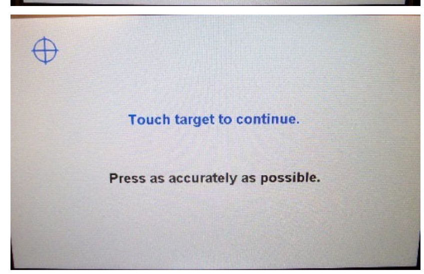
This routine can be used if the area touched on the screen does not seem to match what is intended to be pressed.

Press START key to begin Calibration



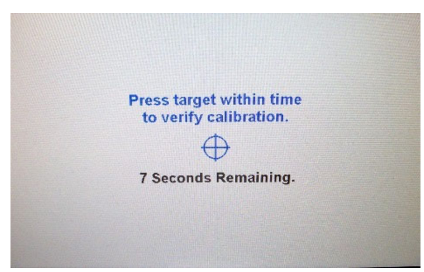
Press each of the 4 Targets when shown, as accurately as possible.

- Upper Left
- Upper Right
- Lower Right
- Lower Left



Press the “Verification Target” to finalize the calibration.

- If not pressed within 10 seconds or if the target is not sensed as being touched (calibration not correct), the Calibration process will be restarted.



“Calibration” Note:

If at any time the Touch Screen does not seem responsive to touches or does not seem to be calibrated correctly, use the following procedure.

Press and Hold both Start and Stop for 5 seconds:

- will start the Touch Screen Calibration routine.

6 PPT CIRCUIT BOARDS & PROGRAMMING

6.1 PPT "Basic" Circuit Board Overview Red 1-Chip (Effective March 2024 -)

Note: Ver B02 Boards started shipping in late March 2024.

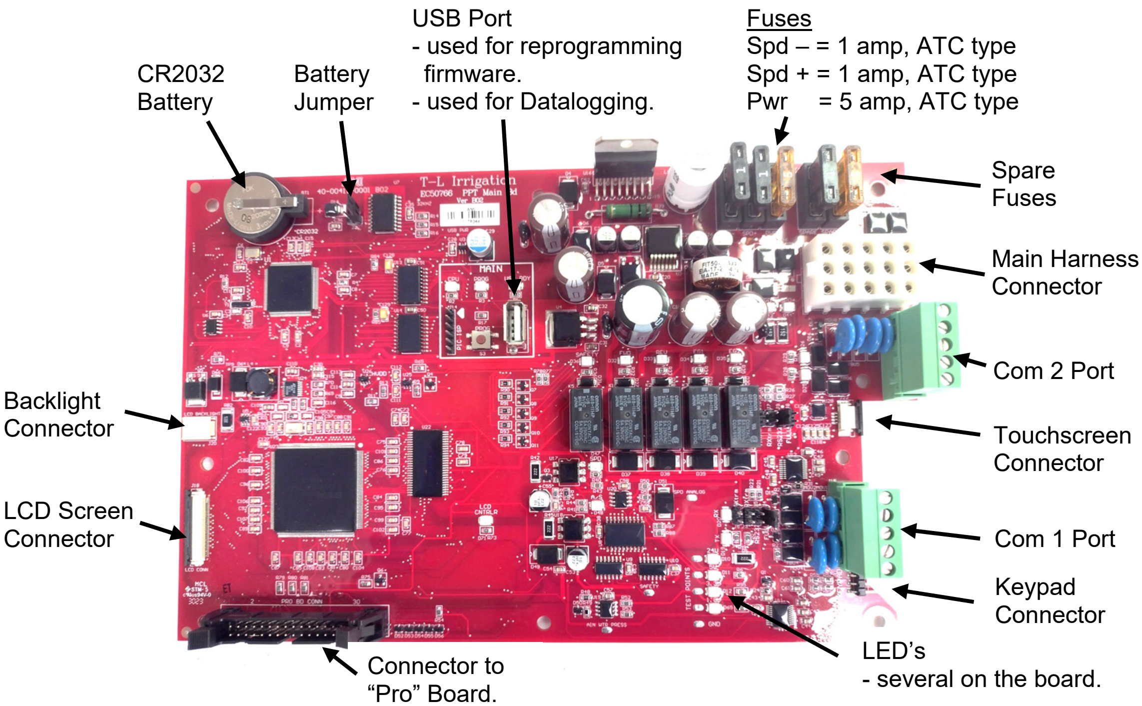


Figure 5. PPT Circuit Board Overview Red 1-Chip.

6.2 PPT "Basic" LED's Red 1-Chip (Effective March 2024 -)

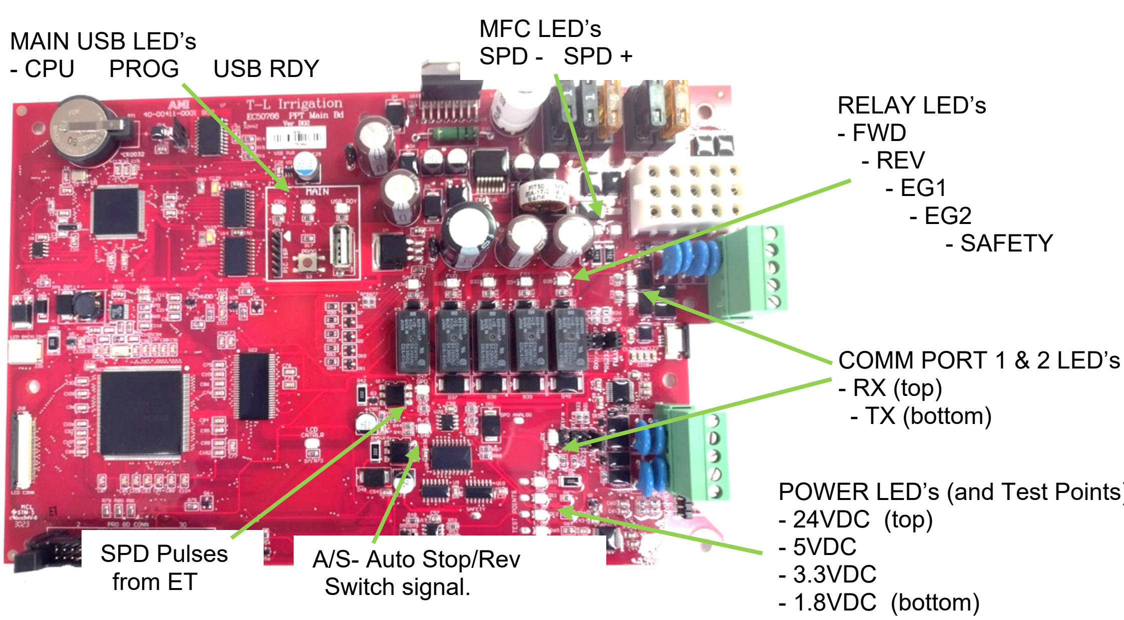


Figure 5.1. PPT Circuit Board LED's Red 1-Chip.

6.3 *PPT "Basic" Circuit Board Overview Green 2-Chip (<u>OBSOLETE March 2024</u>)*

Note: Ver A03 Boards started shipping in late March 2019 (retrofit kits first).
Ver A04 Boards started shipping Feb. 2020. (D07 to 0 Ohm resistor, floating gnd fix.)
Ver A05 Boards started shipping Nov. 2020. (3.3V circuit capacitors, spd pulse circuit, etc.)
Ver A06 Boards started shipping Feb. 2022 (capacitor changes, additions)

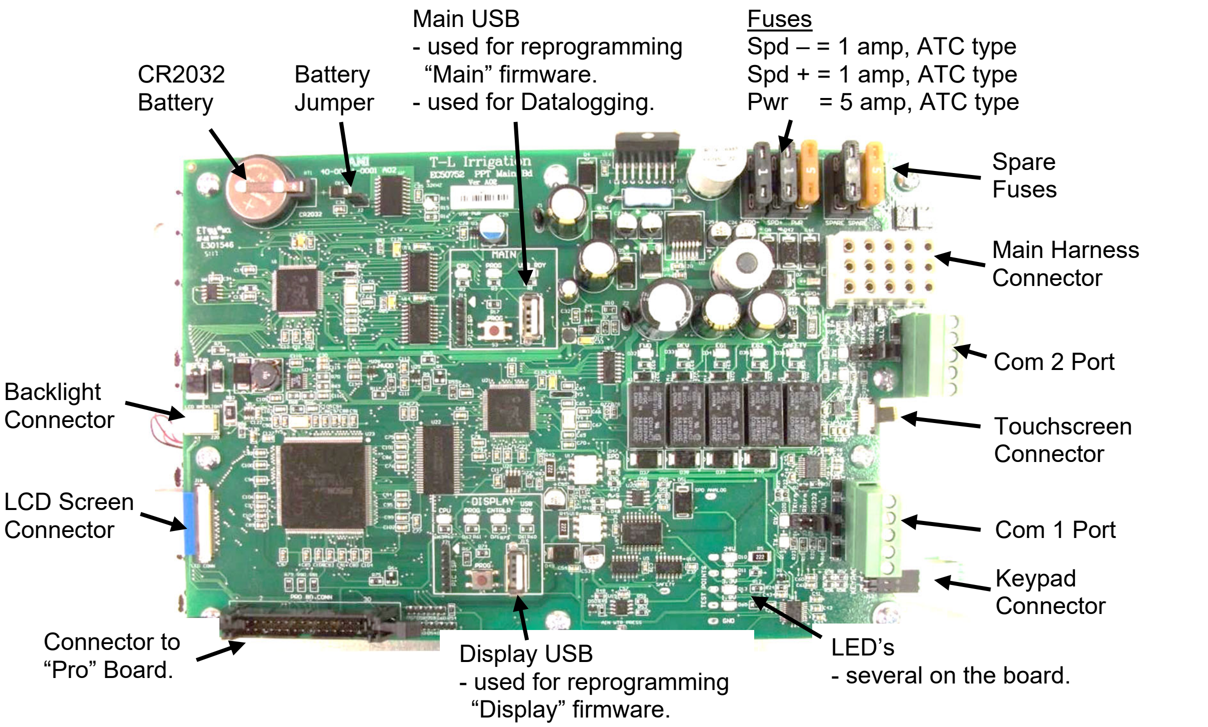


Figure 6 PPT Circuit Board Overview

6.4 *PPT "Basic" LED's Green 2-Chip (<u>OBSOLETE March 2024</u>)*

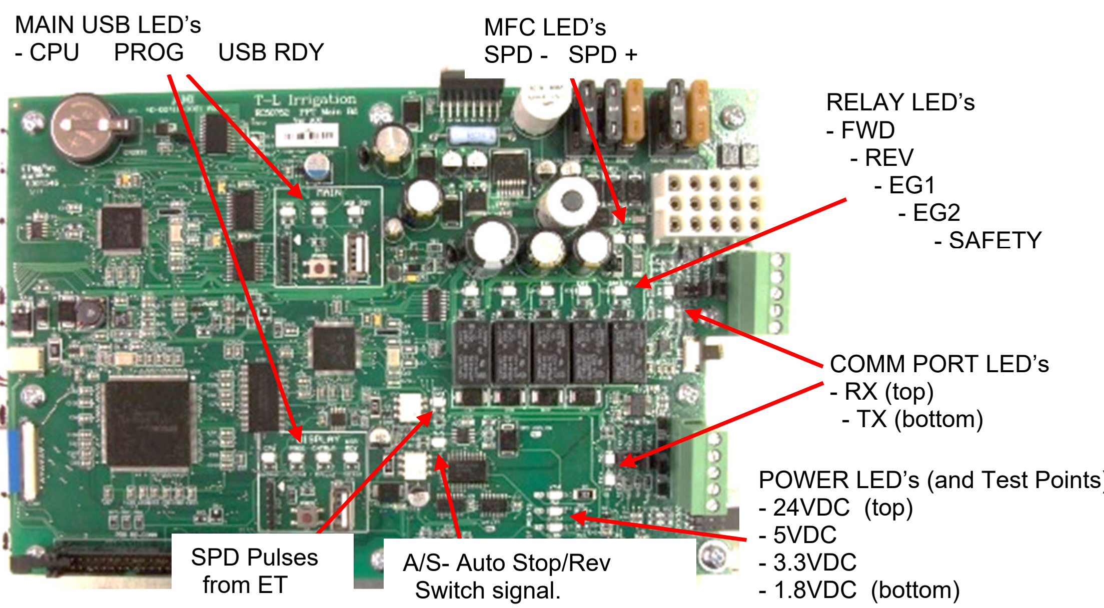


Figure 6.1. PPT Circuit Board LED's.

6.5 PPT "Basic" Comm Port Jumpers

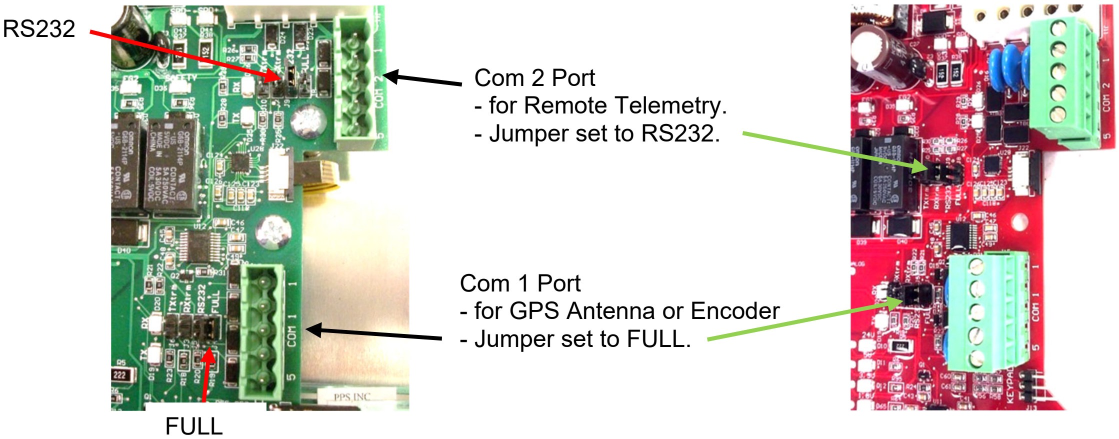
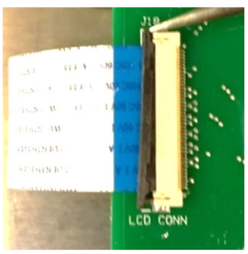


Figure 7. Comm Port Jumpers.

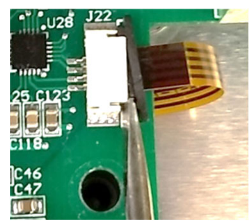
6.6 LCD and Touchscreen Ribbon Cable Connections

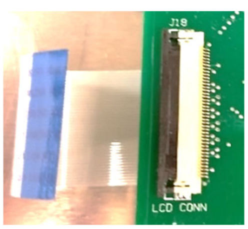
LCD Ribbon Cable

Touchscreen Ribbon Cable

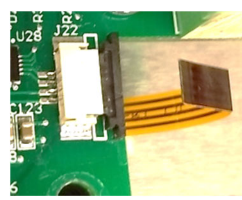


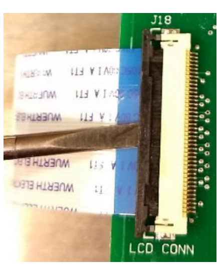
Pry both sides of black insert outwards to release tension on Ribbon Cable.



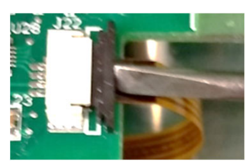


Pull Ribbon Cable out of connector.





Reinsert Ribbon Cable.
- Push Cable in and MAKE SURE IT IS FULLY SEATED INWARD.
- Then push black insert inward to lock Ribbon Cable in place.



6.7 Updating Firmware in the PPT Main Board – Red 1-Chip (Effective March 2024 -)

There will be 1 processor on the circuit board that may need to have firmware programmed.

- The firmware file name on the USB must be renamed as stated below and must be on the root level. It cannot be in a file folder on the USB.
- USB must be FAT32 format or the PPT Board will not recognize it.
- Blue USB from T-L works the best with the Red board.
- Firmware filename must be **ppt.hex**.

Update procedure:

Step 1: Save Settings to USB. (Hand record all settings as a backup.)

Step 2: Reprogram the Main processor. (Only 1 processor to program on Red 1-Chip Board).

Step 3: Perform “Restore Settings” procedure.

Step 4: Load Settings from USB or hand enter. (Double-check settings.)

Step 5: **Cycle Power** to PPT Panel – Power Off and then back On. **MUST CYCLE POWER.**

Step 6: If using Telemetry, go to MENU, go into SETUP, and check that Telemetry is YES.

MAIN

- original file name = “pptv276 Basic 1-Chip.hex” (or similar)
- copy to USB and rename. File name on USB = **ppt.hex**
- Power OFF, insert USB into Board.
- press and hold PROG Button, then Power ON.
- Release PROG button when PROG LED is lit.
- PROG LED will flash rapidly for a couple secs.
- when done the PROG LED will be off (not lit).

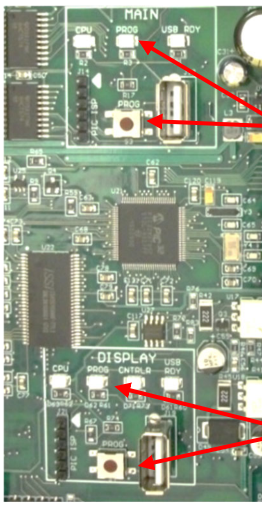
Figure 8. Main & Display USB Ports.

6.8 *Updating Firmware in the PPT Main Board – Green 2-Chip (OBSOLETE March 2024)*

There will be 2 different processors on the circuit board that may need to have firmware programmed.
- If both are to be reprogrammed, **the order does not matter**.
- The firmware file names on the USB must be renamed as stated below and must be on the root level. They cannot be in file folders on the USB.
- USB must be FAT32 format or the PPT Board will not recognize it.

Update procedure:

Step 1: Record all settings, or Save to USB.
Step 2: Reprogram Main and Display (order doesn't matter).
Step 3: Restore Settings according to Release Notes.
Step 4: Enter back all settings, or Load from USB if Restore Settings used.
Step 5: Cycle Power to PPT.



MAIN
- original file name = ppcimage253.hex (or similar)
- copy to USB and rename. File name on USB = **ppcimage.hex**
- Power OFF, press and hold PROG Button, then Power ON. Release PROG button when PROG LED is lit.
- PROG LED will flash slowly for about 20-30 secs, then will flash rapidly for 40-60 secs.
- when done the PROG LED will be off (not lit).

DISPLAY
- original file name = ppcgfx253.hex
- copy to USB and rename. File name on USB = **ppcgfx.hex**
- Power OFF, press and hold PROG Button, then Power ON. Release PROG button when PROG LED is lit.
- PROG LED will flash slowly for about 20-30 secs, then will flash rapidly for 40-60 secs.
- when done the PROG LED will be off (not lit).

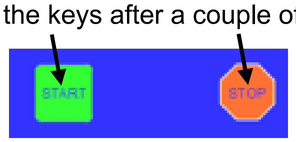
Figure 8.1 Main & Display USB Ports.

6.9 *Restore Settings*

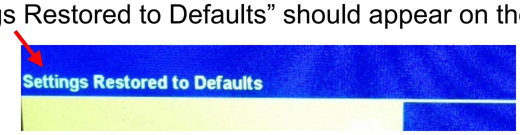
- NOTE: "Restore Settings" will erase all the settings for the Menu items.

A. IMPORTANT – PPT to do before Restore:
- Save Configuration to USB. MENU – arrow down to 2nd screen – USB. See previous section.
- or write down all the Settings in "Appendix 4 – Setup Records." See Firmware Release Notes.

Procedure to Restore Settings
- Power Switch to OFF.
- Press & Hold "START" and "STOP" keys, then switch Power ON.
  Release the keys after a couple of seconds.



- "Settings Restored to Defaults" should appear on the Title screen above the Progress Bar.



B. IMPORTANT – PPT to do after Restore Settings:
1. Load Configuration back from USB. MENU – 2nd screen – USB. "Load from USB"
   - Or manually enter all settings into PPT.
2. Cycle Power to PPT Panel - Off and then back On. Ensures Application speeds are correct.

7 SAFETY MESSAGES

7.1 Safety Messages

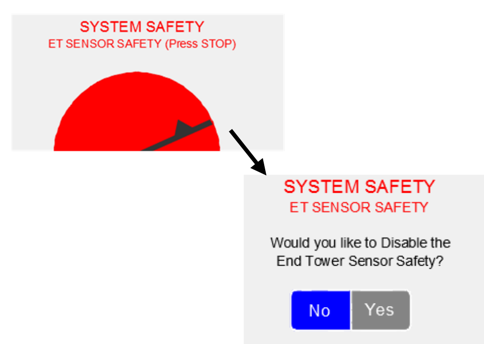
Message:	Initial Delay	Run Delay	Description
ET Sensor Safety: *(explained in more detail following)*	user set in HOME - Safeties		PPT Board not receiving pulses from End Tower, reading speed as 0 IPM for time set (30 to 240 secs). System has stopped moving, speed sensor or cable defective, or wiring problem, or other. (Max changed to 240 in v268.)
Overwater Safety:	user set in HOME - Safeties		Used if the End Tower gets stuck: GPS or Encoder - system has not moved 1.0º in the time set.
Wrong Direction (GPS):	10 minutes	7 minutes	System has moved 5 degrees in the opposite direction set. (Ignore for first 10 minutes. Then if it has moved 5 degrees the opposite direction, delay for 7 minutes and then cause safety.)
Wrong Direction (Enc):	7 minutes	7 minutes	System has moved 5 degrees in the opposite direction set.
Auto Stop Shutdown:			AS Angle reached or End Tower Switch activated. If set "By Angle", AS ignored for first XX minutes as calculated when system 1st started moving.
Position Sensor Safeties:			
GPS Comm Lost:	40 seconds (v239, was 25)	5 minutes	GPS Antenna cannot be detected or communicated with.
Encoder Comm Lost:	6 seconds	6 seconds	Encoder Position Sensor cannot be detected or communicated with (6 second delay).
GPS Error:	10 minutes	6 minutes	Number of Satellites detected less than 3.
Position Stop Shutdown:			Position Stop Angle reached.

7.2 Power Up after End Tower Sensor Safety

NOTE: Only for ET Sensor Safety:

After STOP is pressed to clear the safety, you are given a choice to Disable the ET Sensor Safety.

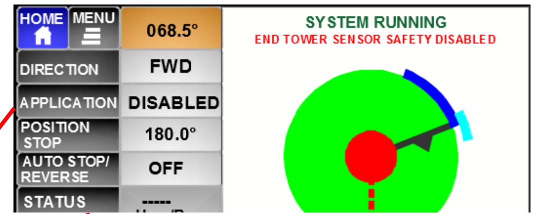
- No = Returns to System Idle screen.
- Yes = Goes to "SAFETIES" screen so the "End Tower Sensor Safety" can be Disabled.



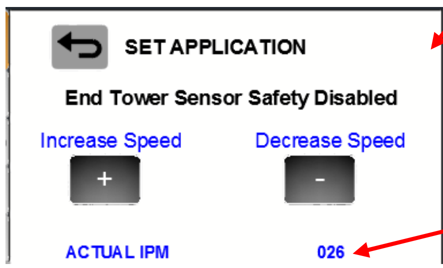
7.3 *End Tower Sensor Safety Disabled*

The End Tower Sensor Safety is used to stop the system if no End Tower Sensor pulses are received in the time set. Disable in HOME - SAFETIES.

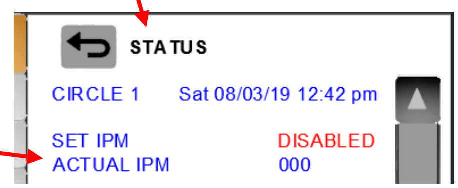
On the HOME Screen with the ET Sensor Safety Disabled, the Application rate will show “Disabled”. Auxiliary & VRI Applications will not function.



Note: With End Tower Sensor Safety Disabled, APPLICATION key functional **ONLY WHEN RUNNING.** The end tower speed can be manually adjusted from the PPT panel by using the “+” and “-“ buttons.



ACT IPM shown if working.



7.4 *Position Sensor Safety Disabled*

If Position Sensor related Safeties are occurring, the Safeties can be Disabled to allow the system to run.

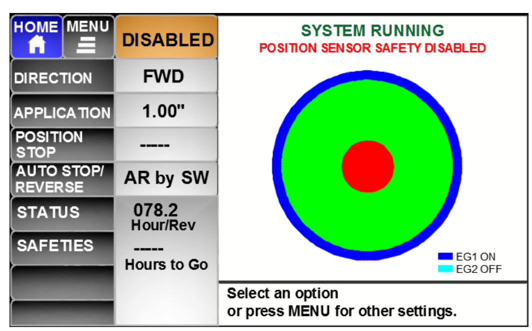
The following safeties will be disabled when the Position Sensor Safety is Disabled.

- GPS Comm Lost
- Encoder Comm Lost
- GPS Error

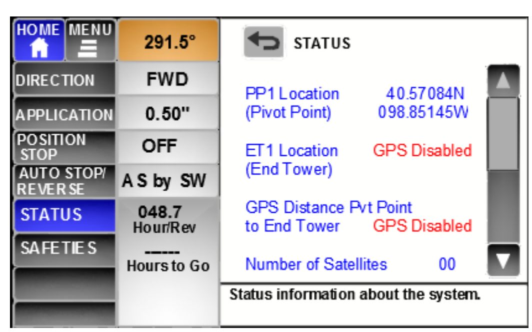
<u>System Running with Position Sensor Safety Disabled:</u>

Cannot tell the angle, no position sensor.
Angle dependent functions will not work.
- Pos Stop, Auto Stop or Reverse angles.
- Auxiliary Application angles.
- Web Based VRI tables.
- End Gun areas.

End Guns can be set to Always On or Off.
Main Application rates can still be set.
Forward and Reverse.
If an End Tower Switch is installed, the system can travel both directions, using the End Tower Switch to Auto Stop/Reverse.
- must be set in Auto Stop/Reverse.

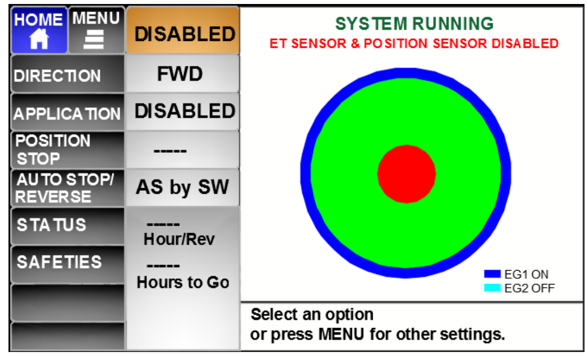


Status 2 Screen GPS information will show “GPS Disabled” for some of the readings.



7.5 *End Tower Sensor Safety & Position Sensor Safety Disabled*

System Running - End Tower Sensor Safety and Position Sensor Safety Disabled:



Cannot tell the angle, no position sensor.
Angle dependent functions will not work.

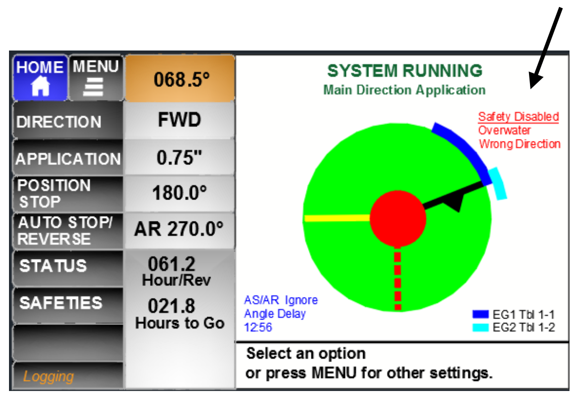
Cannot set Application Rate, no speed sensor.

Can run at speed set manually.
Can Auto Stop or Reverse By Switch.

7.6 *Overwater or Wrong Direction Safety Disabled*

System Running with Overwater Safety Disabled and/ or Wrong Direction Safety Disabled:

Message will show up in the upper right of the pivot graphical area.



These two safeties do not affect the ability to set the Application (speed) or angle dependent functions.

8 TROUBLESHOOTING

"Troubleshooting" Section copied from PPC III.

8.1 Troubleshooting Table

Symptom	Possible Cause	Test Procedure	Possible Remedy
1. System will not move in one direction, moves ok in other.	A. Direction coil defective at end tower B. Open Connection C. Direction Solenoid Contaminated	A.1. Figure 67 - Check coil ohms B.1. Check Voltage C.1. Remove Cartridge Fig 68	A.1. Replace coil, T-L Part No. EC98389 C.1. Blow Cartridge out.
2. System will not move in either direction.	A. No Power from Power Supply B. End Tower Filter plugged C. Motorized Flow Control either fully closed or plugged D. Direction Solenoid Contaminated	A.1. Figures 69 & 70. Check Power Supply B.1. Remove filter element and check C.1. Figures 73, 74, 75 - Check MFC D.1. Remove Cartridge Fig 68	A.1. Replace fuse, see Figure 69 or 70 for size B.1. Replace element, Part HF57778 C.1. Replace MFC if it will not respond. Part HV98384 D.1. Blow Cartridge out.
3. Speed will not adjust.	A. Fuse Blown on Circuit Board B. Continuity lost to MFC at End Tower C. Speed will not slow down, runs wide open	A.1. Figure 65 check Speed control fuses. B.1. Figures 73 and 74. C.1. Figure 77 to 77.6- Check Sensor	A.1. Replace Fuse, see Figure 65 B.1. Check Connections C.1. Replace sensor.
4. ET Sensor Safety or Overwater Safety using Sensor Motor occurs continuously.	A. Speed sensor defective on End Twr or Next to Last Twr B. Collector Ring Brushes dirty or not making contact C. Overwater Speed Sensor is defective D. Pulses not getting to Circuit Board E. High Voltage Spike	A.1. Figure 77 to 77.6 - Check Speed Sensor B.1. Check continuity from brush to ring See Figure 80 C.1. Test Sensor, see Fig. 77 to 77.6 D.1. Check pulses, Fig. 77 to 77.6 E.1. Check ET Manifold Bd	A.1. Replace sensor, ES53218 (M12) B.1. Clean collector ring and brush See Figure 81 C.1. Replace Sensor D.1.Check connection in wiring harness E.1. Replace ET PCB, EH53240.
5. Optional End Gun Control will not switch End Gun On or Off	A. Solenoid coil shorted open at end tower B. Open Connection C. End Gun Solenoid Contaminated	A.1. Check coil ohms Fig. 82 B.1. Check Voltage C.1. Remove Cart. Fig 82.2	A.1. Replace coil, EC98389 B.1 24 VDC C.1. Blow Cartridge out.
6. Auto Reverse Switch will not reverse pivot.	A. Loose or broken wire connections B. Wired improperly C. Reversing Switch at ET not working D. Direction Solenoid Contaminated	A.1. Check wiring from point to end B.1. Review wiring diagrams B.2. Test reversing in panel, Figure 83 C.1. Check operation of switch D.1. Remove Cartridge Fig 68	A.1. Repair, tighten as needed B.1. Rewire as needed C.1. Replace, T-L P/N EA52513 D.1. Clean Cartridge

End Tower Control Manifold Components (Figure 65)

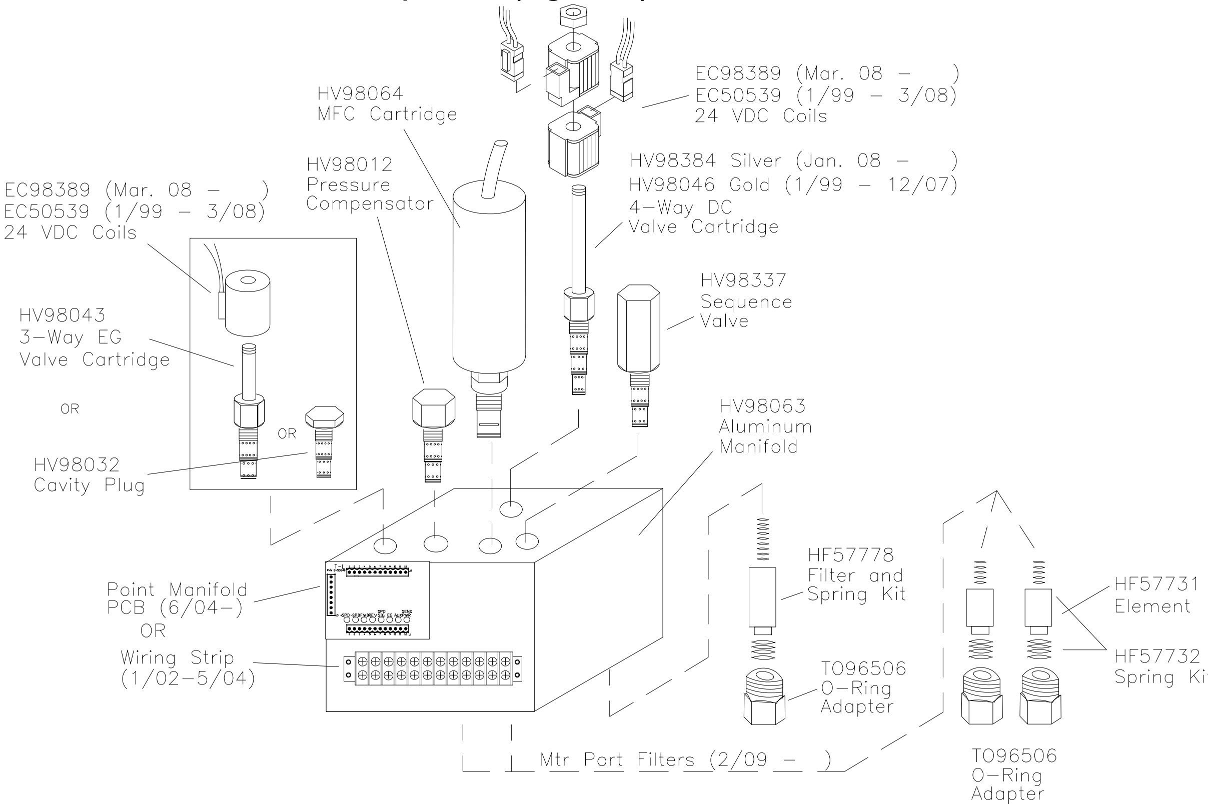
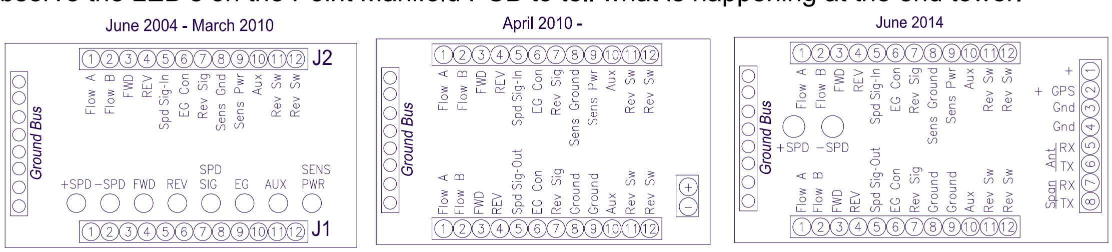


Figure 65.2 End Tower Control Manifold Components

Point Manifold PCB (Printed Circuit Board EH53240) June 2004 on

Observe the LED's on the Point Manifold PCB to tell what is happening at the end tower.



Add Power Port

Add MFC adj./GPS Connector

Figure 66.
Point Manifold PCB.

Ground Wire Checks (Especially if PPC Panel is Not on Pivot Point.)

Make sure panel grounds are in place from the #8 Terminal Block to the Pivot Frame. Use Star Washers.

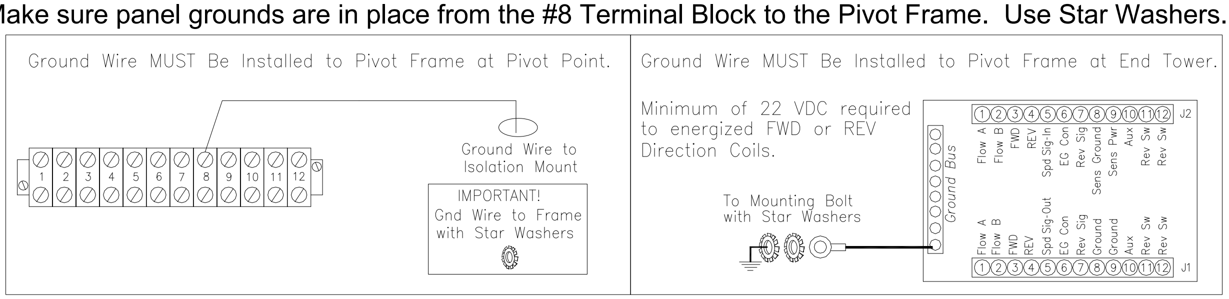


Figure 66.1. Check Ground Wires.

System Will Not Move in One Direction – Check Coils

Check for proper voltage (24 VDC) at the End Tower Terminal Block. See Figure 67.
NOTE: Voltage MUST BE at least 22 VDC for the Direction Solenoid to work properly.
CW = Forward, check between terminals 3 and 8 (ground)
CCW = Reverse, check between terminals 4 and 8 (ground).

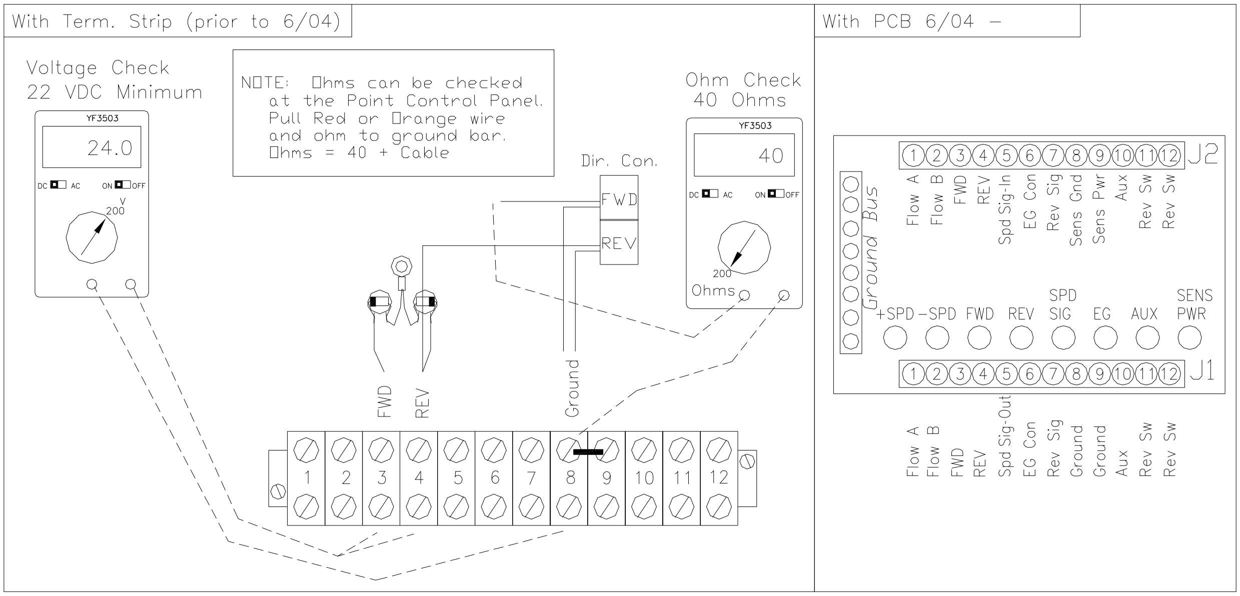


Figure 67. Check End Tower Solenoid Coils

System Will Not Move in One Direction – Check Solenoid Valve Cartridge

If the system will not stop when it should or if both coils can be removed and the system continues to walk; the 4-Way DC Valve Cartridge may be contaminated. Remove and check as follows:

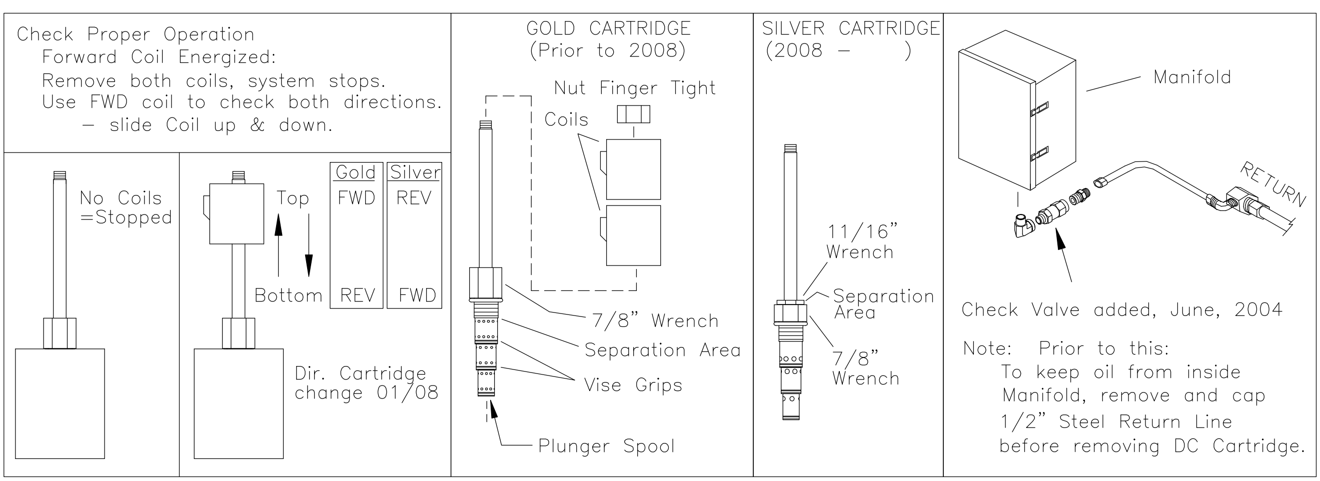


Figure 68. Check End 4-Way Solenoid Cartridge.

Remove the coils from the Valve Cartridge and then remove the Cartridge from the Body by unscrewing it. (To reduce oil loss, close the Ball Valve on the Pressure Line and if the system does not have a check valve, remove and cap the ½" Return Line.) Examine the Solenoid Valve ports for contamination. Blow the ports out, preferably with compressed air. Try to work the plunger, which is located inside the cartridge and can be accessed from the bottom of the cartridge.
The cartridge can be separated for contamination removal. The bottom portion will unscrew from the top portion. The threads have been sealed with lock-tight. See Figure 68.

System Will Not Move in Either Direction – Possible Reasons

Hydraulic reasons:

- No Hydraulic system pressure.
- Pressure Line Ball Valve at End Tower is closed.
- End Tower Filter is Plugged.
- DC Valve Cartridge is stuck, will not change position.

Electrical reasons:

- No 24VDC power to End Tower Coils.
- MFC (Motorized Flow Control) cartridge is in fully closed position.
- Ground Wires not in place at either the Pivot Point or the End Tower.

System Will Not Move in Either Direction – Check Power Supply

Electric Supplies - Figure 69. The power supply is located in the T-L Electric Panel and is regulated in the Precision Point Control III Panel. Power is supplied when the hydraulic pump is running.

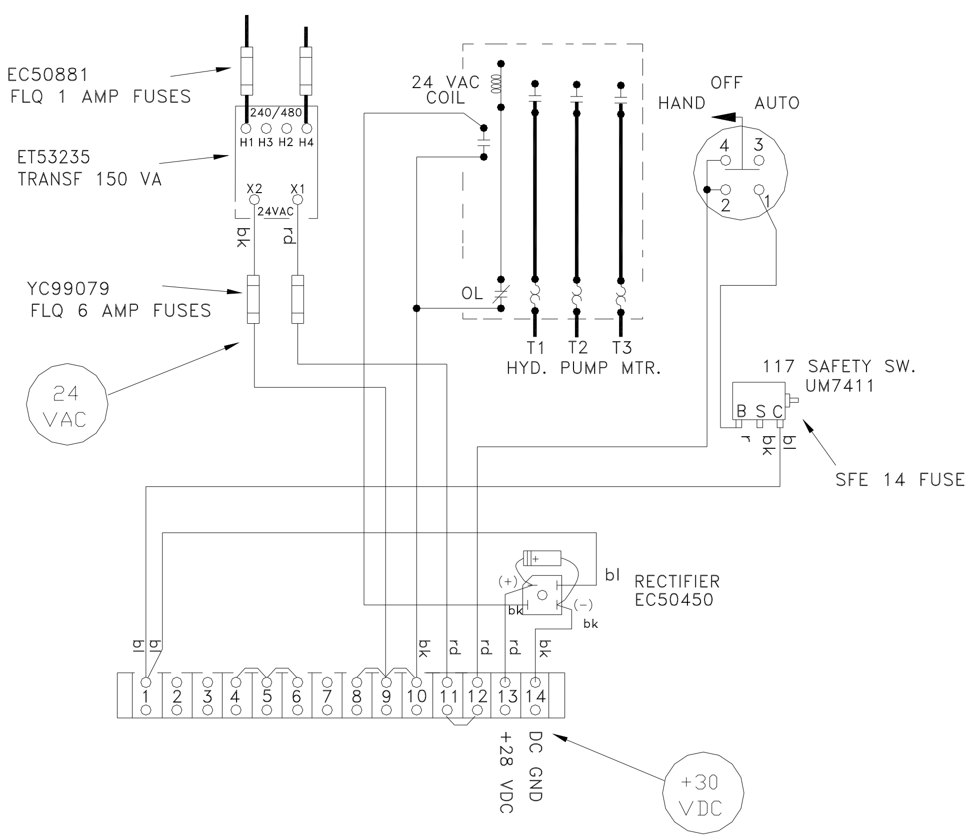


Figure 69. Power Supply Check - Electric Drive

Engine Supplies - Figure 70. The Engine Power Supply is located inside the Precision Point Control III Panel.

+12 VDC in from Murphy 117 “C”, fused with AGC 15 amp fuse
+28 VDC out, MDL 5 amp fuse
T-L Part No. EA51055: DC-DC Power Supply

The voltage input to the DC-DC Power Supply board should be **at least +12VDC**, higher is better. The lower the input voltage, the harder the board has to work to develop the output. The engine alternator must be rated at **15 amps** continuous or higher.

Following are readings that can be taken to determine why the DC/DC Converter Board is defective, once a new board has been installed:

Item	Amps	Voltage
Direction Coil	0.58	24 VDC
End Gun Coil	0.58	24 VDC
Run Light	0.80	24 VDC
Motorized FC	0.25	12 or 24 VDC
DC/DC Bd Rating	2.75 output	28 VDC Maximum Rating on Board (80 watt version DC-DC)

Example: Point Control System, moving FWD, with End Gun On and Run Light on at 24 VDC:
0.58 amps + 0.58 amps + 0.80 amps = 1.96 amps on the output side
Amps into the DC/DC Bd should roughly be 3 times the 24 VDC output current
Amps into Bd = 1.96 x 3 = 5.88 Amps in at 12 VDC

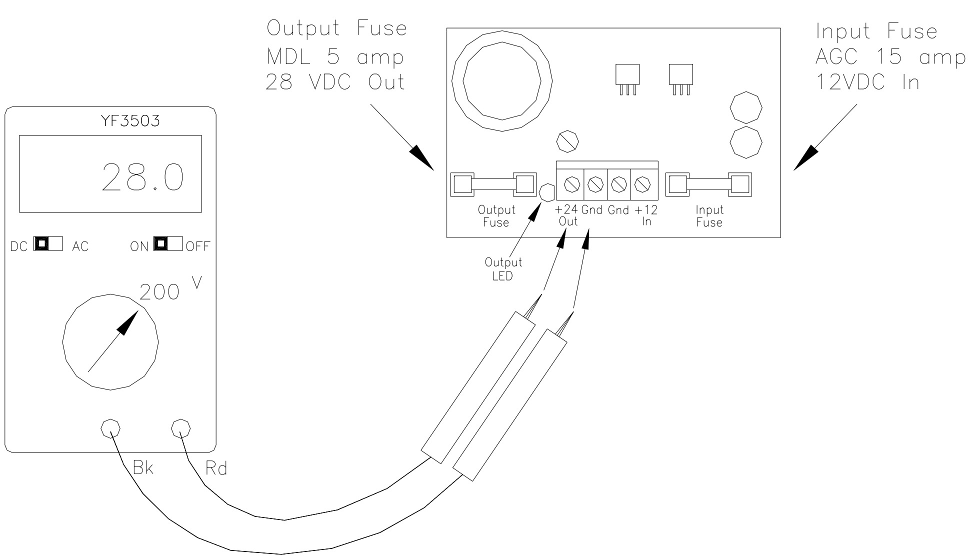


Figure 70. Power Supply Check - Engine Drive

System Will Not Move in Either Direction – Check Motorized Flow Control (MFC)

Make the following checks on the Precision Point Circuit Board:
- Verify CPU LED is flashing and that the Power LED's are lit.
- Verify integrity of F2 "- Spd Fuse" and F3 "+ Spd Fuse".
    - Remove fuse from board and use ohm meter to check continuity.

With the system operational, verify speed change indication on Test Screen:
- Verify that the corresponding LED is lit on the Circuit Board.
- Review "Automatic Speed Control" from Operation section.
- Automatic Speed Adjustment occurs every 20 seconds for the 1st 2 minutes.
    - Note the "+" or "-" on the Test Screen every 20 seconds for the 1st 2 minutes.

SET IPM	038	+
ACT IPM	036	

- Power to Terminals 1 & 2 of the 12 terminal block in the PPC panel will only be present when speed adjustments are being made.

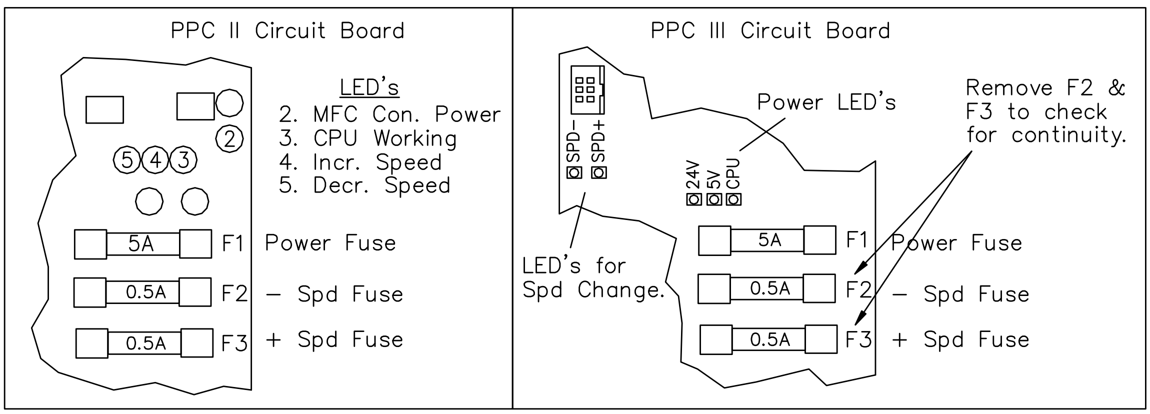


Figure 72. Check LED's and Fuses on PPC Board.

Check Voltages to MFC In PPC Panel and at End Tower.

NOTE: If either F2 or F3 fuse is blown, you may get voltage readings on both terminals 1 & 2. You also may see both "+ Spd" and "- Spd" LED's lit on the End Tower Manifold board.. .

Verify that 24 VDC is present at terminals 1 or 2 in the Precision Point Control Panel while the speed is being adjusted. See Figure 73. Then verify that 24 VDC is present at terminals 1 or 2 in the End Tower Panel while the speed is being adjusted.

It may be easier to Disable the "ET SEN SPD CON" in the System Setup and use the "+" and "-" keys on the keypad so you can make the speed changes manually instead of waiting every 20 seconds for the automatic speed change to occur.

If the voltage to the MFC checks good, then remove the Motorized Flow Control (MFC) from the aluminum body for visual inspection and operation. See Figure 74.

Manual speed adjustments at the End Tower Manifold can also be performed to check operation of the MFC Cartridge. See Figure 75.

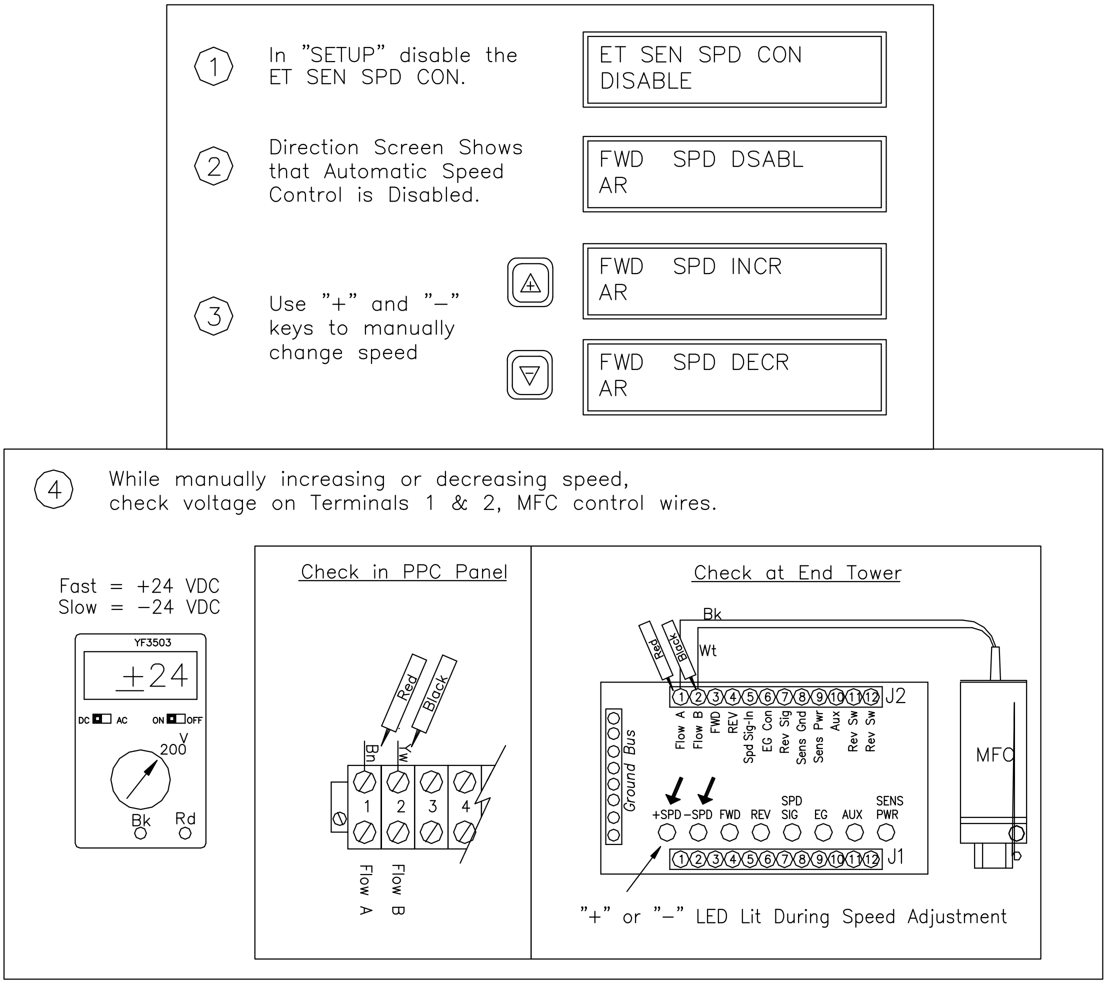


Figure 73. Check Voltage for MFC Operation in PPC II Panel and at End Tower.

Visually check the response of the Motorized Flow Control. Connect the 2 wires to a battery +12 VDC or to +24 VDC to see if the MFC adjusts. As the speed is being adjusted, you should visually see the internal sleeve rotating, until either full open or full closed position is achieved.

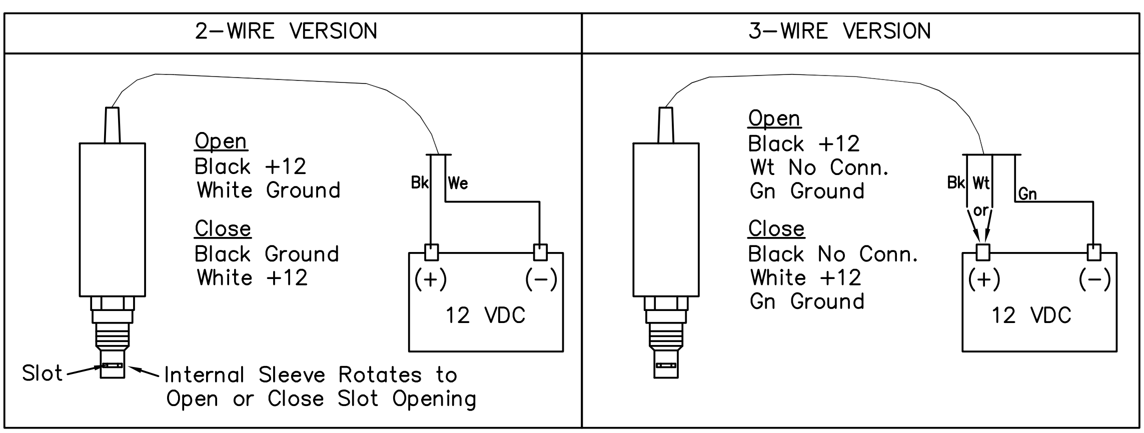


Figure 74. Check Motorized Flow Control by Removing and Connecting to a Battery.

Manually Adjust Speed at End Tower, System Moving Either FWD or REV

- Pushbuttons added to ET Manifold Board June 2014.
- Green Wire added to MFC in 2008. Green Wire disconnected from Gnd Bar June 2020.
- If Green Wire not connected, temporarily connect Green Wire to Gnd as troubleshooting aid to manually adjust the MFC End Tower Speed.

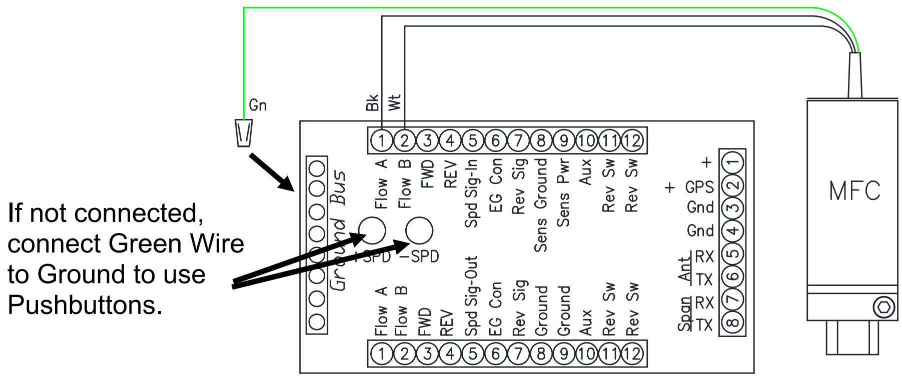


Figure 74.1 Use +SPD and –SPD Buttons to adjust.

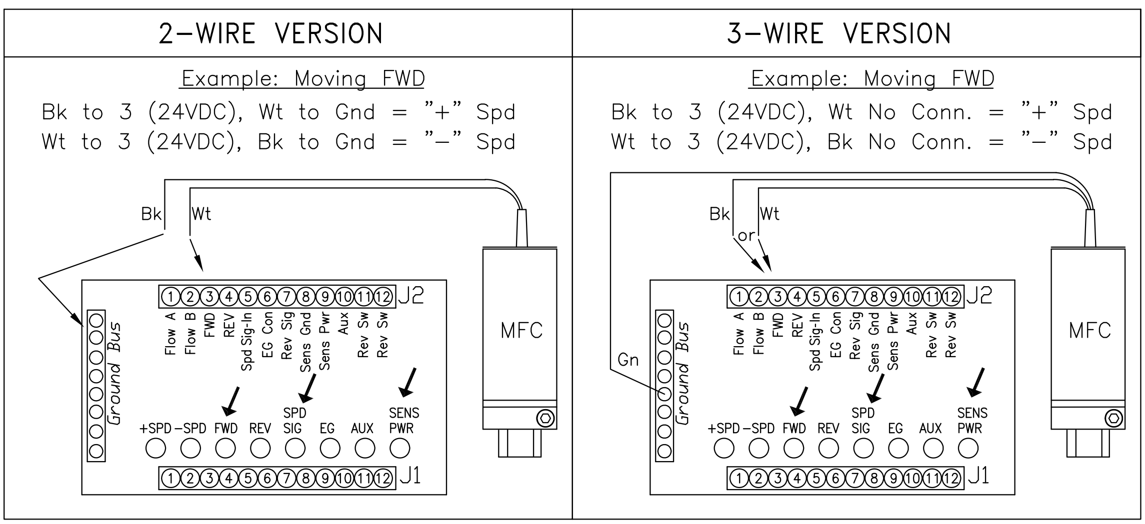


Figure 75. Manually Adjust End Tower Speed using Bk & Wt wires of MFC Valve.

Mechanically set MFC so End Tower Creeps Slowly

- If the end tower must run really slow, the MFC can be mechanically adjusted so it cannot close all the way.

- Electrically close MFC.
- Loosen Set Collar.
- Manually rotate MFC Body CW.
- Tighten Set Collar.

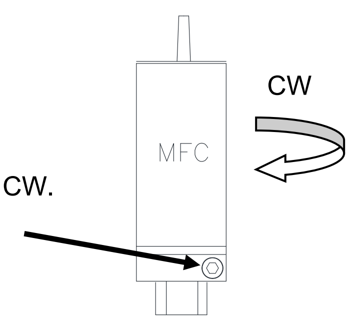


Figure 75.1. Mechanically set MFC.

Intermediate Tower in Bypass, ET Sensor Safety

NOTE: Use Caution, refer to Pivot Operators Manual.
Option 1: End Tower movement not needed to re-align:

1. Do not use PPC, all toggle switches in PPC Panel down.
2. Start Hydraulic Pump, safety switches to “Start” position.
3. Re-align tower, start PPC and operate normally.

Option 2: End Tower movement desired as tower is re-aligned:

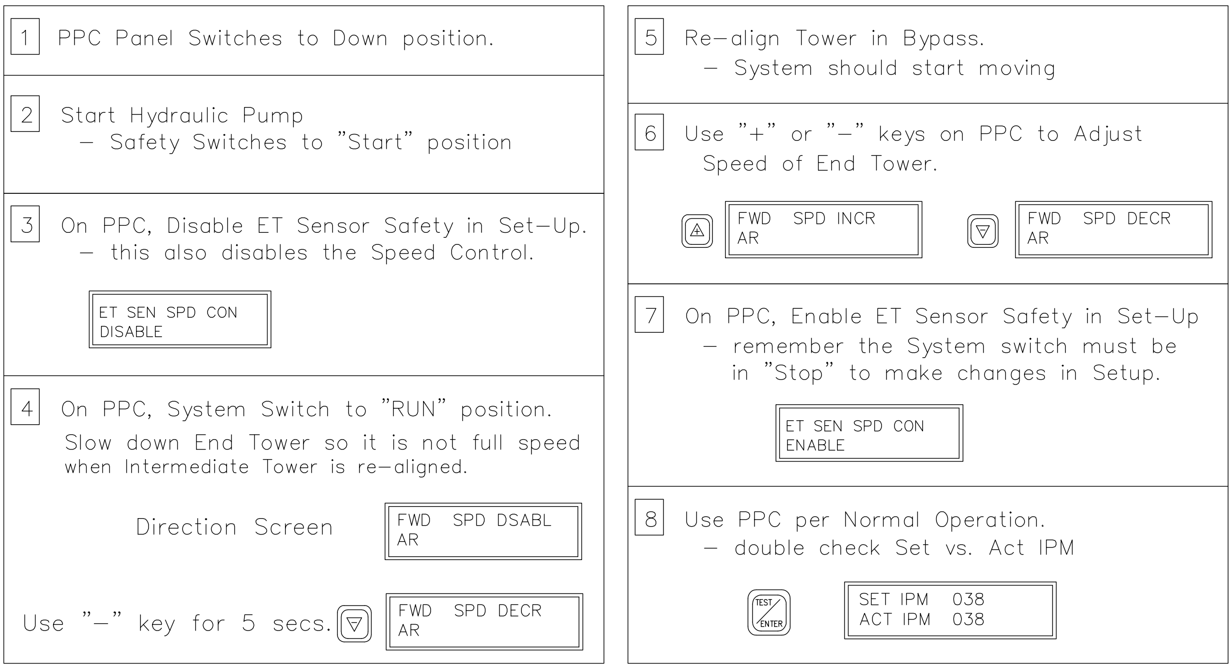


Figure 76. Procedure to Re-align Tower in Bypass.

ET Sensor Safety (Actual Speed Displays "0" When "TEST" Button is Pressed)

- Precision Point Control Ill Board not receiving pulses from the End Tower Speed Sensor
  - for 60 secs. (vers 2.02 on) with "System" switch in the "RUN" position,
- The end tower may be stopped, or it may be moving with a defective sensor or wiring problem.

OPERATION NOTE: If Overwater Speed Sensor works, those pulses can also be used as the End Tower Sensor Signal until the ET Sensor is changed. (Jumper 9 to 5 in PPC Panel, remove blue wire from 5).

STEP 1: Initial Checks at PPC Control Panel
Result: If no pulses, disable ET Sensor Safety.
Note: Need to slow End Tower Speed to 6 – 10 IPM.

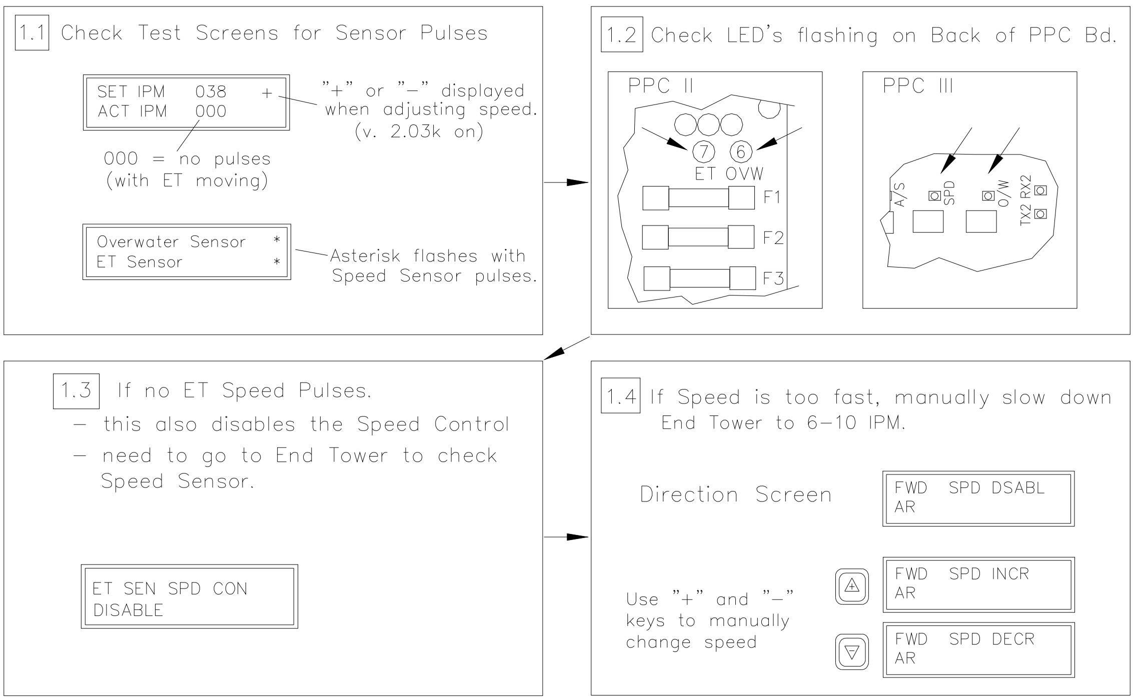


Figure 77. Initial Check's to Make at Precision Point Control Panel.

Speed Pulse Testing Notes for the following Steps:

- Make sure that End Tower is moving 6 - 10 IPM.
- Some "Auto-Ranging" meters cannot react quickly enough to correctly measure the pulsing voltage during the above tests. The use of Auto-Ranging meters is not recommended.
- If sensor pulses are present but the low voltage is above 1.0 VDC during any of the tests, the typical cause is poor system grounding. Be sure to check all grounds at components and if there are any available wires in the span cable add them to ground.

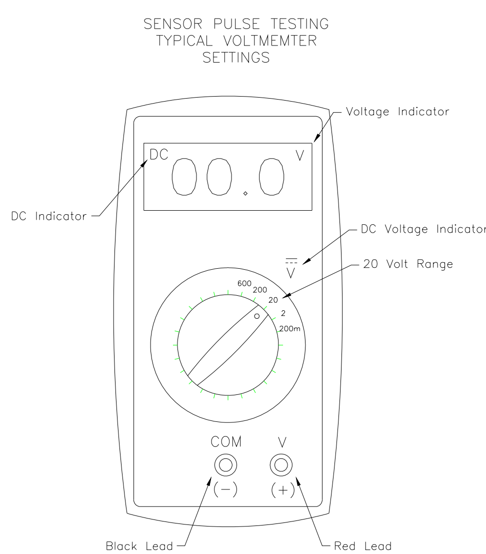


Figure 77.a. Voltmeter Settings

STEP 2: Check Pulse Voltage in PPC Control Panel – Blue Wire Connected to #5.

Result: If Pulse Voltage is good at Term #5, Speed Sensor and wiring back to PPC must be good and PPC Board must be defective.
- Replace PPC Board.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

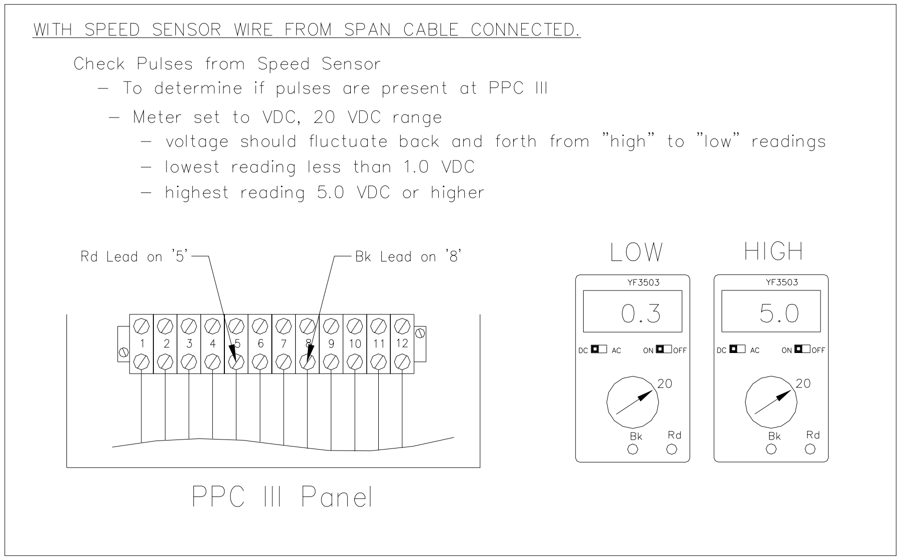


Figure 77.b Check Speed Pulses at PPC III

STEP 3: Check Pulse Voltage in PPC Control Panel – Blue Wire NOT Connected to #5.

Result: If Pulse Voltage is good at Blue wire when NOT connected to Term #5, Speed Sensor and wiring back to PPC must be good and PPC Board must be defective.
- Replace PPC Board.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

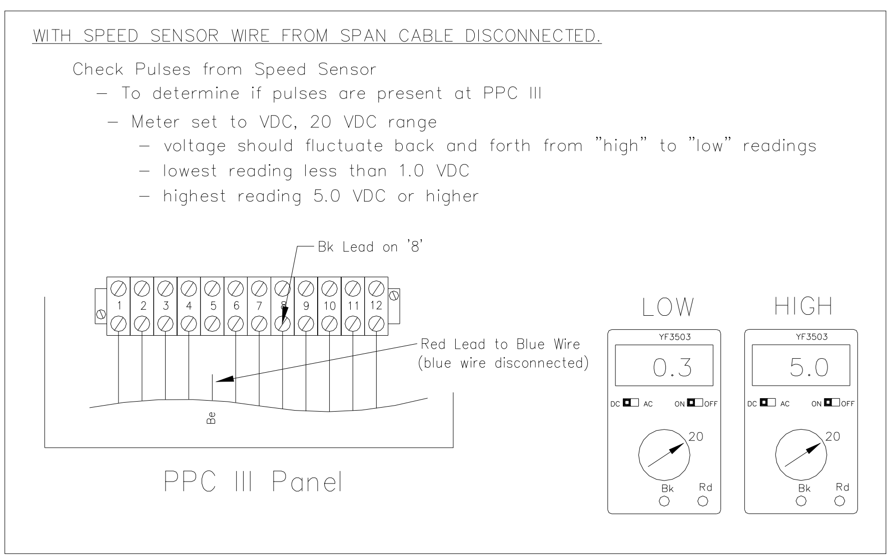


Figure 77.c Check Speed Pulses at PPC III – Blue Wire Disconnected

STEP 4: At End Tower, Check Power to Speed Sensor

Result: If 19-22 VDC supplied to Speed Sensor, continue to Step 5.
If no voltage, verify Fwd or Rev Voltage, replace defective component.

Notes: Use respective diagram for type of wiring at End Tower.

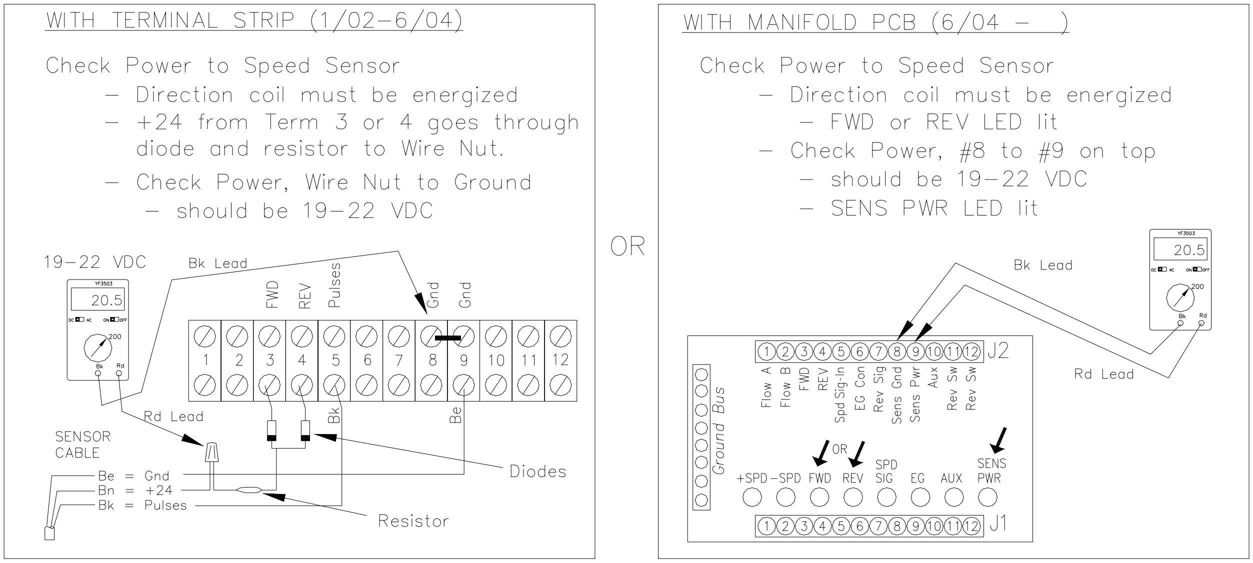


Figure 77.1 Check Speed Sensor Power at End Tower.

STEP 5: Check Speed Sensor Pulses at End Tower

Result: With Terminal Strip: If no pulses, go to Step 6. If pulses, check wiring back to PPC.
With Manifold PCB: If no pulses on top, go to step 6. If pulses on top and not on bottom, go to step 6. If pulses both top & bottom, check wiring back to PPC.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

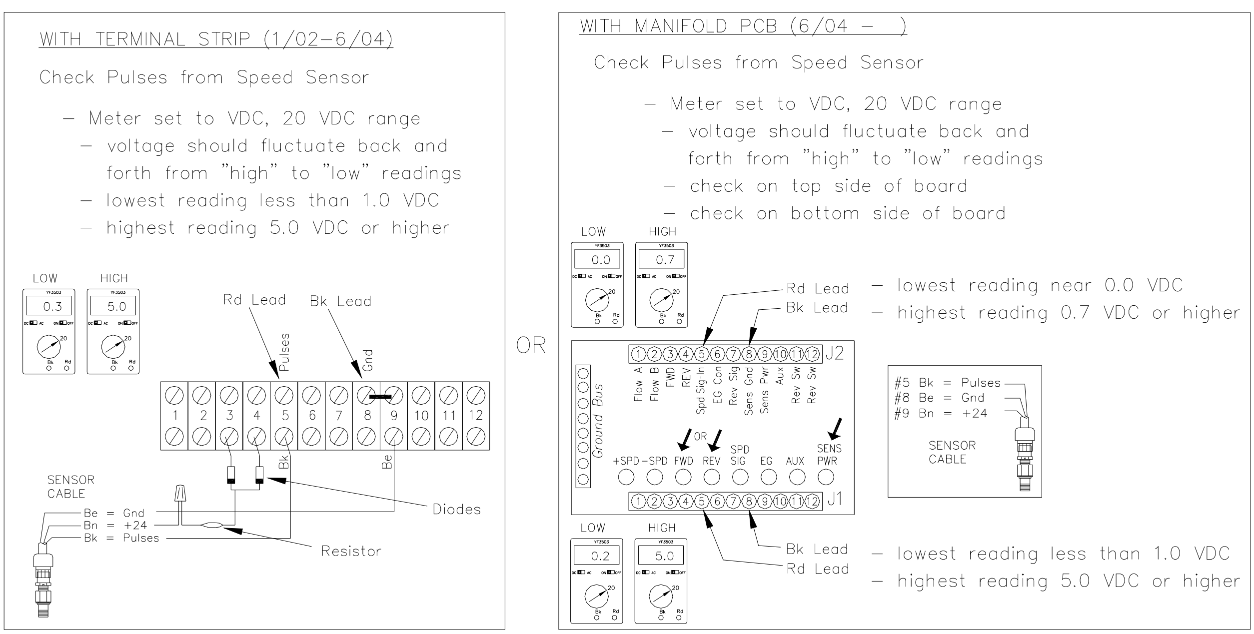


Figure 77.2 Check Speed Pulses from Sensor at End Tower.

STEP 6: Check Speed Sensor Pulses at End Tower, Black Pulse wire NOT Connected to Term 5.

Result: If no pulses, then Speed Sensor or Cable is defective, go to Step 8. If pulses when not connected but no pulses when connected, then pulses must be shorted somewhere – check all connection points back to PPC Panel.
If Manifold PCB, go to Step 7.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

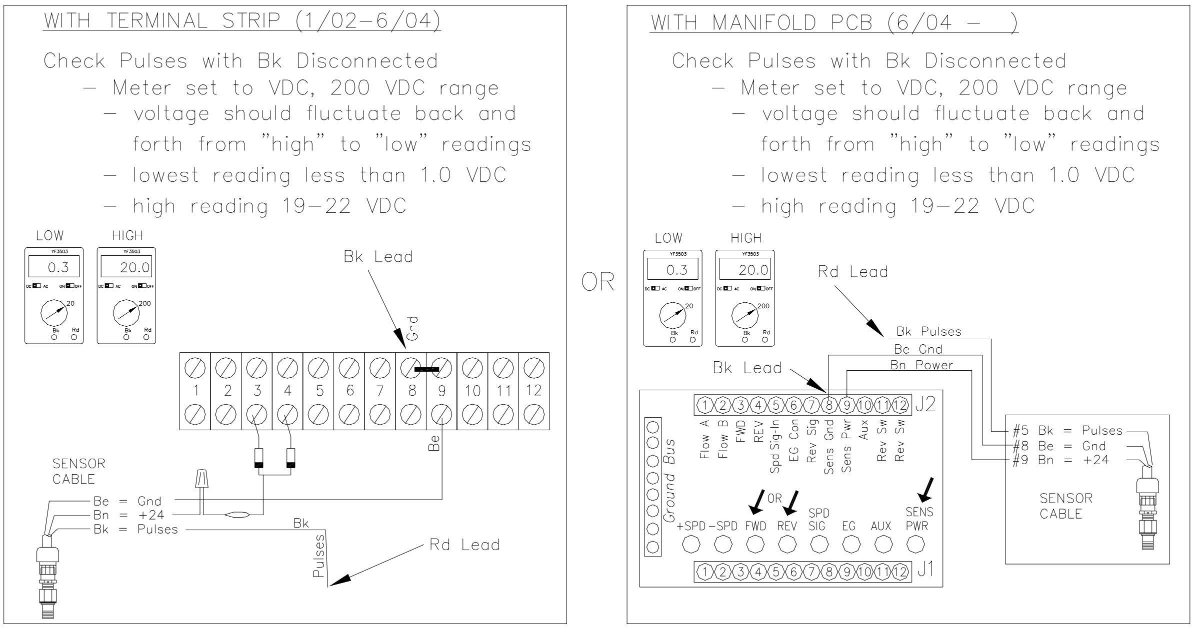


Figure 77.3 Check Speed Pulses from Sensor with Bk Signal Wire Disconnected at End Tower.

STEP 7: With Manifold PCB – Connect Bk Pulse Wire direct to Span Cable, bypass Circuit Board.

Result: If no pulses, then pulses must be shorted somewhere – check all connection points back to PPC Panel. If pulses, then replace Manifold PCB, T-L PN EH53240.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

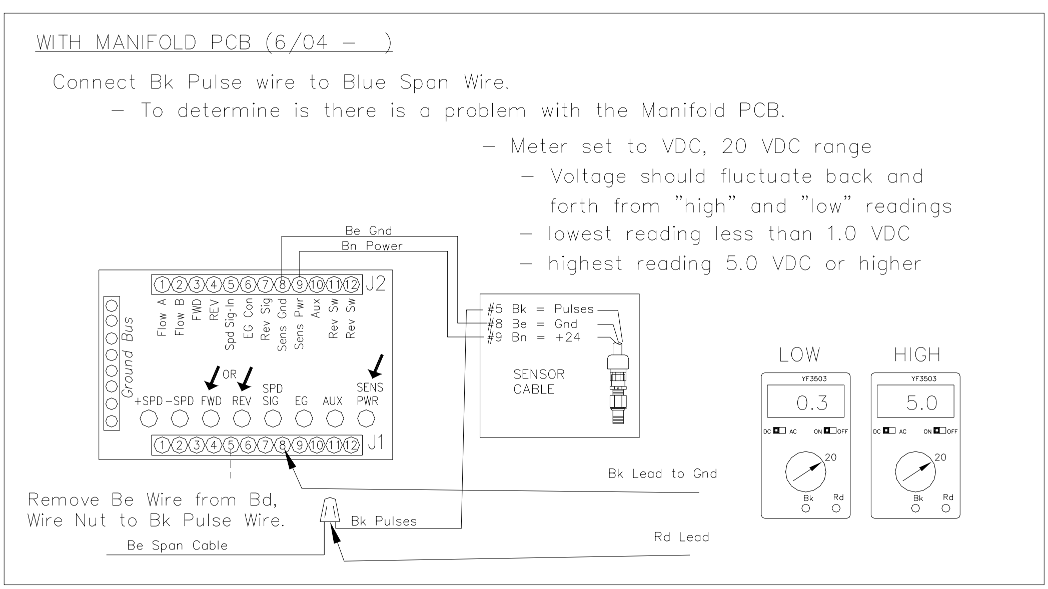


Figure 77.4. Bypass Manifold PCB with Signal Pulse Wire.

STEP 8: Check Speed Sensor on system using T-L Speed Sensor Tester – EA532180.

Result: If Speed Sensor checks good, replace Sensor Cable (ES53219)

If no sensor pulses, and alignment is good, replace Speed Sensor – see Step 10.

Note: See Sensor Tester Notes below.

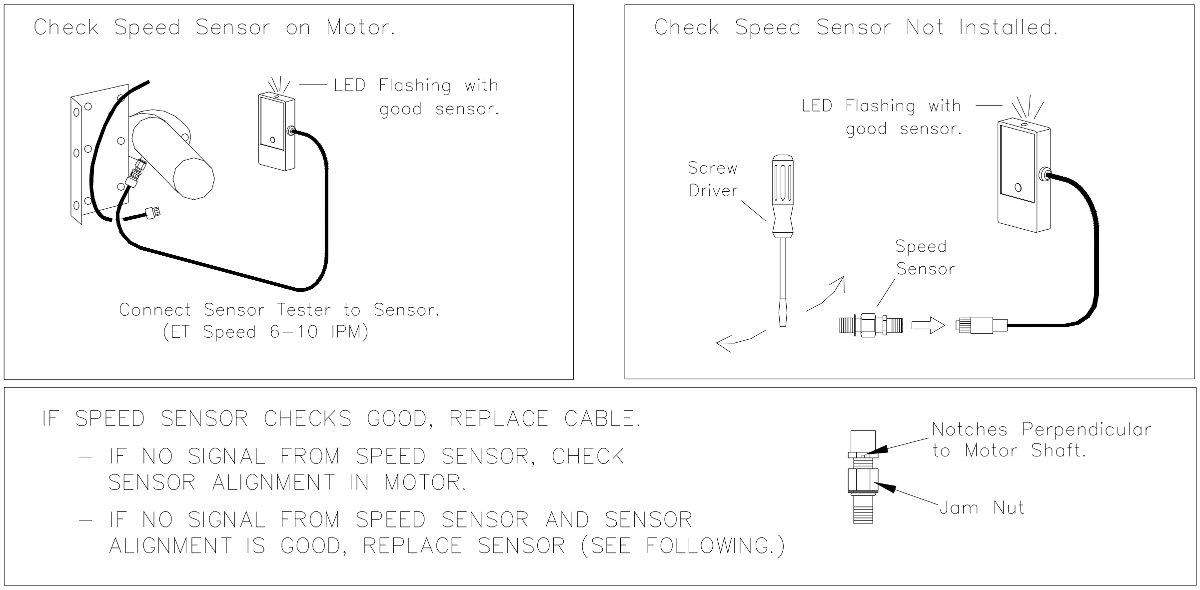


Figure 77.5. Use Sensor Tester to Check Speed Sensor.

T-L Sensor Tester (EA532180) Notes:

- Make sure that End Tower is moving 6 - 10 IPM.
- Press the Battery Test button on tester before connecting to sensor.
  - LED on top of Tester should light.
- Sensors that are not installed can be tested by connecting sensor to Tester and then placing "Ferrous" metal (screwdriver, pliers, etc) next to sensing face of sensor. Then move back and forth across the face of sensor to simulate motor shaft teeth going by sensor.

STEP 9: Check for additional Ground wire if Remote Panel, offset from Pivot Point.

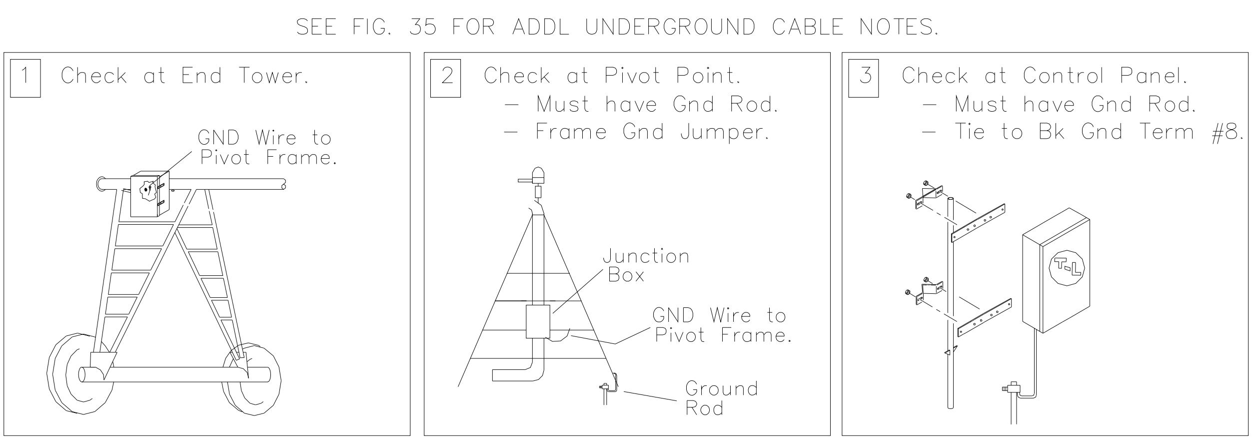


Figure 77.6. Remote Panel Sensor Checks.

STEP 10: Replace Speed Sensor.

If Speed Sensor does not generate pulses with +24 power to the sensor, it will need to be replaced using the following procedure:

1. Remove existing speed sensor wires from end tower junction box OR if separate yellow cable comes off sensor, unscrew cable from sensor
2. Unscrew existing speed sensor from hydraulic motor.
3. Install new speed sensor.
   - turn speed sensor in till it bottoms out on shaft (do NOT force) (be sure jam nut is backed off sufficiently)
   - back speed sensor out at least ¼ turn CCW plus the amount necessary to align the notches. Notches on speed sensor hex nut must end up perpendicular to motor shaft.
   - back speed sensor out only enough to align notch (at least ¼ turn)
   - hold in place and tighten jam nut.
   - Parker Sensor motors do not require alignment

NOTE: If the Speed Sensor at the End Tower is defective or pulses are not received by the Precision Point Control III Board, the Board will see this as the pivot not moving and thus keep trying to increase the speed of the pivot until the "ET Sensor Safety" occurs. (0 IPM actual vs. Set speed)

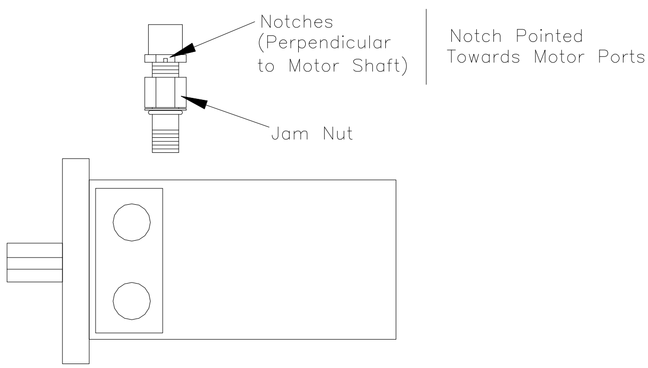


Figure 78. Speed Sensor Replacement.

Overwater Safety Occurs Continuously (Sensor Motor)

If the system is moving and Sensor Motor is being used for Overwater Timing, and "Overwater Safety" occurs, try the following:

a. Verify Overwater Pulses on Test Screen or check "OVW SIG" LED on PCB.
   - see "Overwater Timer Operation" in the Operation Section
b. Lengthen the "Overwater Timer" value in "Set-Up"
c. Check the Overwater Speed Sensor for proper operation and pulses
   - see Figures 77 to 77.6.
   - if the Overwater speed sensor is defective, replace as shown previously.
   - Check ground jumper wires to frame. Use Star Washers for good grounding.

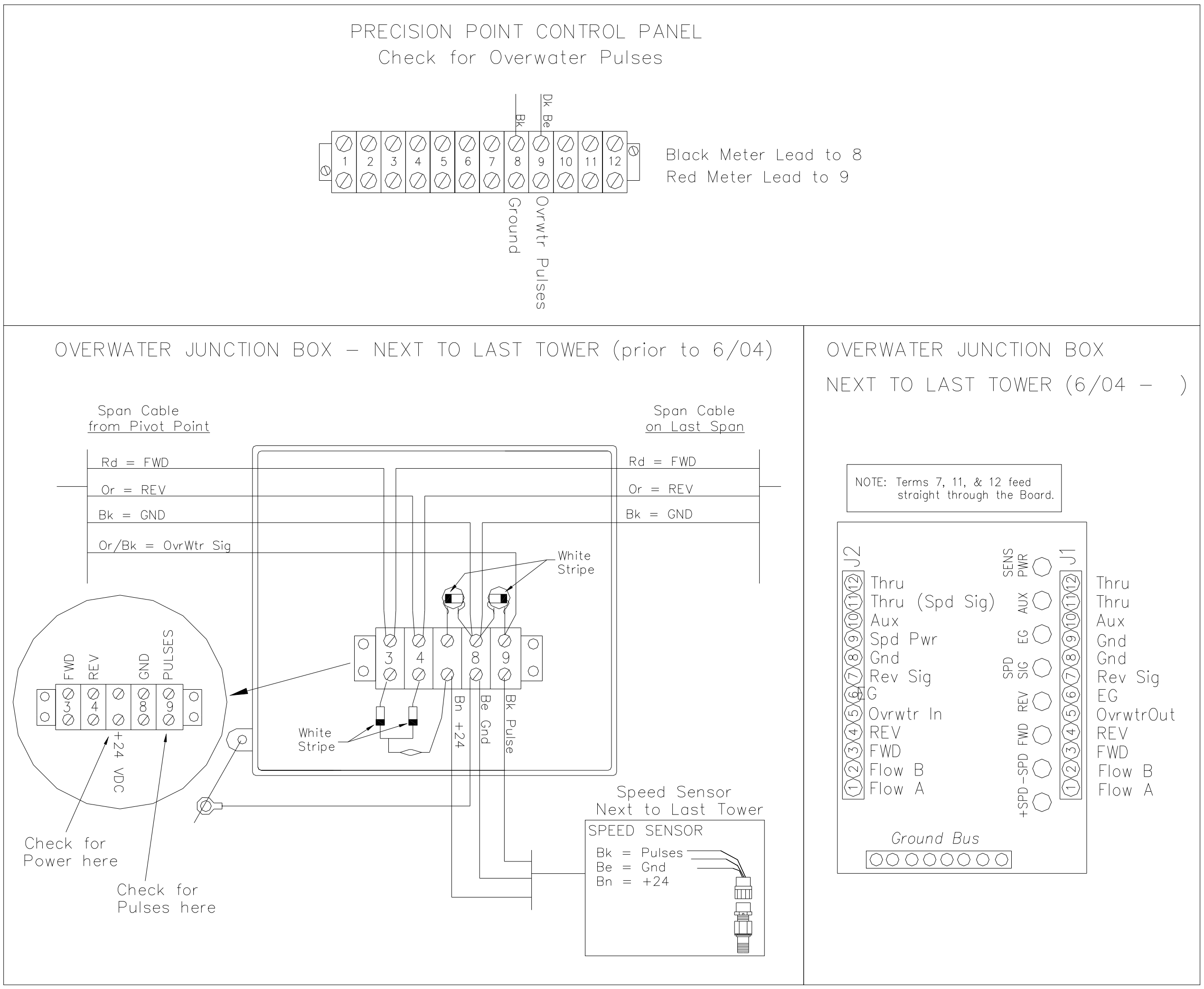


Figure 79. Check Overwater Speed Sensor Operation

Checking and Cleaning Collector Ring

Over time, buildup can occur on the Collector Rings and Brushes, causing a high resistance. This may cause erratic speed readings. See the following diagrams for checking and cleaning the Collector Ring.

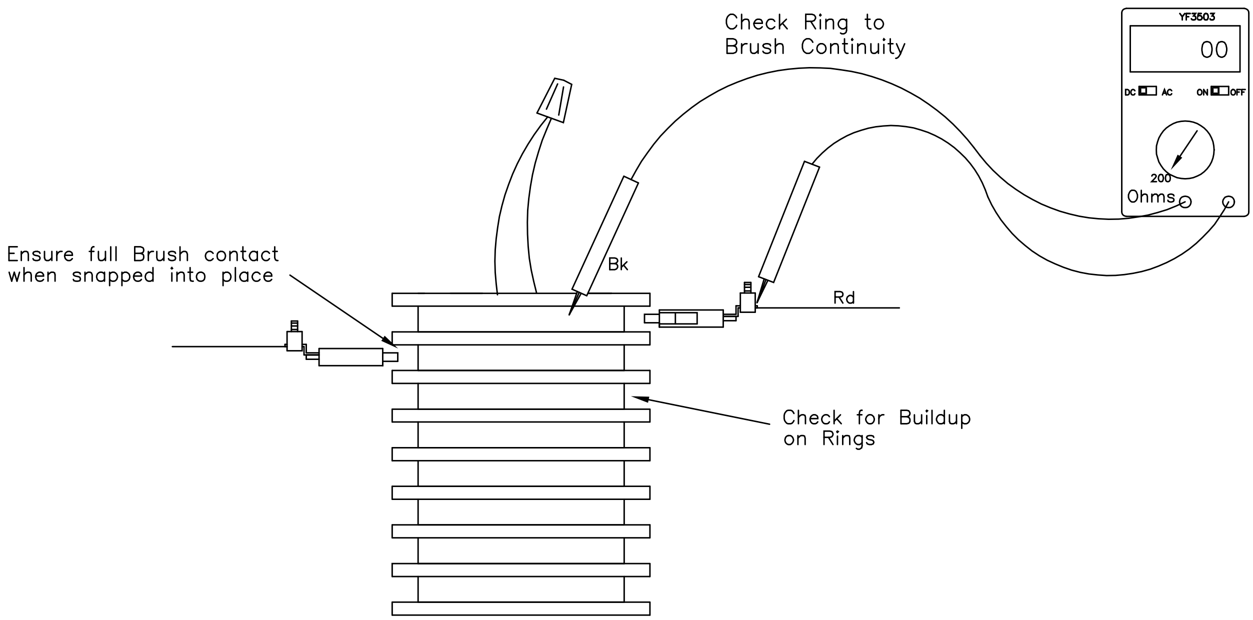


Figure 80. Check Collector Ring Continuity

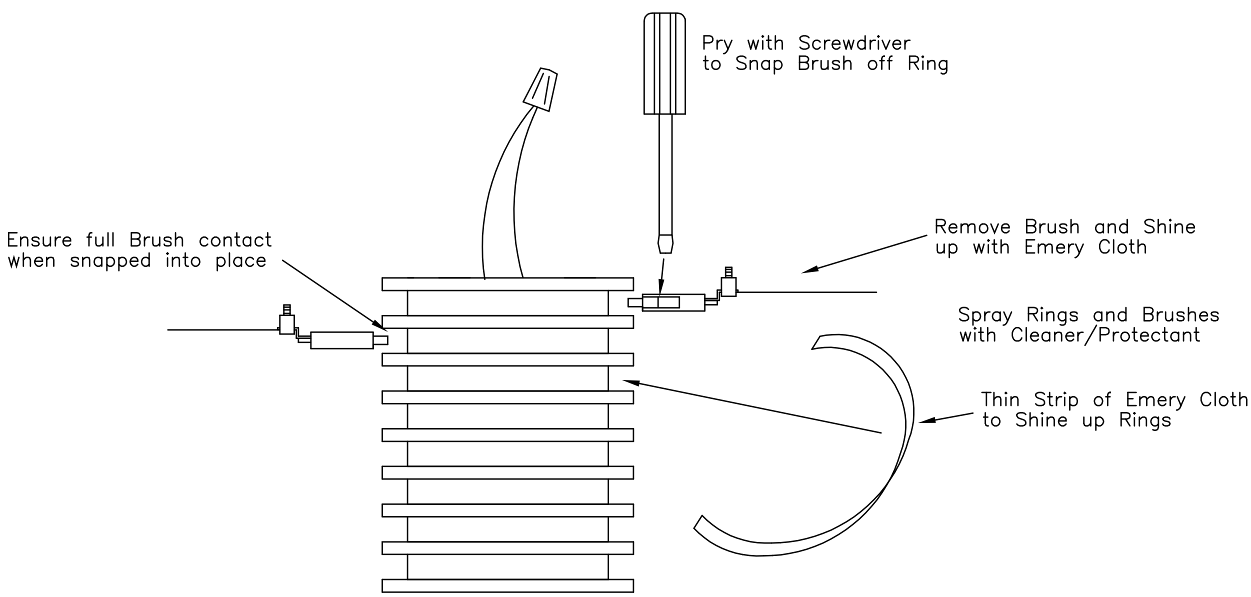


Figure 81. Cleaning Collector Ring and Brushes

End Gun Not Working

1. Check Wiring, Voltages, and LED's on Circuit Boards.
   - With Position Sensor, use End Gun Relay on PPC III Board:

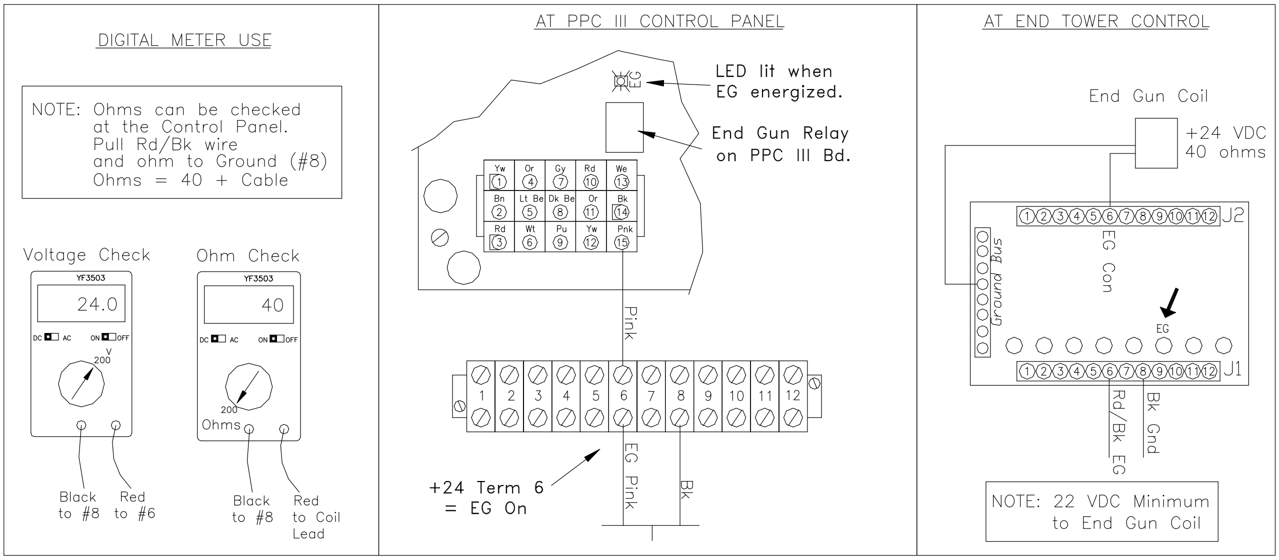


Figure 82. Check EG Operation, PPC III with Position Sensor.

- Without Position Sensor, use Roller Switch at Pivot Point>

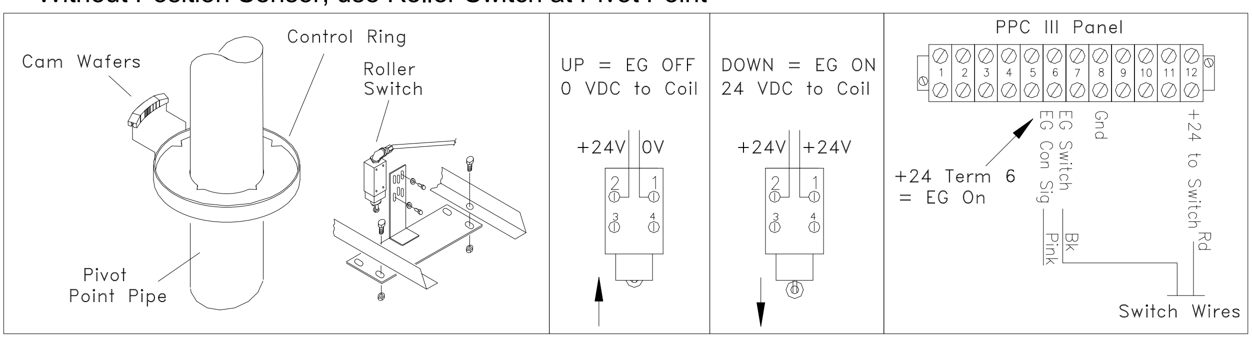


Figure 82.1. Check EG Operation, PPC II or III without Position Sensor.

2. Check Proper Operation at End Tower:
   - proper voltage to End Gun Coil (+24 VDC)
   - ohm the Coil (40 ohms)
   - Nut should be on FINGER TIGHT (3 ft. lbs. torque)
   - remove the Cartridge and check for contamination
   - Separate top from bottom if Plunger Spool will not shift.

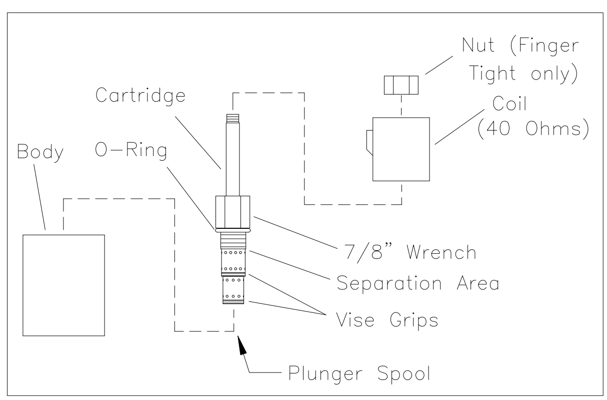


Figure 82.2. Check End Gun Solenoid.

3. Check Operation of Cylinder Shutoff Valve (without Booster).

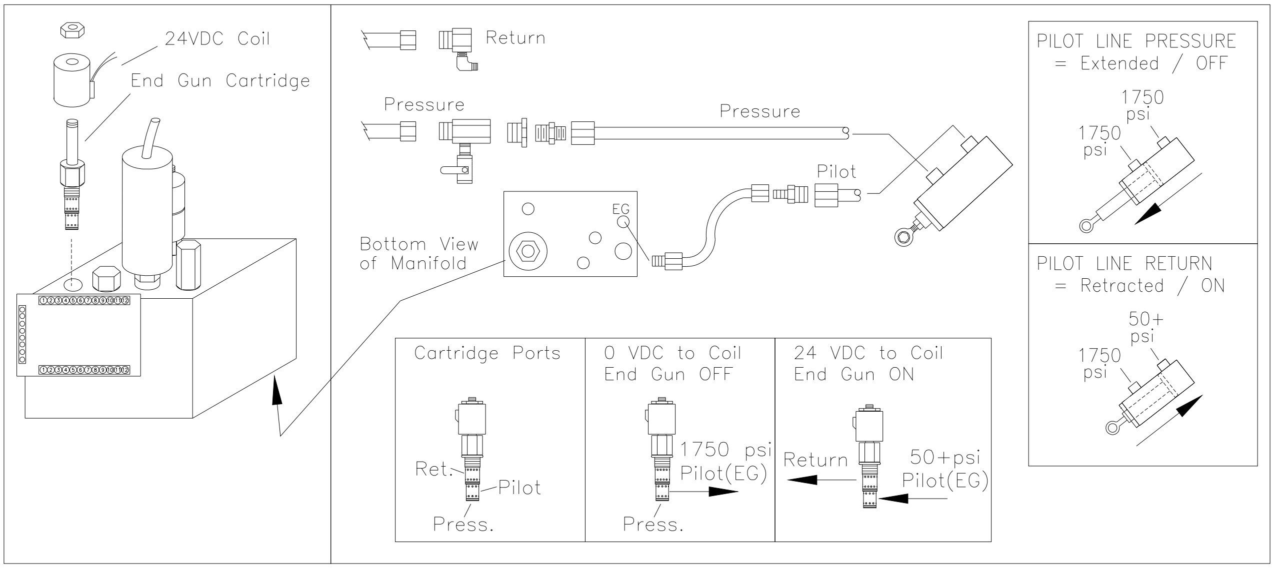


Figure 82.3. Check Operation of Cylinder Shutoff Valve.

4. Check Operation of Cylinder Shutoff Valve (with Booster).

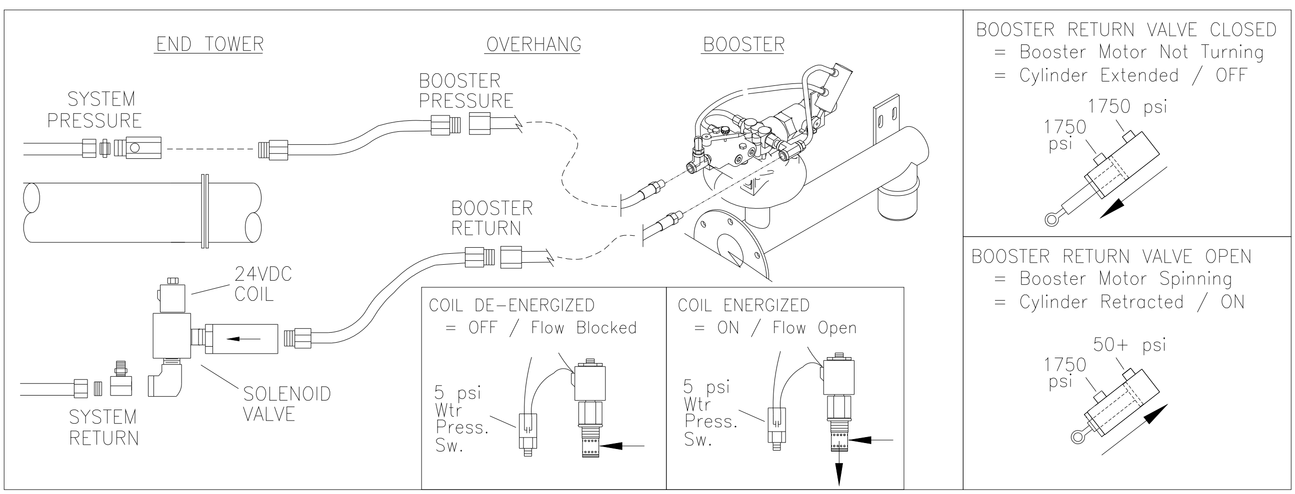


Figure 82.4. Check End Gun with Booster Solenoid.

Checking Auto Reverse Operation (by Switch)

Determine if the problem is hydraulic or electrical.

- Check that the direction coils are being activated correctly (see previous sections).
- If the coils are activated correctly, the problem is more than likely hydraulic and probably contamination in the directional control solenoid valve cartridge.
- If the coils are not switching, the problem can be assumed to be electrical. Start at the Control Panel and trace the direction control wires from the Control Panel to the End Tower.
- Trace the Reversing Signal from the Reversing Switch to the Control Panel. The Auto Reverse or Auto Stop function of the Precision Point Control III Panel can be checked by using a jumper wire in the control panel as shown in Figure 83.
- The A/S LED on the PPC III Bd will be lit when the Auto Reverse Signal is present.

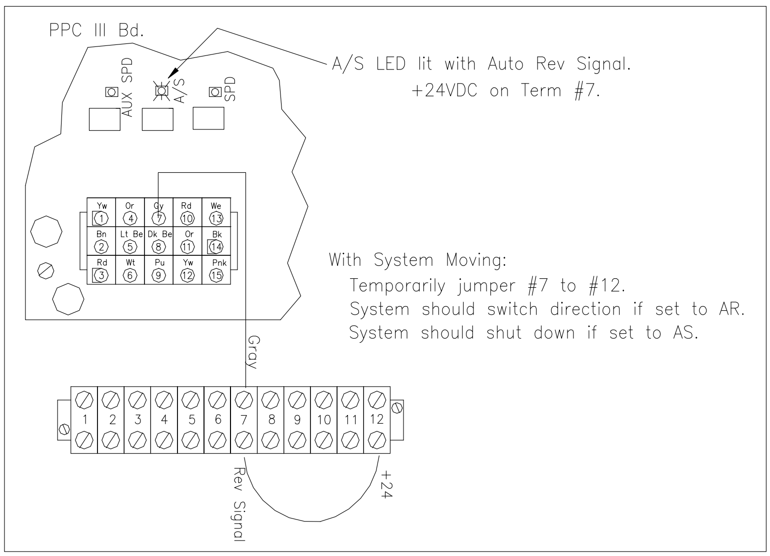


Figure 83. Check Reversing Operation in Precision Point Control III Panel.

Verify functionality of switch.

- The function of the switch can be checked by verifying continuity as described below.
- Once activating the A/R function of the PPC III panel, you must wait at least 30 seconds before attempting to activate the function again.

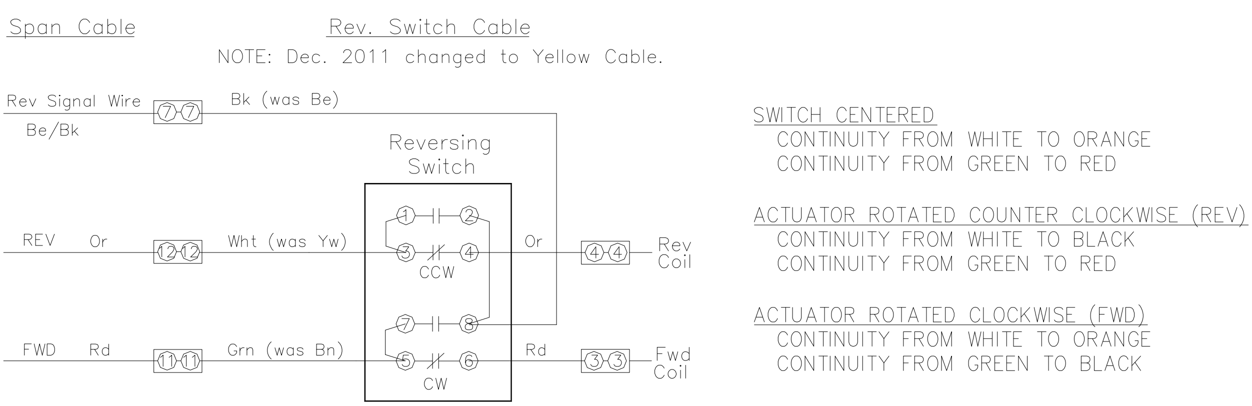


Figure 83a. Check Reversing Switch Functionality.

Communications Error (Encoder Comm Lost or GPS Comm Lost)

Check all electrical connections:

- power to Encoder or GPS.
- comm wires connected properly at Com 1 Term Block on PPC III Bd.
- comm wires connected properly at Encoder or GPS Antenna..
- System must have good grounding for comm. to work. Use ET sensor low pulse voltage to verify good system grounding (less then 1.0 VDC low) at PPC III panel. Check Panel, J-box, ET manifold, grounding wire's with Star washers, etc..
- Com1 Jumpers set correctly on PPC III Bd (Encoder = FDPLX, GPS = RS232, FDPLX After 06/14)
- Check RX1 LED is flashing for Com 1 of PPC III Bd.
  - RX1 LED flashes on 1 second, off 1 second, etc. GPS only flashes when system is in RUN.

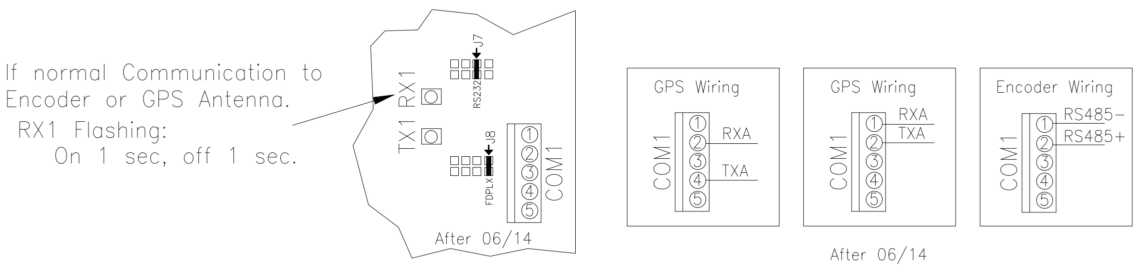


Figure 83.1 Check RX1 LED and Comm. wiring.

Perform "Com Loop Test" to check the system wires.

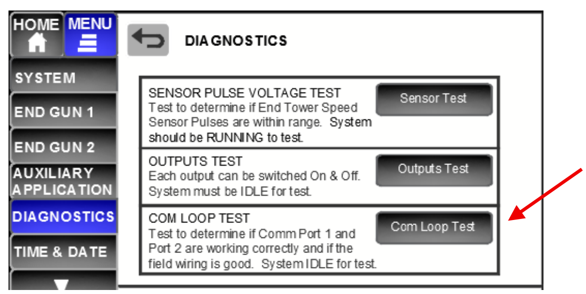
PPC III - System Setup
"Com Loop Test"

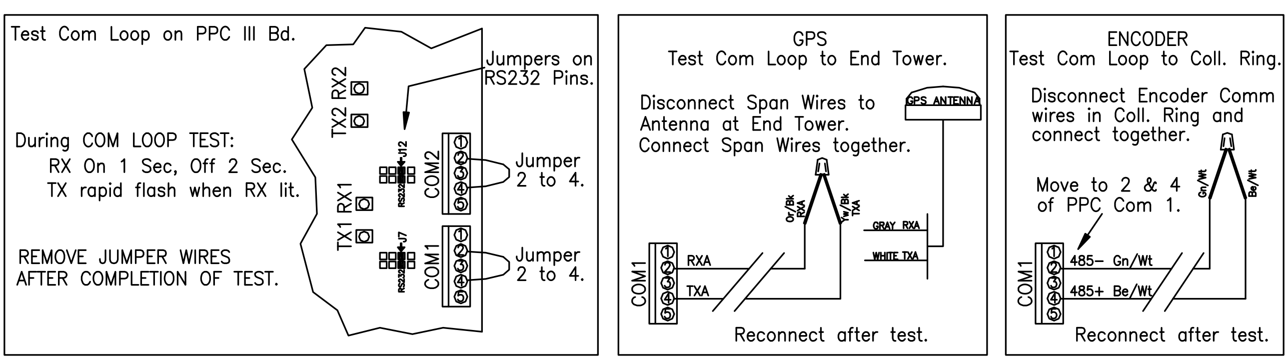
COM LOOP TEST
Yes

RS232 Jumper
Yes

CONFIG LOOP
Yes

PPT (Touch)
Menu – Diagnostics – Com Loop Test





TX1 014 TX2 014
RX1 014 RX2 000

Com Port 1 Com Port 2

Compare TX1 to RX1 (Comm 1)

Compare TX2 to RX2 (Comm 2)

Packets sent
Packets back

Should be equal. If not, Com Port or Wiring issue.

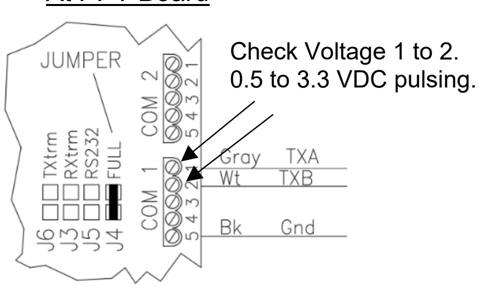
GPS Comm Lost Testing

Check Voltage on Com Port 1 of Board with System Running

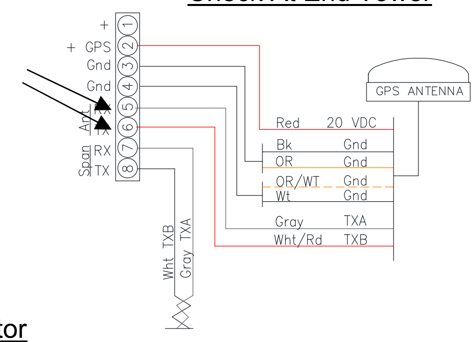
NOTE: System MUST be running to power the GPS Antenna at the End Tower Manifold.

- Voltage should pulse (fluctuate) between 0.5 to 3.3VDC.
- A steady voltage usually means there is no communication.
- Voltage on the low side not less than 1.5 VDC means grounding not good enough.

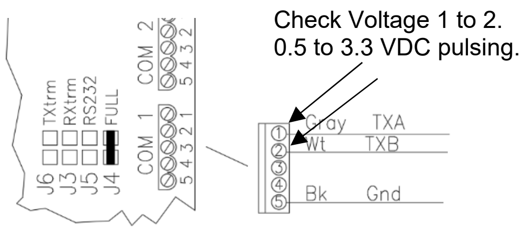
At PPT Board



Check At End Tower

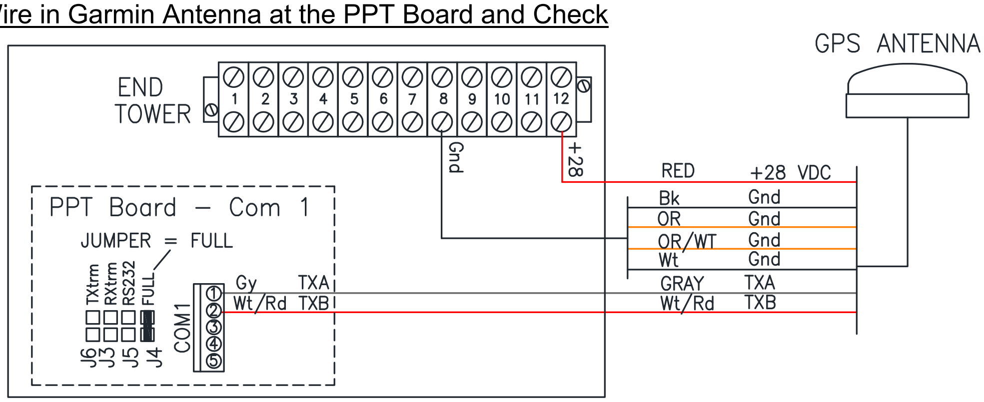


Remove Com Port 1 Connector of Board and Test at Connector



- If Pulsing, then the GPS Antenna and wiring is good.
- If voltage does not change when connected, but does pulse on the Connector when disconnected, then most likely an issue with Com 1 on PPT Board.

Wire in Garmin Antenna at the PPT Board and Check



End Tower Speed Limit – Orifice Disk Installation

To limit the top end speed, an Orifice Disk can be placed underneath the MFC Cartridge. The standard MFC flow rating is 3 GPM. Orifice sizes are available in 1.5, 2.0, and 2.5 GPM.

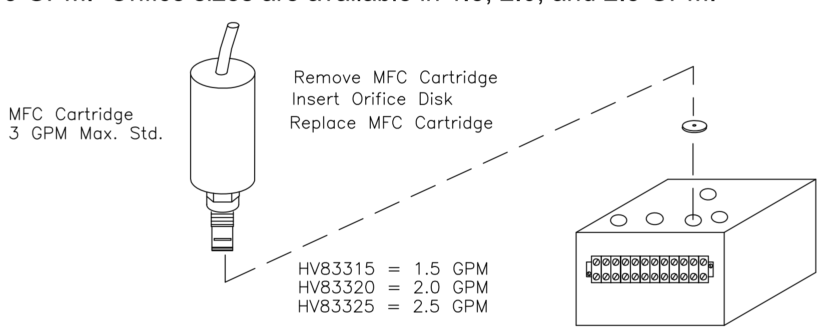


Figure 84. MFC Orifice Disk Installation.

8.2 Annual Maintenance Items

In addition to the regular maintenance items required for the pivot system (checking tire pressure, gear lube, filter changes, greasing pivot point, etc.), there are a few maintenance items required for the Point Control:

All Precision Point Control Systems
- check Control Panel at pivot point for loose wires
- check electrical cables and eliminate wear points
- open the End Tower Junction Box and inspect
- inspect Speed Sensor Cable and insure that it is not damaged

Reversing Point Control Systems
- check slide tube at End Tower for free movement in both directions

Collector Ring Systems
- Inspect Rings and Brushes for corrosion or buildup (see previous Figures)

Wrap Cable at Pivot Point
- check Plug and Socket contacts at pivot point for corrosion.
- clean and use electrician's di-electric grease to prevent corrosion.

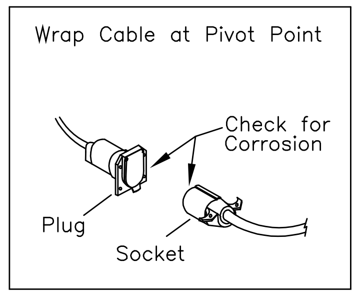


Figure 85. Check for Corrosion.

8.3 Service and/or Maintenance Parts Listing

Parts in PPT Panel:

AP50754	PPT LCD Screen Mount Strap
EA51055	DC-DC Converter, Engine 12 to 24 VDC (Green Bd)
EC53051	Electric Supply 28VDC Clipper Kit (Blue Bd)
EC50721	PPT LCD Touchscreen
EC50757	PPT Basic Faceplate Assy with PCB & Screen (Green 2-Chip)
EC50752	PPT Main PCB (Green 2-Chip)
EC50768	PPT Basic Faceplate Assy with PCB & Screen (Red 1-Chip)
EC50766	PPT Main PCB (Red 1-Chip)
EC50759	PPT Main Wire Harness
EC50764	Fuse 15 amp AGC - Input for DC-DC Green Board or Clipper Blue Board
EC50953	Fuse 5 amp MDL slow acting - Output for DC-DC Green Board or Clipper Blue Board
EC50964	Fuse 1 Amp ATC (MFC Spd– or Spd+)
EC50965	Fuse 5 Amp ATC (Main Power Fuse)
ES5277	PPT Power Rocker Switch Off-On
EW53680	#15 Pink Wire for Wiring Harness, to add End Gun Control to existing PPC II Harness.

Parts in Collector Ring:

EC53204	Brush Assy for HDC Collector Ring
EC5322	Encoder
EH53614	Collector Ring Spray Cleaner

Parts at End Tower:

EA52513	Rev. Switch Assy - Yellow Cable (Dec. 2011 -)
EC53231	GPS Antenna
EC98370	Coil Wires 36" Long with Deutsche Connector
EC98371	Coil Only with Deutsche Connector.
EH53240	Point Manifold PCB (6/04 -)
ES53218	Speed Sensor for Eaton Motor – M12
ES532180	Sensor Tester
ES53219	Sensor Cable for M12
HF57731	Replacement Element for Motor Port Filters in Manifold Body (2/09 -)
HF57778	Replacement Element for Pressure Filter in Manifold Body
HV98012	Pressure Compensator Cartridge
HV98032	Cavity Plug 8-3 (for 3-way cartridge)
HV98043	End Gun Solenoid Valve Cartridge
HV98063	Aluminum Manifold, Body Only, for End Tower
HV98064	MFC Cartridge Only – 3 GPM
HV98065	MFC Cartridge Only – 5 GPM
HV98337	Pressure Sequence Valve Cartridge
HV98382	Sterling Cartridge 2-Way for Booster Control Valve
HV98384	DC Solenoid 4-Way Valve Cartridge - Silver Plated (1/08 -)
HV98386	Aluminum Body with ¾" Ports for 2-Way Booster Control Valve
MB53243	Retrofit Kit – Manifold PCB & Snap Track for Manifold system with 12 terminal block.

Misc:

EC50450	Rectifier Assy for Electric Power Supply
EC53204	Brush Assy for HDC Collector Ring
EH53613	Weather Guard Corrosion Spray Protectant
FW56512	Star Washer 1/4"
FW56513	Star Washer 5/16"

9 APPENDIX 1: MAXIMUM SPEED TABLES

Gear Type	Planetary			**
Pumping Unit	7 1/2 HP Electric			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	90	104	95	108
8 SPANS	51	58	54	61
12 SPANS	36	42	36	42
16 SPANS	28	32	28	31
20 SPANS	22	26	22	26

End Gun Boosters N/A w/ 7½ HP

Gear Type	Worm			**
Pumping Unit	7 1/2 HP Electric			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	50	58	61	68
8 SPANS	32	36	37	42
12 SPANS	22	24	25	28
16 SPANS	17	19	19	21
20 SPANS	14	16	16	18

End Gun Boosters N/A w/ 7½ HP

Gear Type	Planetary			**
Pumping Unit	10 HP Electric, Isuzu 3CB1			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	102	117	121	137
8 SPANS	70	80	72	81
12 SPANS	49	56	49	56
16 SPANS	38	43	38	42
20 SPANS	30	35	31	35

End Gun Boosters N/R w/ 10 HP

Gear Type	Worm			**
Pumping Unit	10 HP Electric, Isuzu 3CB1			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	71	82	85	96
8 SPANS	43	49	50	57
12 SPANS	29	33	34	39
16 SPANS	23	25	26	29
20 SPANS	19	21	22	25

End Gun Boosters N/R w/ 10 HP

Gear Type	Planetary			**
Pumping Unit	15 HP Electric, Isuzu 3CE1			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	102	117	121	137
8 SPANS	102	117	108	122
12 SPANS	70	80	77	85
16 SPANS	56	65	57	64
20 SPANS	47	54	46	53

With 2 HP End Gun Booster = 66% of Table

Gear Type	Worm			**
Pumping Unit	15 HP Electric, Isuzu 3CE1			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	71	82	85	96
8 SPANS	63	72	75	84
12 SPANS	44	50	51	58
16 SPANS	34	38	39	44
20 SPANS	28	31	32	36

With 2 HP End Gun Booster = 66% of Table

Gear Type	Planetary			**
Pumping Unit	20 HP Electric, Belt Drive 1800 RPM			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	102	117	121	137
8 SPANS	102	117	121	137
12 SPANS	90*	104*	94*	105*
16 SPANS	70*	80*	72*	81*
20 SPANS	56*	65*	57*	65*

With 2 HP End Gun Booster = 75% of Table
With 5 HP End Gun Booster = 55% of Table

Gear Type	Worm			**
Pumping Unit	20 HP Electric, Belt Drive 1800 RPM			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	71	82	85	96
8 SPANS	71	82	85	96
12 SPANS	54	63	66	74
16 SPANS	43	49	50	57
20 SPANS	34	38	39	44

With 2 HP End Gun Booster = 75% of Table
With 5 HP End Gun Booster = 55% of Table

(Table continued on next page)

Table 1. Maximum End Tower Speed, Inches per Minute (continued)

Gear Type	Planetary			**
Pumping Unit	25 HP Elec, Belt 2200 RPM, Isuzu 4LE			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	102	117	121	137
8 SPANS	102	117	121	137
12 SPANS	102*	117*	112*	127*
16 SPANS	80*	93*	86*	97*
20 SPANS	68*	78*	69*	78*

With 2 HP End Gun Booster = 80% of Table
With 5 HP End Gun Booster = 65% of Table

Gear Type	Worm			**
Pumping Unit	25 HP Elec, Belt 2200 RPM, Isuzu 4LE			
	End Tower Tire Size			
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38
4 SPANS	71	82	85	96
8 SPANS	71	82	85	96
12 SPANS	66	76	79	89
16 SPANS	50	58	61	65
20 SPANS	40	46	47	54

With 2 HP End Gun Booster = 80% of Table
With 5 HP End Gun Booster = 65% of Table

*Speed may need to be reduced if operating at temperatures below 60 degrees F
**If the system uses an End Gun Booster, the maximum speed will be less than listed in the tables. Use notes at bottom of each table to calculate maximum speed.

NOTE: Maximum speeds calculated as follows: Worm gear systems = All P Motors
Planetary with 16.9 x 24 or 11.2 x 38 tires = All G Motors
Planetary with 11.2 x 24 or 14.9 x 24 tires = G Motors on End Tower, all others = E's

10 APPENDIX 2: CALIBRATION OF ROLLING RADIUS

Problem: Time it takes to make 1 Revolution does not match Hr/Rev on screen.
Solution: Rolling Radius needs to be adjusted in MENU - SYSTEM to match actual field conditions.

Step 1. Disable the End Tower Sensor Safety so automatic speed adjustment does not occur.
- this will allow system to travel at set IPM, without any changes.
- Disable in HOME - SAFETIES

Step 2. Find the "Measured IPM", actual end tower speed as measured with a tape measure and watch.
- Start the system moving. System can travel either FWD or REV.
- Take an average of the Inches per Minute (IPM) or cm/min over a 5 minute period.
- The End Tower speed can be manually adjusted if needed if the system is not traveling at a usable speed. Do this before taking the measurement.

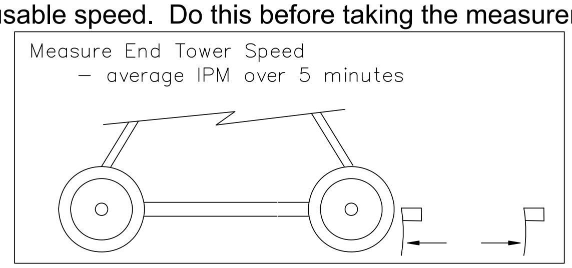
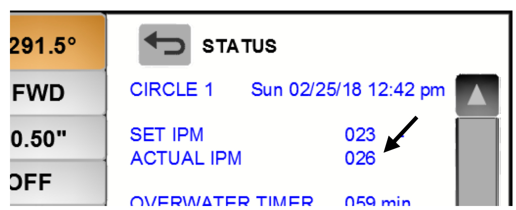


Figure XX. Find the Average IPM for 5 minutes.

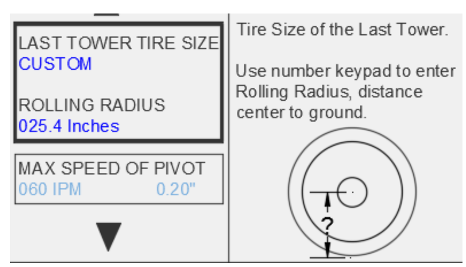
Measured IPM Example:
116 inches in 5 minutes.
= 23 IPM

Step 3. In HOME – STATUS, note what the "Actual IPM" of the End Tower shows on the screen.
(with the system running)
Example: Status Actual IPM = 26



Step 4. In MENU - SYSTEM, perform the following:
- Set the Last Tower Tire Size to "Custom"
- Adjust the Rolling Radius so the Actual IPM of the Status screen matches the measured 5 minute average.
  If the measured IPM is high, increase RR.
  If the measured IPM is low, decrease RR.

Note system must be stopped/Idle to make the change.



Example:
"Measured" IPM = 29
Status Actual IPM = 26
Adjust the RR up from 24.0 to 24.5 as a trial.
- see how that changes the Status Actual IPM

Continue to adjust the Rolling Radius until the Measured closely matches the Status Actual IPM.

11 APPENDIX 3: WIRING DIAGRAMS

(Taken from CD90443 PPT Installation Manual)

11.1 PPT with Swivel or Span Cable Wiring of PPT End Tower Terminal Block (Figure 200)

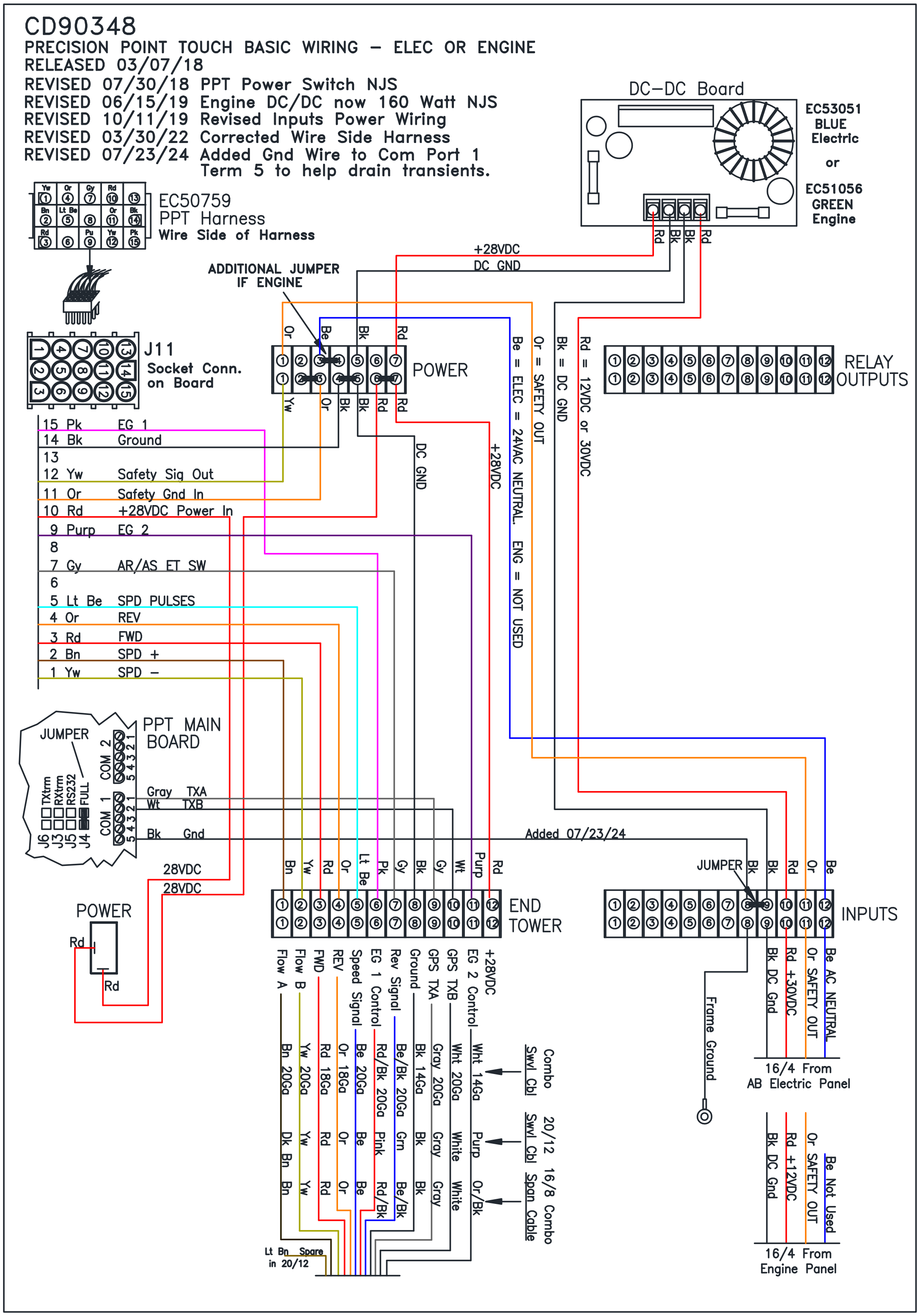


Figure 200. Swivel or Span Cable Wiring of End Tower Terminal Block in PPT Panel.

11.2 PPT – Retrofit Wiring, replacing existing PPC III Panel

Refer to "CD91221 PPT Board Retrofit Installation Sheet" for full instructions.

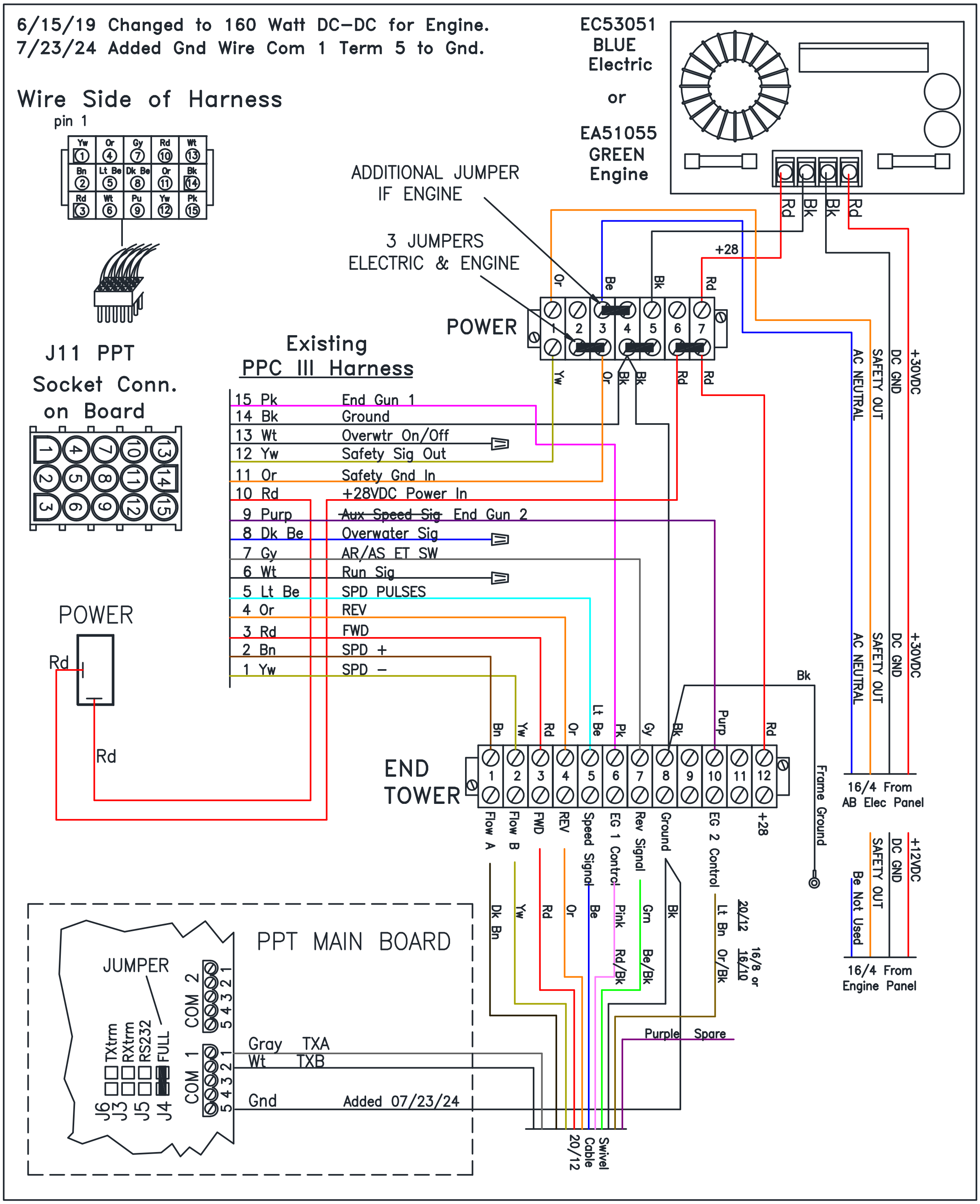
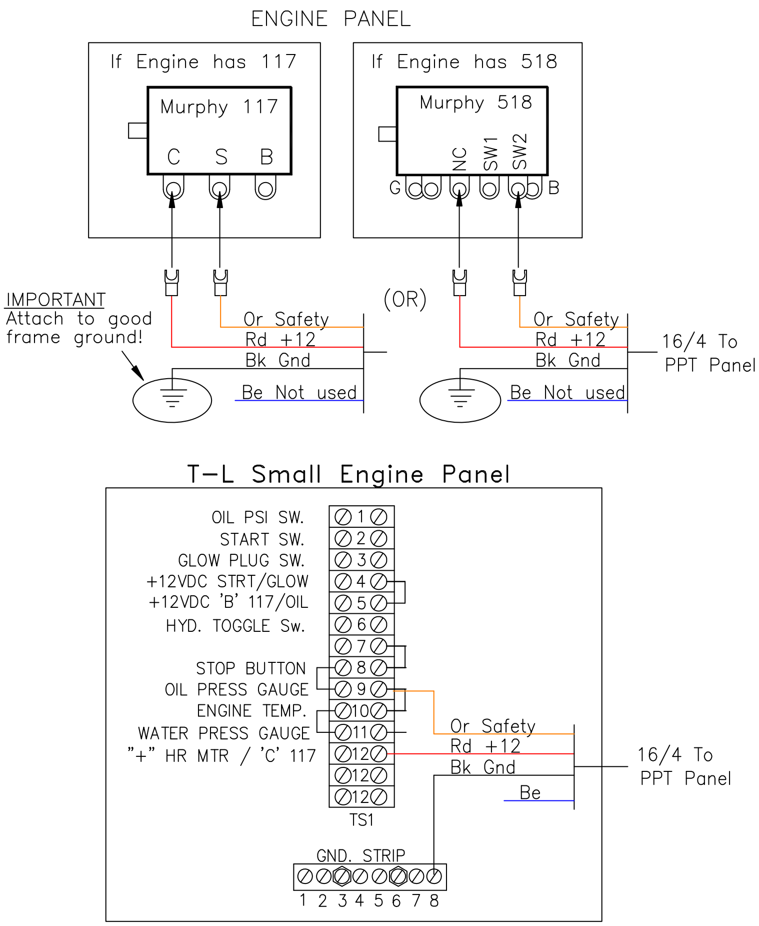


Figure 201. PPT Basic Retrofit from PPC III Wiring.

11.3 Power Cable Wiring to PPT Panel (Figures 202 & 203)

Engines



Part No.	Description	# Req'd.
EW53744	Cable 16/4 TC	45 ft

Figure 202. PPT Engine Power Wiring.

Electric – Allen Bradley

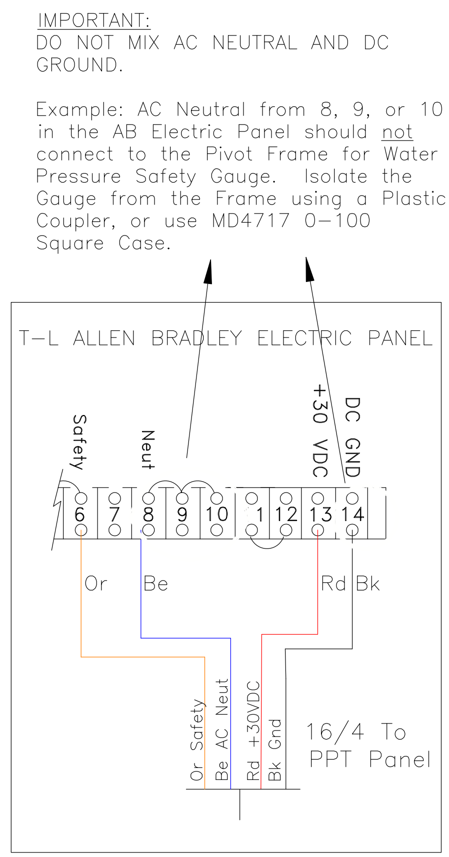


Figure 203. PPT Electric Power Wiring.

11.4 Wiring PPT Panel – Swivel Cable with Encoder 20/15 Cable (Figure 208)

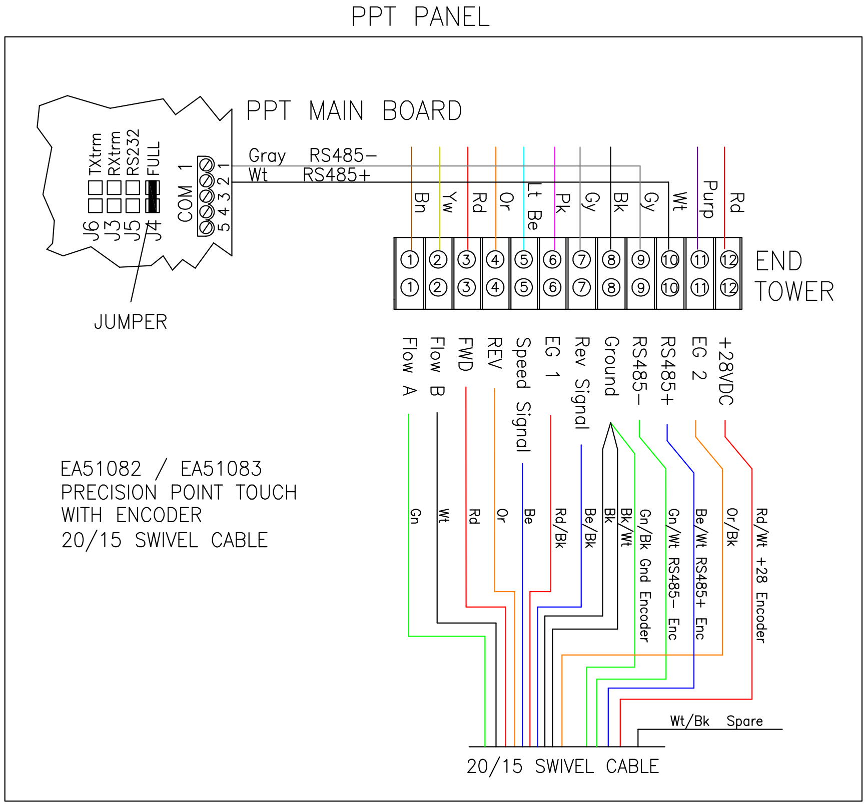


Figure 208. Wiring of Precision Point Touch Panel to 20/15 Gray Swivel Cable with Encoder.

11.5 Collector Ring Wiring with Encoder Position Sensor (Figure 209)

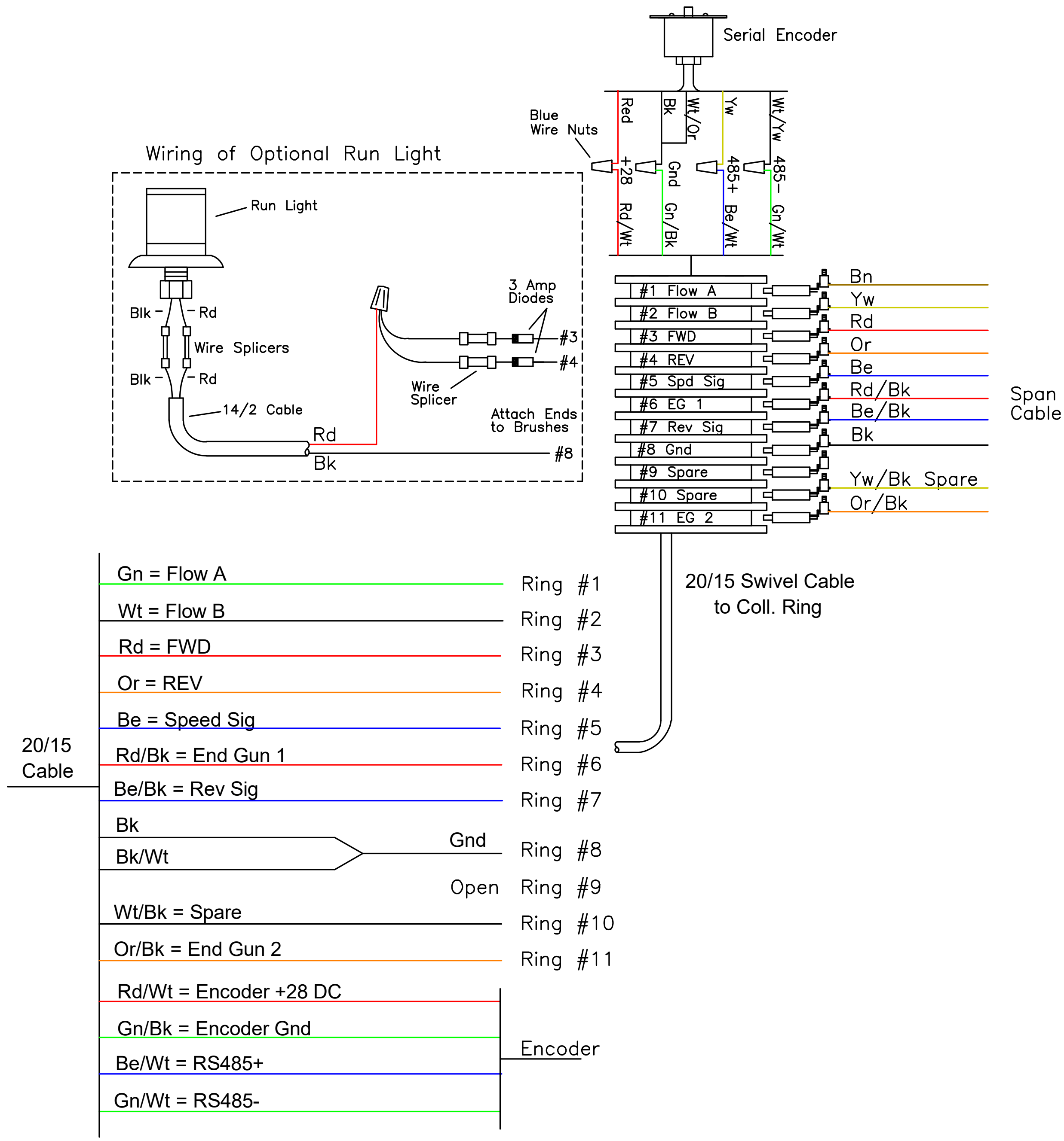


Figure 209. Collector Ring Wiring with Encoder.

11.6 Collector Ring Wiring with GPS Position Sensor (Figure 210)

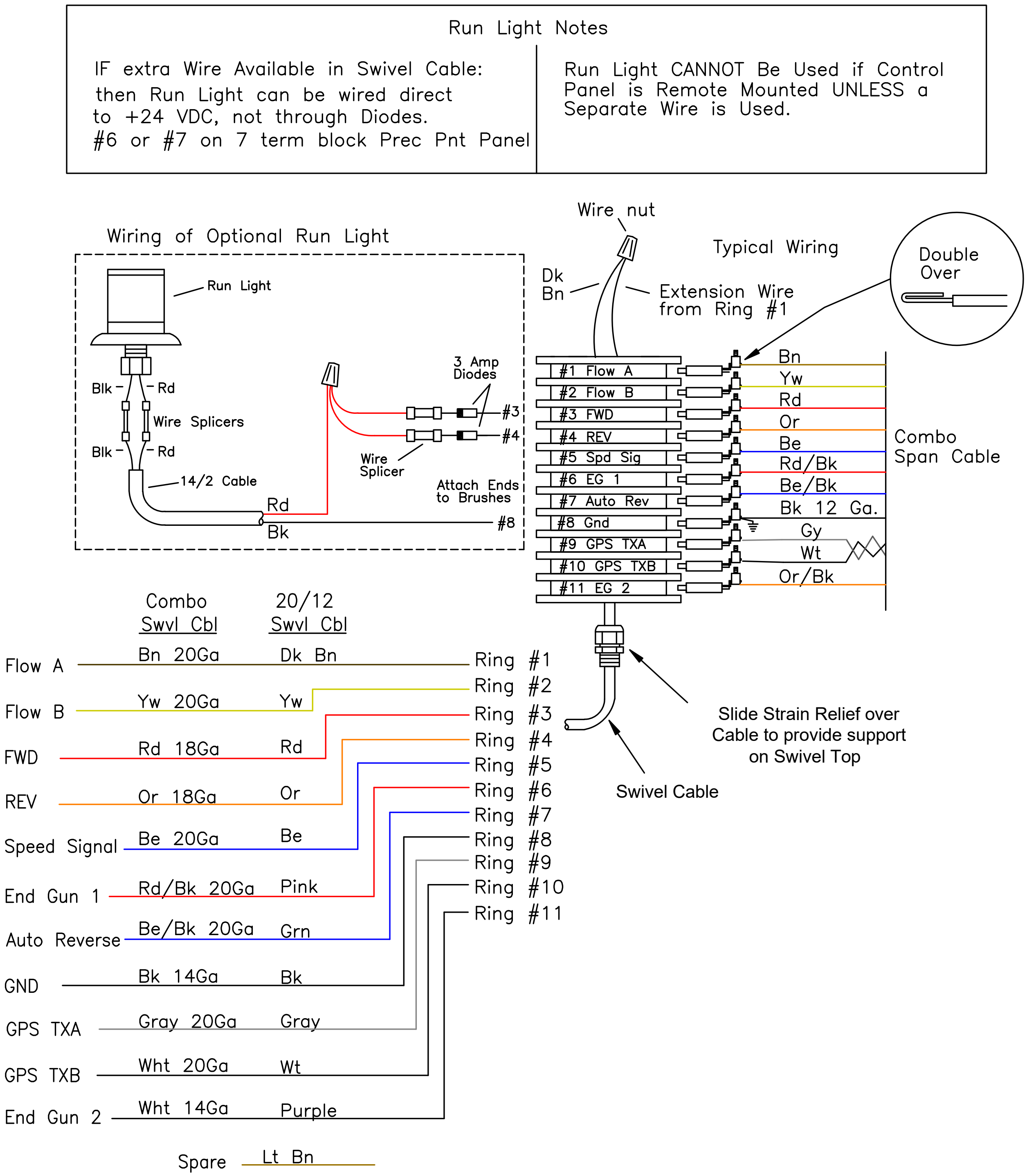


Figure 210. Collector Ring Wiring with GPS.

11.7 End Tower Control Wiring (Figure 211)

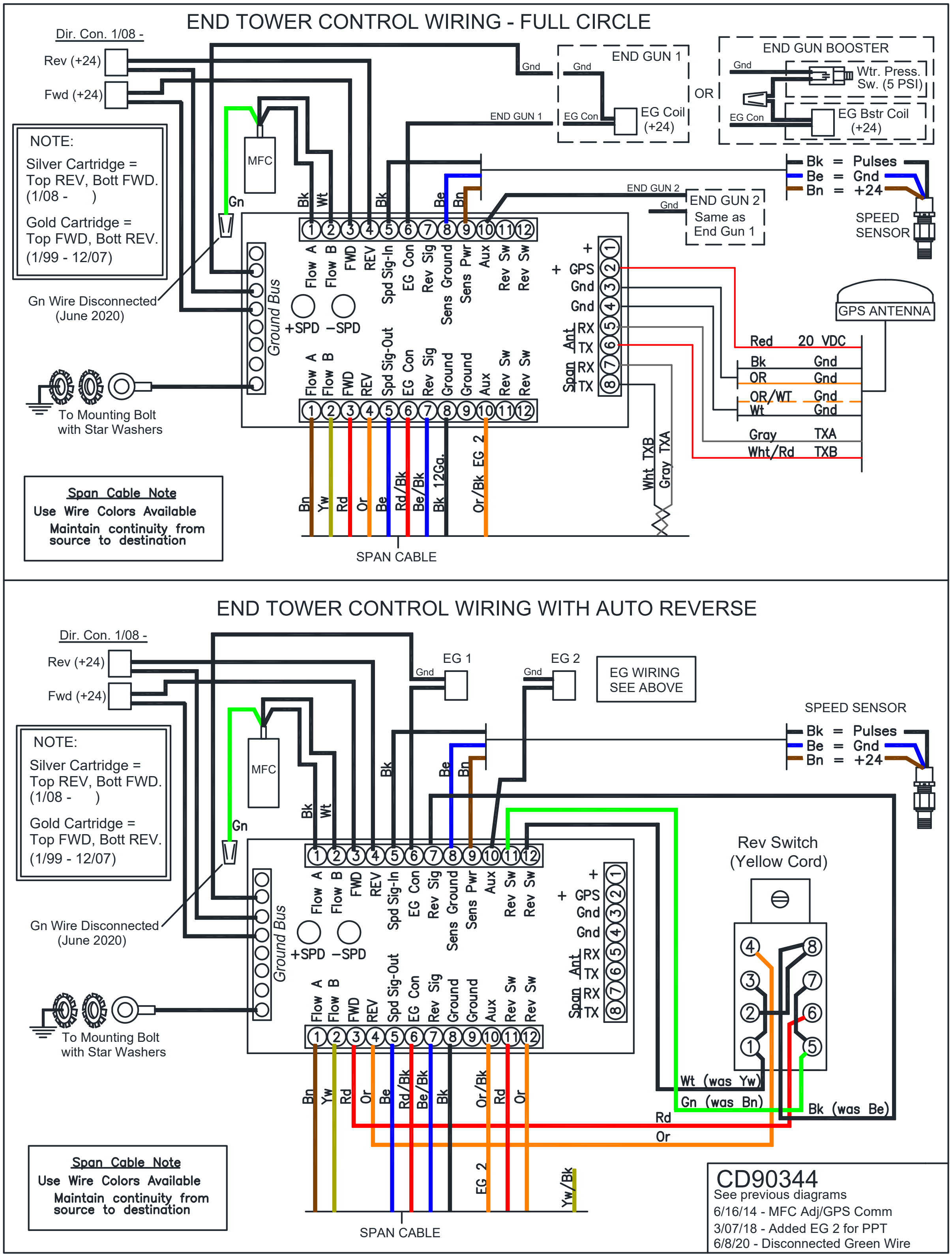


Figure 211. End Tower Control Wiring.

12 APPENDIX 4: SETUP RECORDS

System: ______________________

Date: ______________________

MENU - SYSTEM

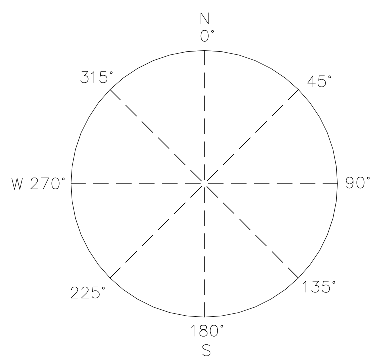
Item	Example	Setting	Choices/Units
Circle to Use	*Circle 1*	______	Circle 1, Circle 2, Circle 3
Units	*English*	______	English or Metric
Last Wheel Track	1246	______	Ft (101 – 3000 ft, 30 to 914m)
Irrigated Radius	*1350*	______	Ft (101 – 3200 ft, 30 to 975m)
Water Flow	*800*	______	GPM (50 – 4000 GPM, 3 to 250 l/s)
Last Tower Gear	*68:1*	______	Worm 50:1, Planetary 68:1, Planetary 24:1
Last Tower Tire	*14.9 X 24*	______	Recap 11.2, 14.9X24, 16.9X24, 11.X 38, Custom
If Custom, RR		______	inches (or cm)
Max Speed of Pivot	*96*	______	IPM (or cm/min)
	.20”	______	in. (or mm)
AR/AS Angle Ignore	*Auto Calculate*	______	Auto Calculate, Fixed 5 min.
Auto Restart	*No*	______	Yes or No
Start Delay Timer	*00*	______	minutes (0 to 10 mins.)
Speed Adj Rate	0.015	______	sec/IPM (0.003 to 0.050 sec/IPM or .001 to .020 s/cm)

MENU – END GUN 1

____________ Use Tables, Always Off, Always On

EG1 - Table 1

EG1 Table 1	Area 1	_______°	To	_______°
EG1 Table 1	Area 2	_______°	To	_______°
EG1 Table 1	Area 3	_______°	To	_______°
EG1 Table 1	Area 4	_______°	To	_______°
EG1 Table 1	Area 5	_______°	To	_______°
EG1 Table 1	Area 6	_______°	To	_______°
EG1 Table 1	Area 7	_______°	To	_______°
EG1 Table 1	Area 8	_______°	To	_______°



EG1 - Table 2

EG1 Table 2	Area 1	_______°	To	_______°
EG1 Table 2	Area 2	_______°	To	_______°
EG1 Table 2	Area 3	_______°	To	_______°
EG1 Table 2	Area 4	_______°	To	_______°
EG1 Table 2	Area 5	_______°	To	_______°
EG1 Table 2	Area 6	_______°	To	_______°
EG1 Table 2	Area 7	_______°	To	_______°
EG1 Table 2	Area 8	_______°	To	_______°

EG1 - Table 3

EG1 Table 3 Area 1 _______° To _______°

EG1 Table 3 Area 2 _______° To _______°

EG1 Table 3 Area 3 _______° To _______°

EG1 Table 3 Area 4 _______° To _______°

EG1 Table 3 Area 5 _______° To _______°

EG1 Table 3 Area 6 _______° To _______°

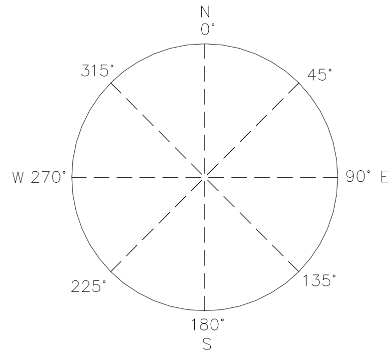
EG1 Table 3 Area 7 _______° To _______°

EG1 Table 3 Area 8 _______° To _______°

MENU – END GUN 2

_____________ Use Tables, Always Off, Always On

EG2 - Table 1



EG2 Table 1 Area 1 _______° To _______°

EG2 Table 1 Area 2 _______° To _______°

EG2 Table 1 Area 3 _______° To _______°

EG2 Table 1 Area 4 _______° To _______°

EG2 Table 1 Area 5 _______° To _______°

EG2 Table 1 Area 6 _______° To _______°

EG2 Table 1 Area 7 _______° To _______°

EG2 Table 1 Area 8 _______° To _______°

EG2 - Table 2

EG2 Table 2 Area 1 _______° To _______°

EG2 Table 2 Area 2 _______° To _______°

EG2 Table 2 Area 3 _______° To _______°

EG2 Table 2 Area 4 _______° To _______°

EG2 Table 2 Area 5 _______° To _______°

EG2 Table 2 Area 6 _______° To _______°

EG2 Table 2 Area 7 _______° To _______°

EG2 Table 2 Area 8 _______° To _______°

EG2- Table 3

EG2 Table 3 Area 1 _______° To _______°

EG2 Table 3 Area 2 _______° To _______°

EG2 Table 3 Area 3 _______° To _______°

EG2 Table 3 Area 4 _______° To _______°

EG2 Table 3 Area 5 _______° To _______°

EG2 Table 3 Area 6 _______° To _______°

EG2 Table 3 Area 7 _______° To _______°

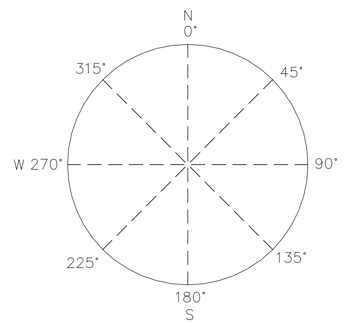
EG2 Table 3 Area 8 _______° To _______°

MENU – AUXILIARY APPLICATION

______________ Auxiliary Application - Disable or Enable
______________ Web VRI Table - Disable or Enable
Note: Web VRI Table settings are sent from Wagnet, not viewable on PPT.

Auxiliary Application Areas:

Fwd Area 1 Appl	_____	in. (mm)	______	º	to	______ º
Fwd Area 2 Appl	_____	in. (mm)	______	º	to	______ º
Fwd Area 3 Appl	_____	in. (mm)	______	º	to	______ º
Fwd Area 4 Appl	_____	in. (mm)	______	º	to	______ º
Rev Area 1 Appl	_____	in. (mm)	______	º	to	______ º
Rev Area 2 Appl	_____	in. (mm)	______	º	to	______ º
Rev Area 3 Appl	_____	in. (mm)	______	º	to	______ º
Rev Area 4 Appl	_____	in. (mm)	______	º	to	______ º



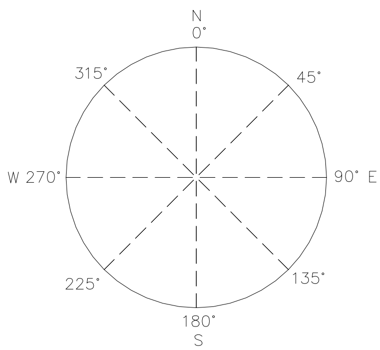
MENU - TIME & DATE

Time	_______	HH:MM AM/PM
Date	_______	MM:DD:YYYY

MENU - POSITION SENSOR

If Encoder

Current Angle ______ º
Direction Offset ______ º



If GPS: NOTE DD.dddddº

Circe 1 Pivot Point: ______.____________° N or S
______.____________° W or E

Circe 2 Pivot Point: ______.____________° N or S
______.____________° W or E

Circe 3 Pivot Point: ______.____________° N or S
______.____________° W or E

HOME - APPLICATION

Forward ______ inches or mm

Reverse ______ inches or mm

HOME – POSITION STOP

__________ Off or On

__________ Set Angle

HOME - AUTO STOP/REV

AUTO STOP / REVERSE ______ Yes or No

AUTO REVERSE DELAY TIMER ______ Min (0 to 45 mins.)

FORWARD ____________ Auto Stop or Auto Reverse

____________ By Switch or By Angle _______º Angle (if by Angle)

REVSERSE ____________ Auto Stop or Auto Reverse

____________ By Switch or By Angle _______º Angle (if by Angle)

HOME - STATUS

________ Total Run Time (HHHHH:MM)

MENU - SAFETIES

Item	Setting	
End Tower Sensor Safety	__________	Disable or Enable
	________	sec (30-120, Default is 075 secs)
Overwater Safety	__________	Disable or Enable
	________	min (10-120, Default is 060 min)
Wrong Direction Safety	__________	Disable or Enable
Position Sensor Safety	__________	Disable or Enable

NOTES